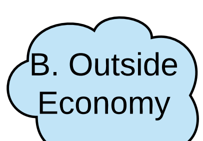
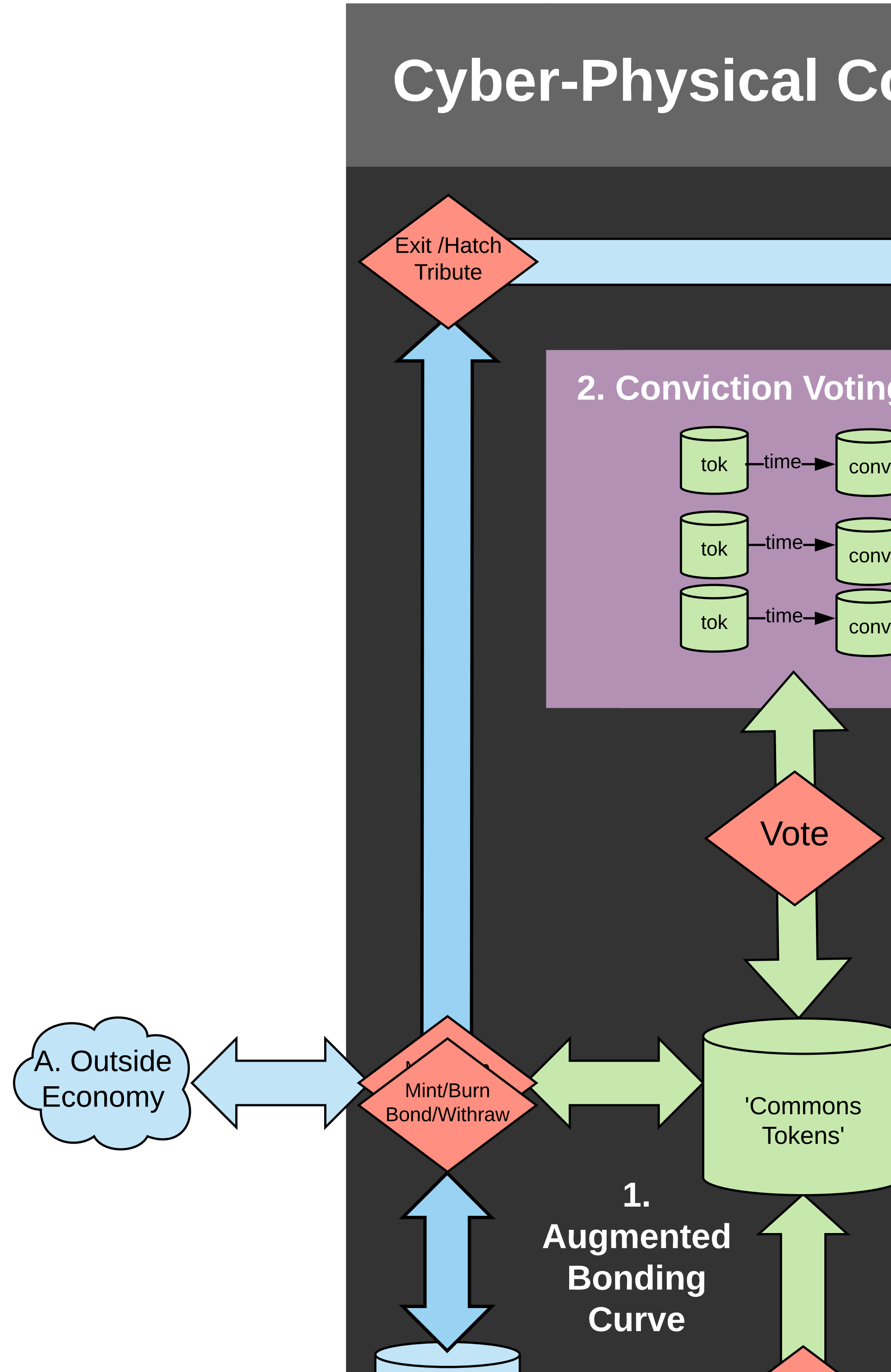
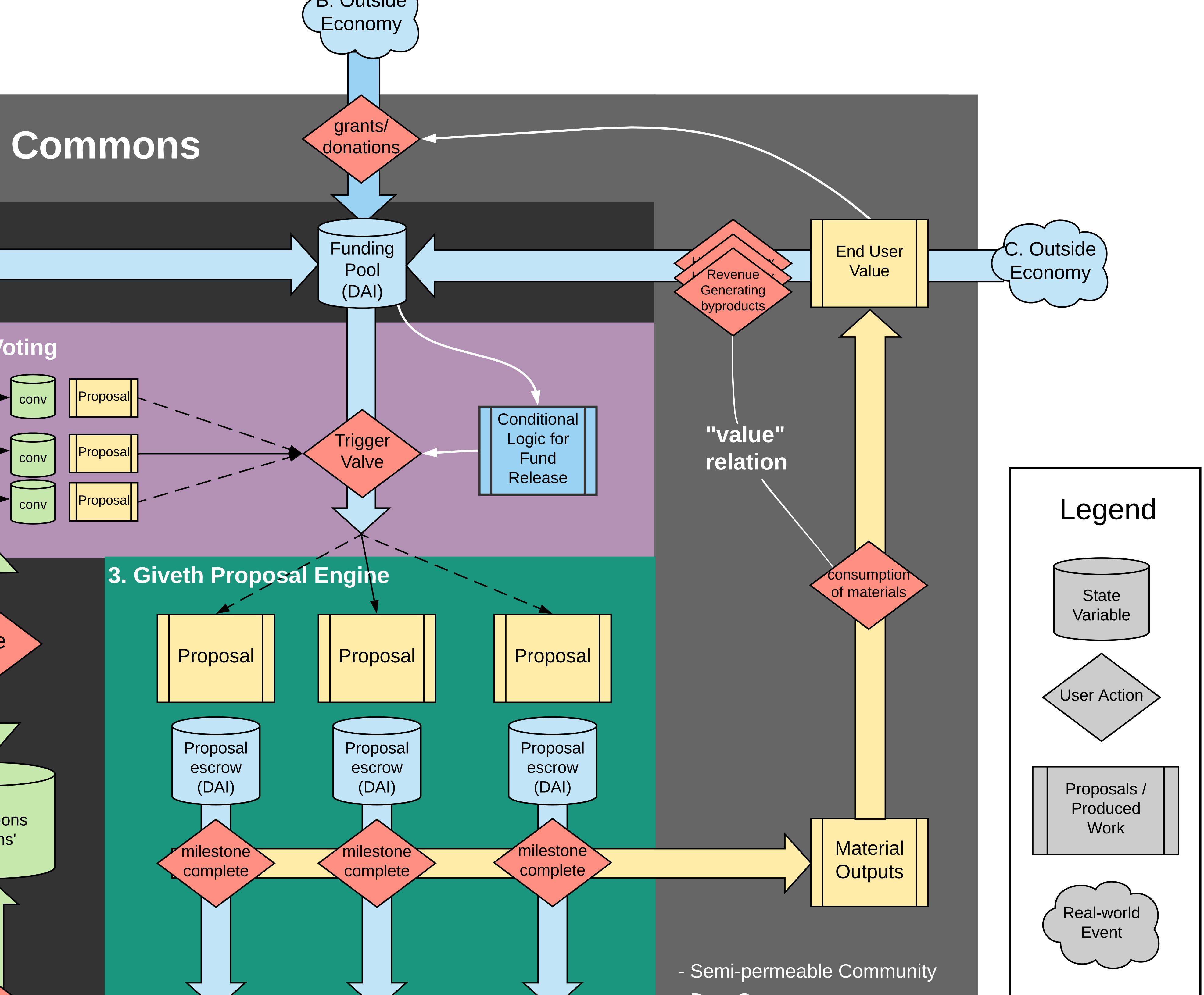

B. Outside
Economy



Cyber-Physical
Exit /Hatch Tribute
2. Conviction
tok
time
tok
time
tok
time
Vote
A. Outside Economy
Mint/Burn Bond/Withraw
'Commons Tokens'
1. Augmented Bonding Curve


B. Outside Economy
Commons
grants/ donations
Funding Pool (DAI)
Revenue Generating byproducts
End User Value
C. Outside Economy
conv
Proposal
conv
Proposal
conv
Proposal
Trigger Valve
Conditional Logic for Fund Release
"value" relation
consumption of materials
3. Giveth Proposal Engine
Proposal
Proposal
Proposal
Proposal escrow (DAI)
Proposal escrow (DAI)
Proposal escrow (DAI)
milestone complete
milestone complete
milestone complete
Material Outputs
Legend
State Variable
User Action
Proposals / Produced Work
Real-world Event
- Semi-permeable Community

Notes on Revenue

What have is a sort of mixture of the 3 types above

blockchain provenance really changes the relationship between club goods and common-pool resources. we should talk aboutit

Excludable

Rivalrous

Private goods food, clothing, cars, parking spaces
Club goods cinemas, private parks, satellite television

Non-rivalrous

https://en.wikipedia.org/wiki/Club_good

"Discount Token Model"
'Co-op' fee for license/use for non-members, that can be offset by members

Harberger tax for unique items

Digital

commons source software
(discount token model)

Implicit Observable common resources

(allows a macro excludability: anyone can participate but no-one can extract)

We're dealing with 2 "Hybrids"

Blockchain

forces to talk about

the observability dimension

directly enables the impact dependent 'costs"

"internal the externali
state dependent cost mo

enhance the after the fact visibility to strengthen normative response

open source software

Commons produced public goods;
Unobservable use common resource

s above

outit

"Harberger Tax Model"
pay for use, possession, can flow to
whomever needs it most

	Non-excludable
	Common-pool resources fish stocks, timber, coal
	Public goods free-to-air television, air, national defense

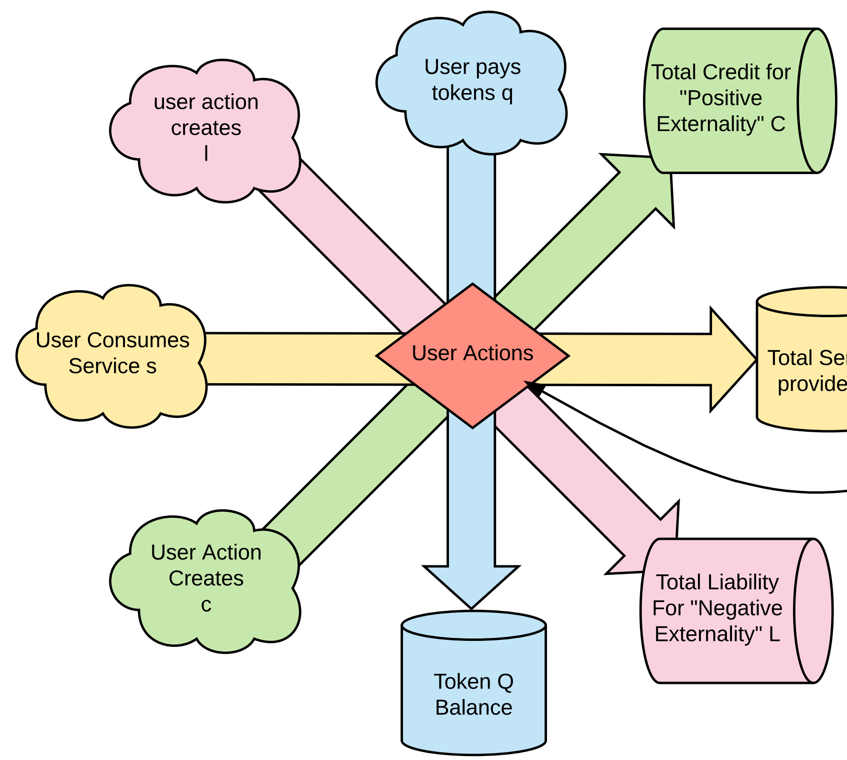
Grants from those who can
afford it, likely free use by
those who cannot

x for
ms

Physical

r:

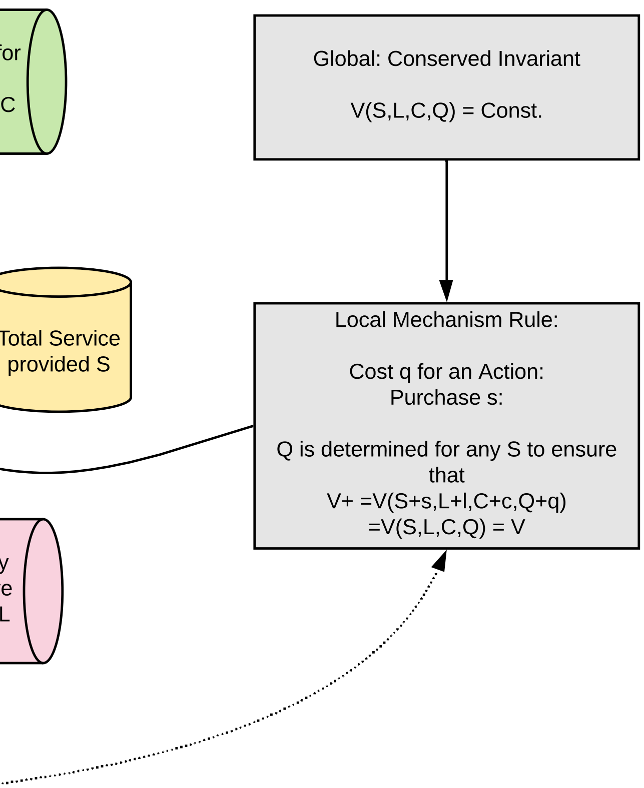
timber if truly tracked
could have its externalies internalized on the consumer
in which case it has some properties of a club good



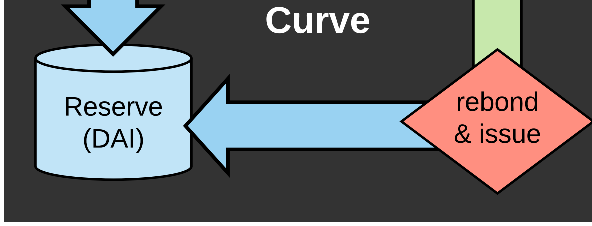
externality"
: cost models

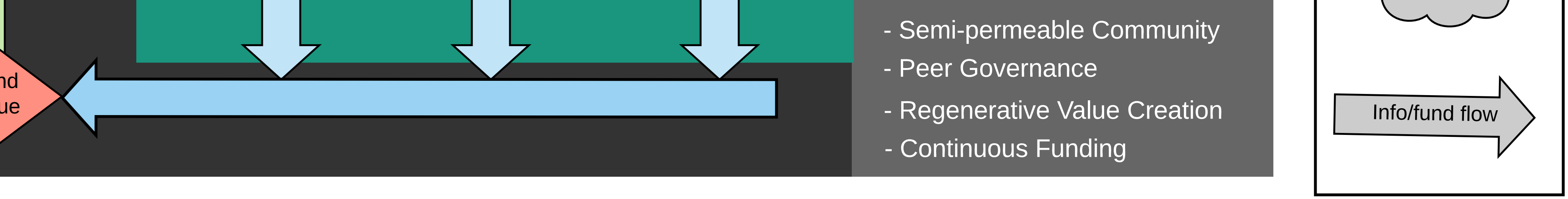
untracked or unobservable
externalities associate with with
common pool resources such
as environmental resources

oods;
resources

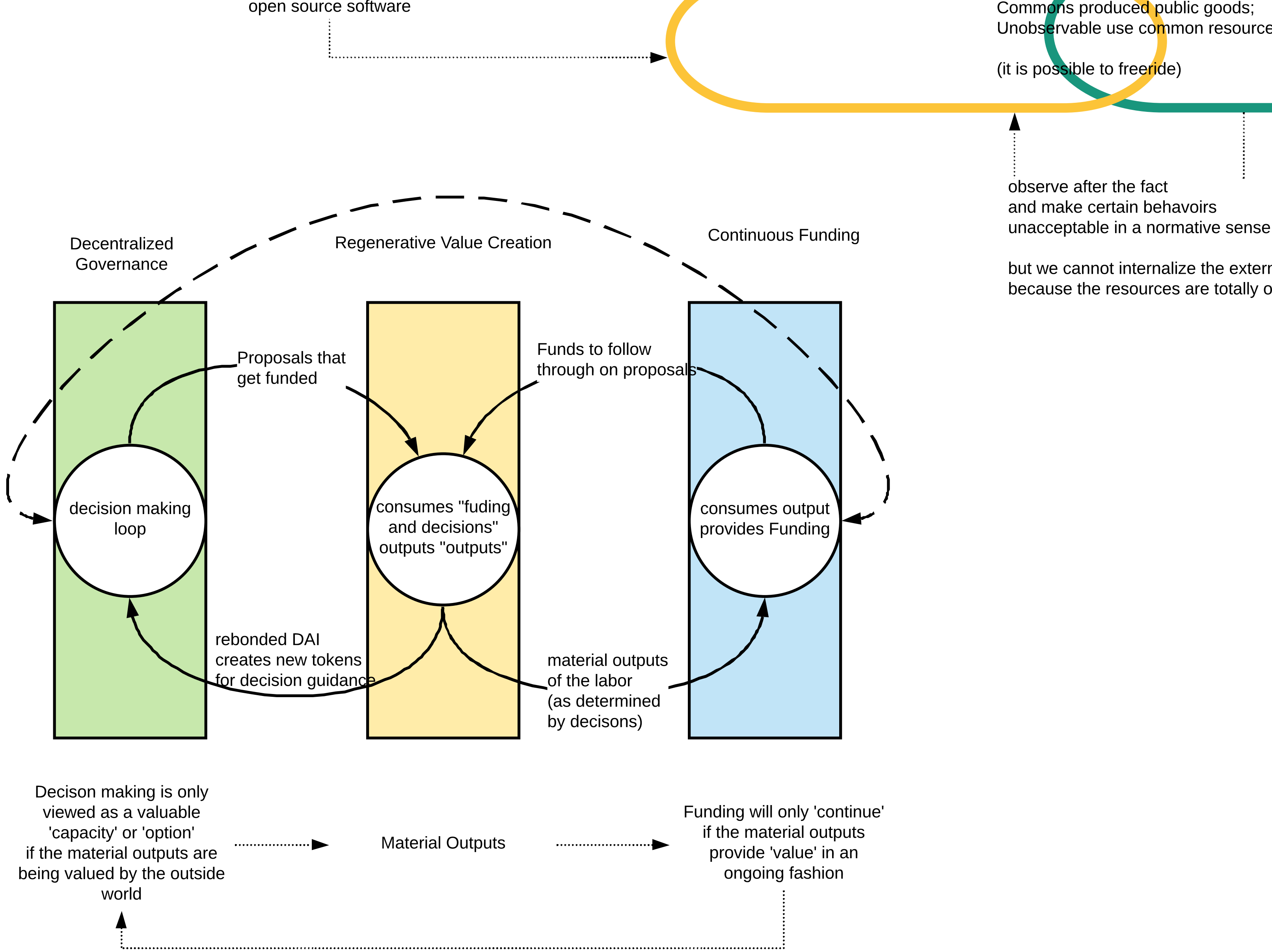


for
C
Global: Conserved Invariant
V(S,L,C,Q) = Const.
Total Service provided S
Local Mechanism Rule:
Cost q for an Action:
Purchase s:
Q is determined for any S to ensure that
V+ =V(S+s,L+l,C+c,Q+q)
=V(S,L,C,Q) = V
y
e
L

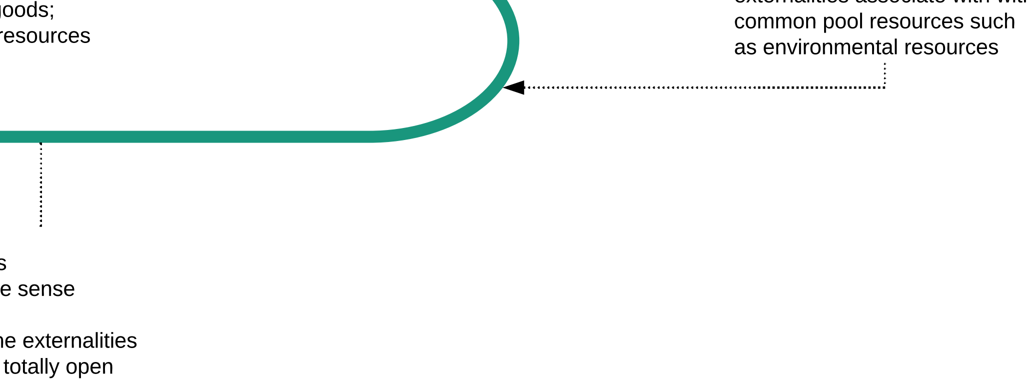

Curve
Reserve
(DAI)
rebond
& issue



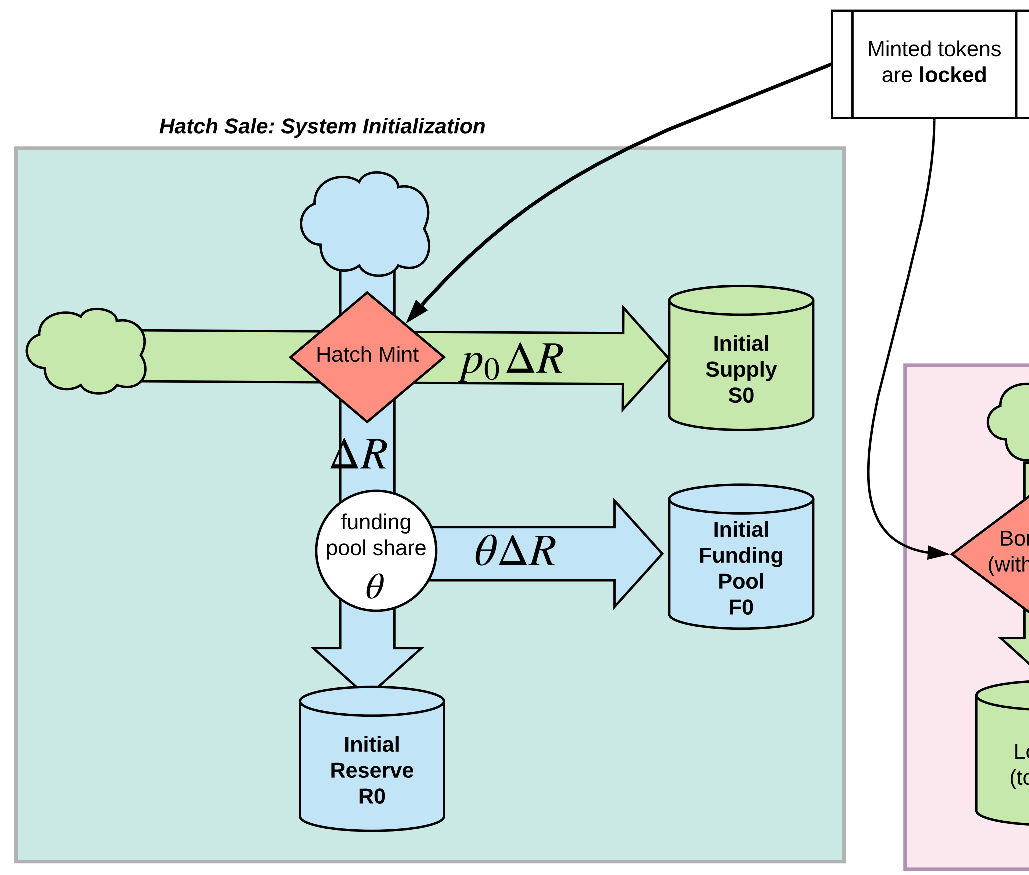
- Continuous funding is limited to philantrophy, value creation, and investment (crowdfunding vs. traditional)



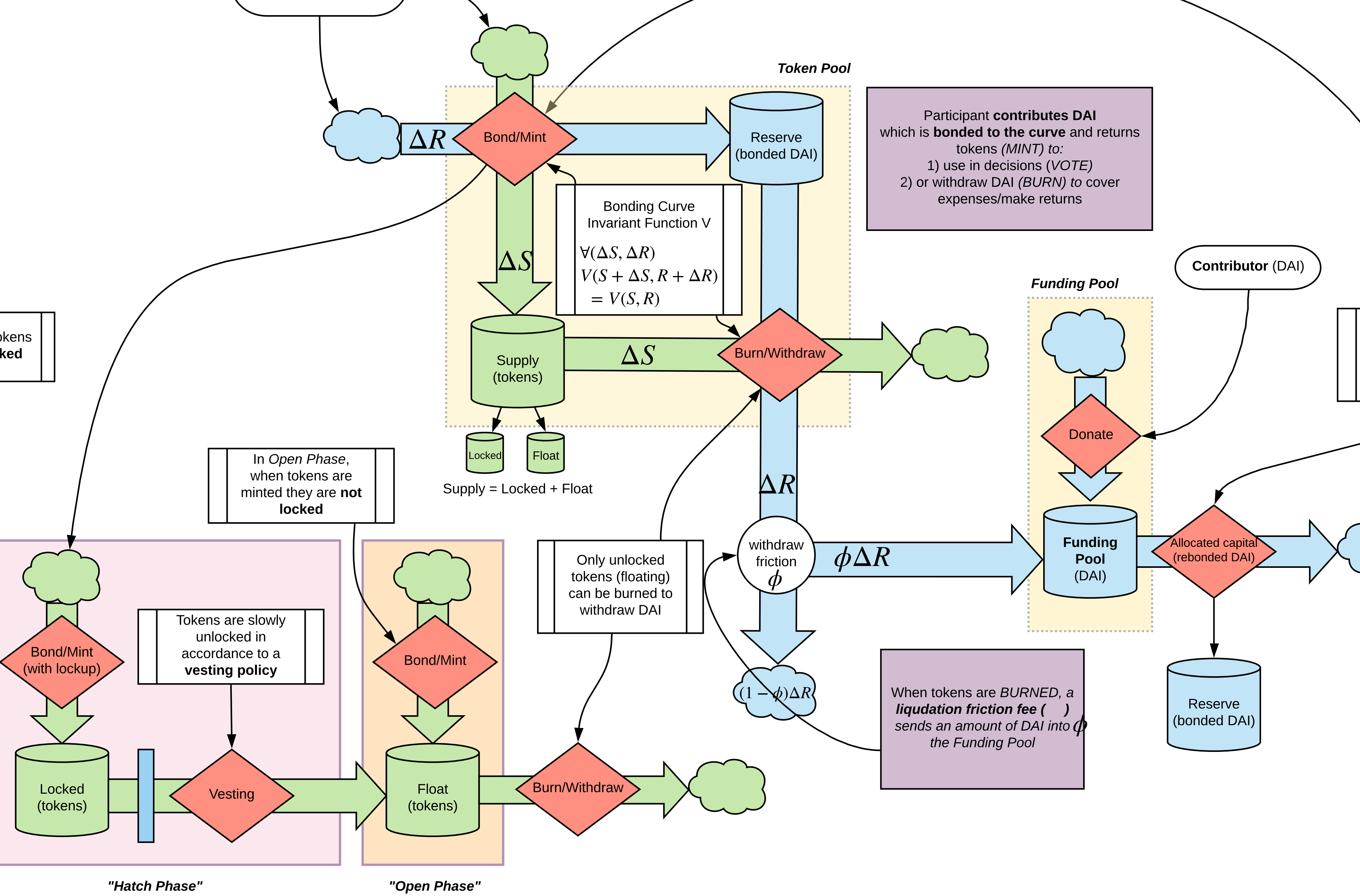
open source software
Commons produced public goods;
Unobservable use common resource
(it is possible to freeride)
observe after the fact
and make certain behavoirs
unacceptable in a normative sense
but we cannot internalize the extern
because the resources are totally o
Decentralized
Governance
Regenerative Value Creation
Continuous Funding
Proposals that
get funded
Funds to follow
through on proposals
decision making
loop
consumes "fuding
and decisions"
outputs "outputs"
consumes output
provides Funding
rebonded DAI
creates new tokens
for decision guidance
material outputs
of the labor
(as determined
by decisons)
Decison making is only
viewed as a valuable
'capacity' or 'option'
if the material outputs are
being valued by the outside
world
Material Outputs
Funding will only 'continue'
if the material outputs
provide 'value' in an
ongoing fashion

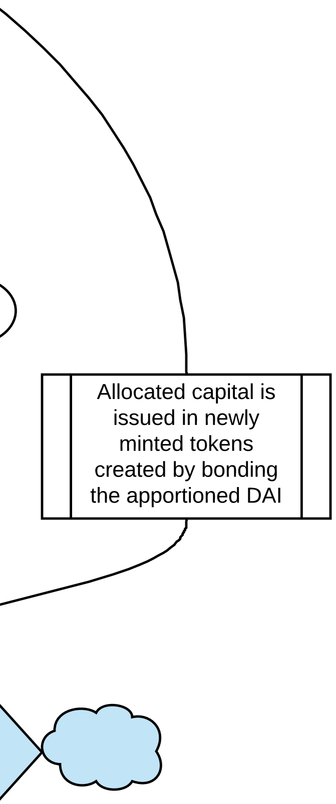

oods;
resources
common pool resources such
as environmental resources
s
e sense
ne externalities
totally open

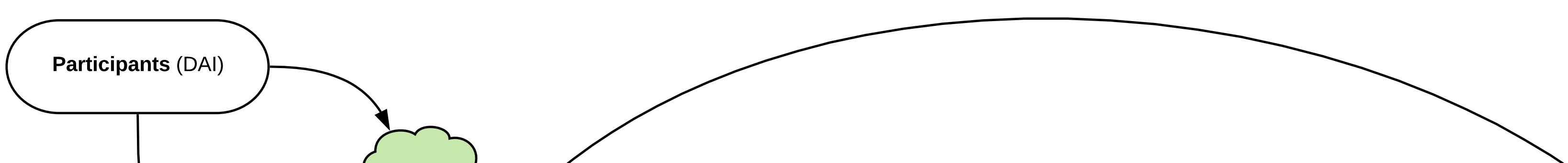

Participants (DAI)

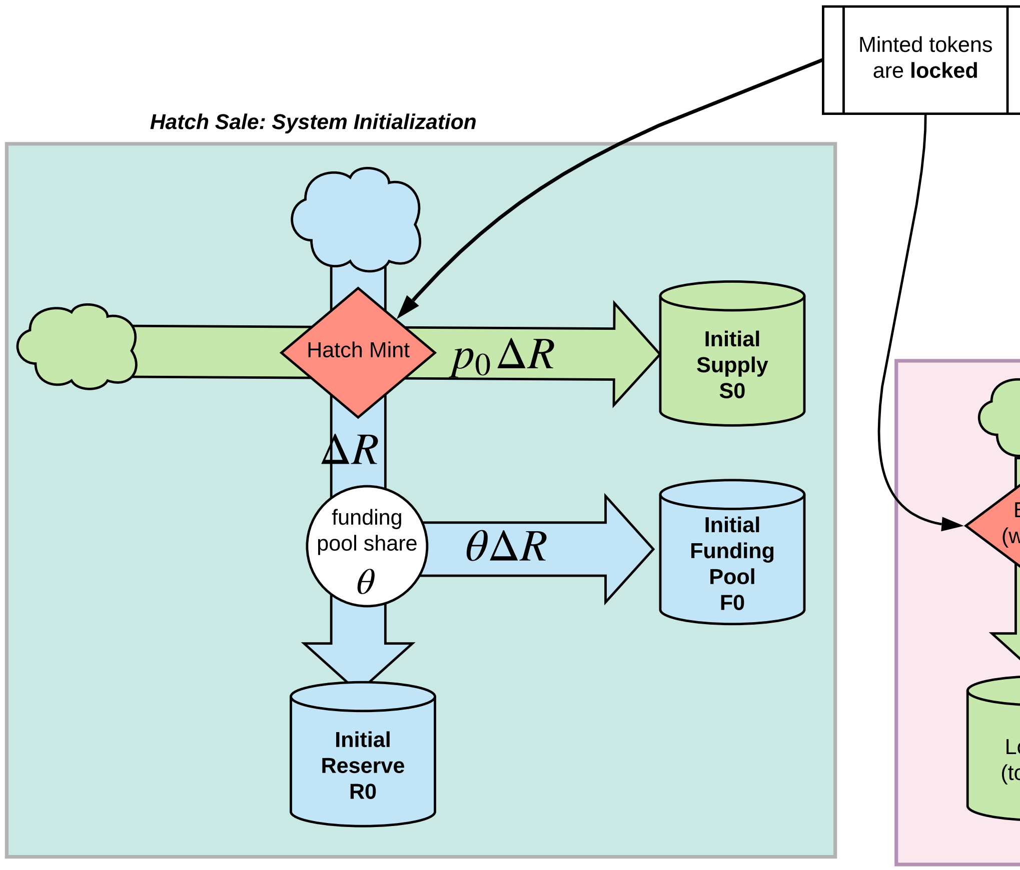


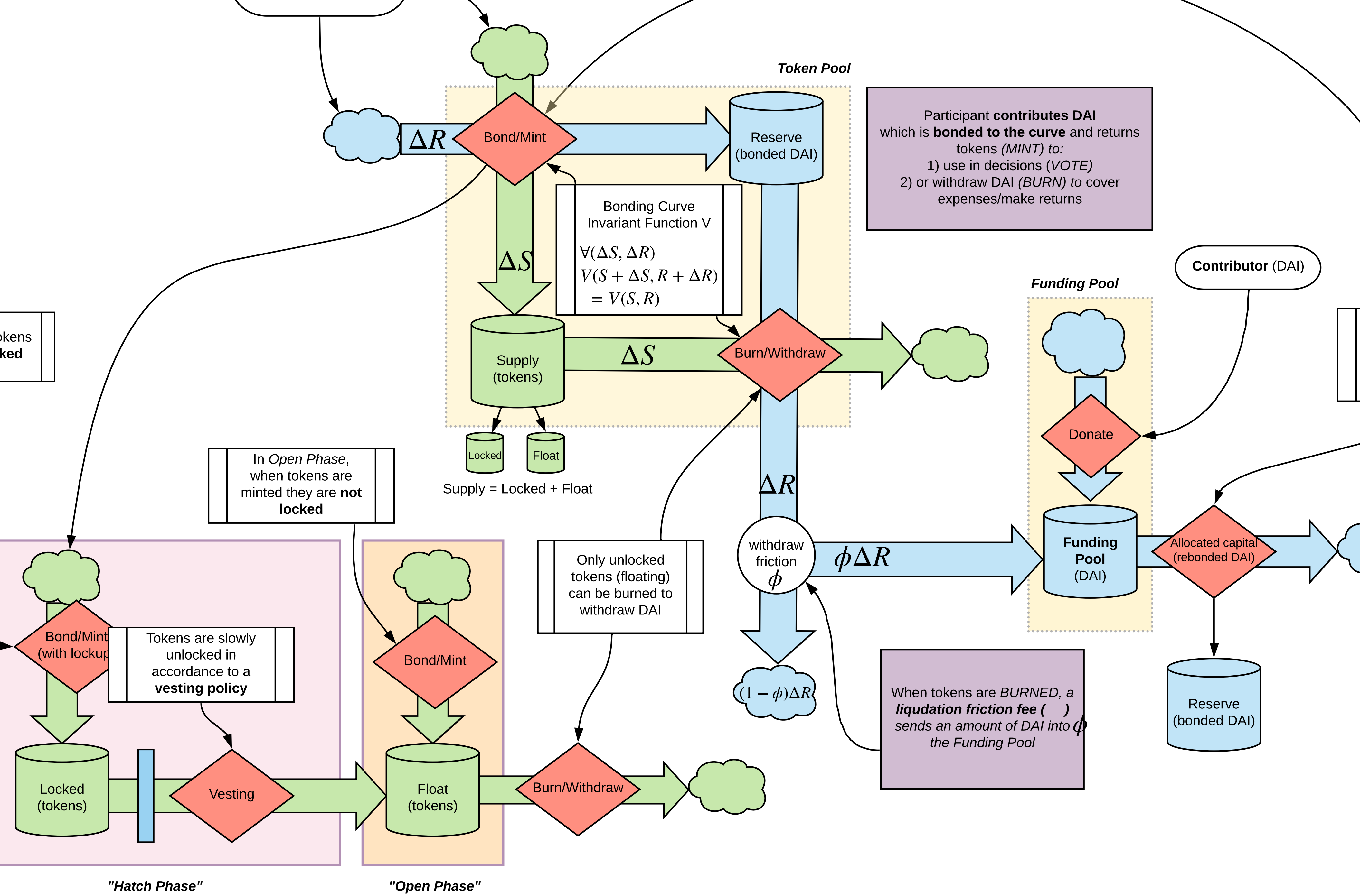
Minted tokens are **locked**
Hatch Sale: System Initialization
Hatch Mint
$p_0 \Delta R$
Initial Supply S0
ΔR
funding pool share θ
$\theta \Delta R$
Initial Funding Pool F0
Initial Reserve R0


Token Pool
Bond/Mint
ΔR
Reserve (bonded DAI)
Participant **contributes DAI** which is **bonded to the curve** and returns tokens *(MINT) to:*
1) use in decisions (*VOTE)*
2) or withdraw DAI *(BURN) to* cover expenses/make returns
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S+\Delta S, R+\Delta R)$
$= V(S,R)$
ΔS
Supply (tokens)
Burn/Withdraw
ΔS
Locked
Float
Supply = Locked + Float
okens
ked
Contributor (DAI)
Funding Pool
Donate
In *Open Phase*, when tokens are minted they are **not locked**
ΔR
withdraw friction ϕ
$\phi\Delta R$
Funding Pool (DAI)
Allocated capital (rebonded DAI)
Only unlocked tokens (floating) can be burned to withdraw DAI
Bond/Mint (with lockup)
Tokens are slowly unlocked in accordance to a **vesting policy**
Bond/Mint
$(1-\phi)\Delta R$
When tokens are *BURNED, a* ***liqudation friction fee (*** ϕ ***)*** *sends an amount of DAI into the Funding Pool*
Reserve (bonded DAI)
Locked (tokens)
Vesting
Float (tokens)
Burn/Withdraw
"Hatch Phase"
"Open Phase"


Allocated capital is issued in newly minted tokens created by bonding the apportioned DAI


Participants (DAI)


Minted tokens
are **locked**
Hatch Sale: System Initialization
Hatch Mint
$p_0\,\Delta R$
Initial
Supply
S0
ΔR
funding
pool share
θ
$\theta\Delta R$
Initial
Funding
Pool
F0
Initial
Reserve
R0


Token Pool
ΔR
Bond/Mint
Reserve (bonded DAI)
Participant contributes DAI which is bonded to the curve and returns tokens (MINT) to:
1) use in decisions (VOTE)
2) or withdraw DAI (BURN) to cover expenses/make returns
ΔS
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S+\Delta S, R+\Delta R)$
$= V(S,R)$
Contributor (DAI)
Funding Pool
Supply (tokens)
ΔS
Burn/Withdraw
Donate
Locked
Float
Supply = Locked + Float
In Open Phase, when tokens are minted they are not locked
ΔR
withdraw friction ϕ
$\phi\Delta R$
Funding Pool (DAI)
Allocated capital (rebonded DAI)
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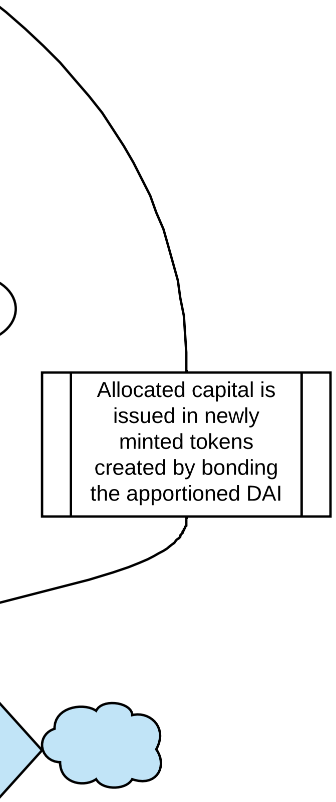
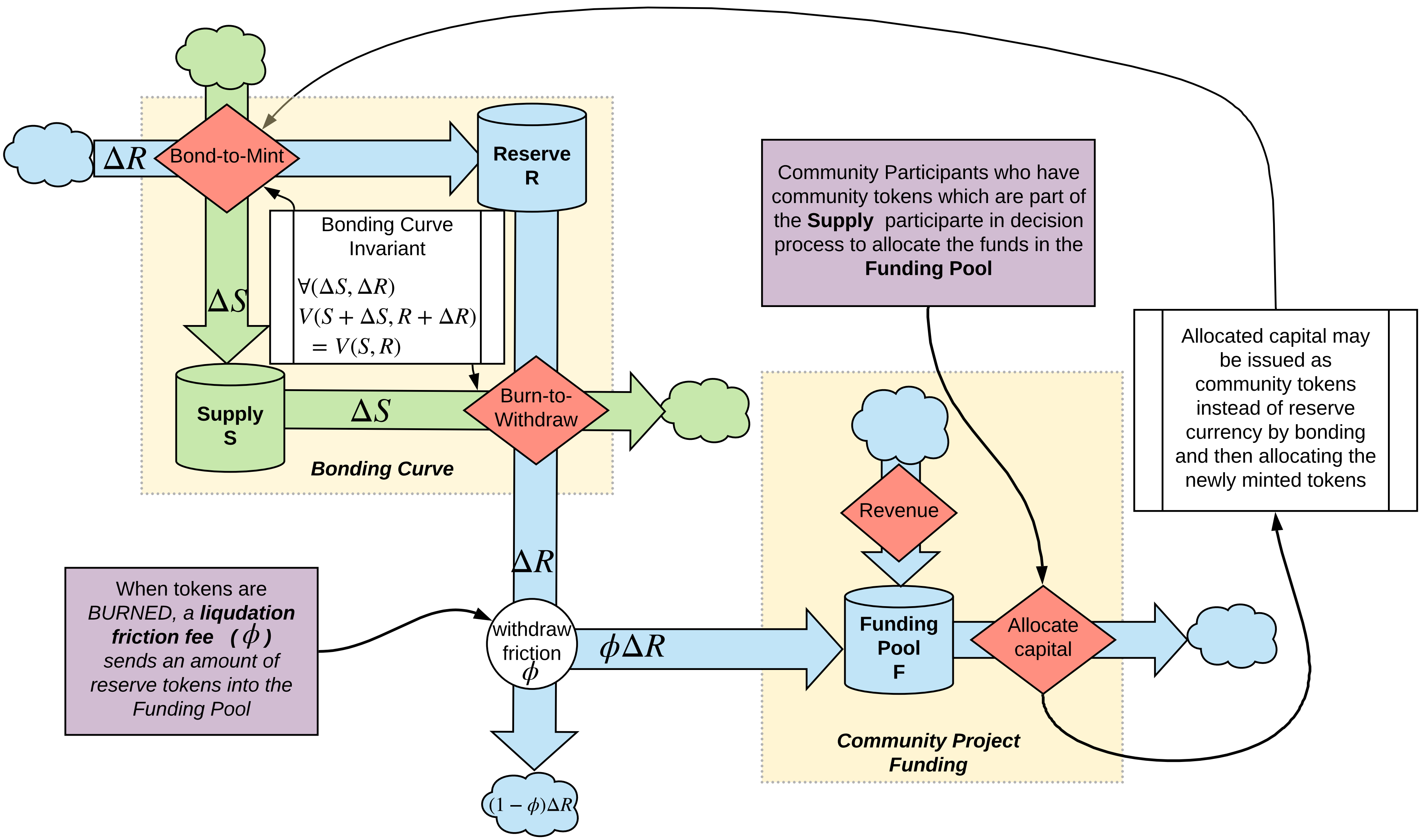
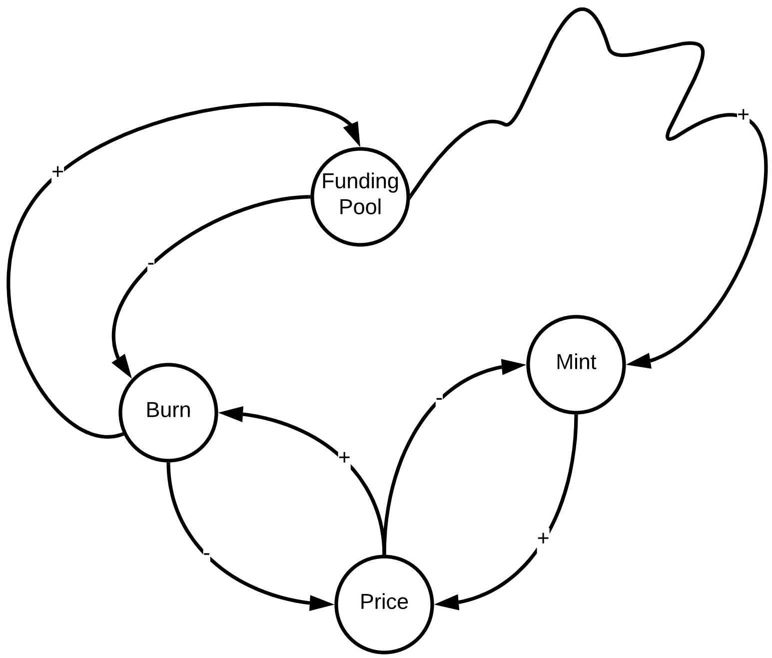
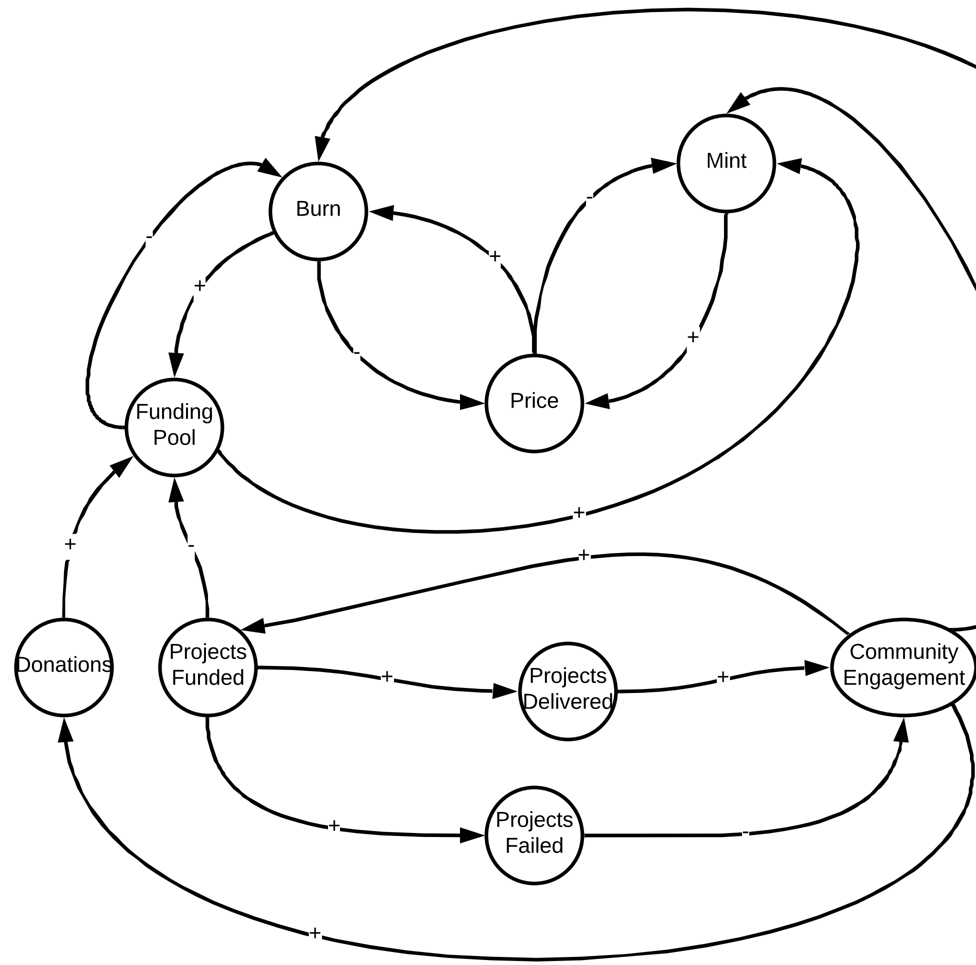
Bond/Mint (with lockup
Tokens are slowly unlocked in accordance to a vesting policy
Bond/Mint
$(1-\phi)\Delta R$
When tokens are BURNED, a liqudation friction fee (ϕ) sends an amount of DAI into the Funding Pool
Reserve (bonded DAI)
Locked (tokens)
Vesting
Float (tokens)
Burn/Withdraw
"Hatch Phase"
"Open Phase"


Allocated capital is issued in newly minted tokens created by bonding the apportioned DAI

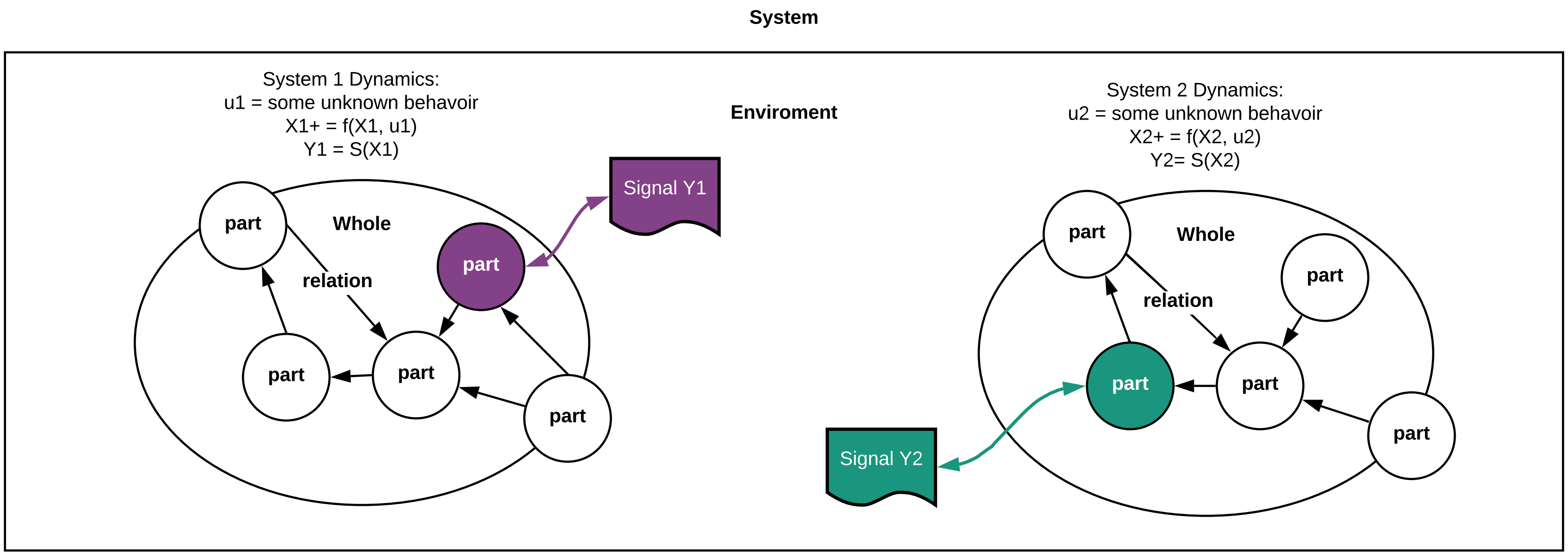


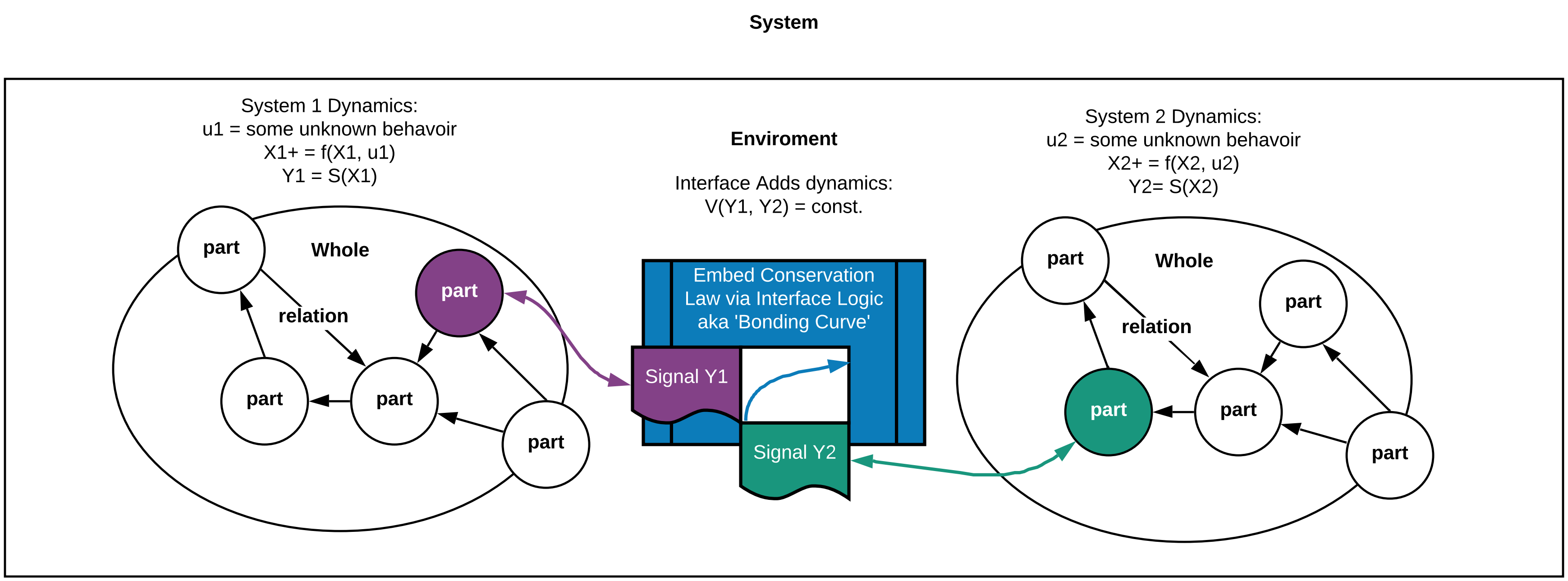
ΔR
Bond-to-Mint
Reserve R
Bonding Curve Invariant
$\forall(\Delta S, \Delta R)$
$V(S+\Delta S, R+\Delta R)$
$= V(S,R)$
ΔS
Supply S
ΔS
Burn-to-Withdraw
Bonding Curve
Community Participants who have community tokens which are part of the **Supply** participarte in decision process to allocate the funds in the **Funding Pool**
Allocated capital may be issued as community tokens instead of reserve currency by bonding and then allocating the newly minted tokens
Revenue
ΔR
When tokens are *BURNED, a* ***liqudation friction fee*** (ϕ) *sends an amount of reserve tokens into the Funding Pool*
withdraw friction ϕ
$\phi\Delta R$
Funding Pool F
Allocate capital
Community Project Funding
$(1-\phi)\Delta R$


Funding Pool
Burn
Mint
Price
+
-
+
-
+
-
+


Burn
Mint
Price
Funding Pool
Donations
Projects Funded
Projects Delivered
Community Engagement
Projects Failed
+
-
+
-
+
-
+
+
-
+
+
+
+
-
+

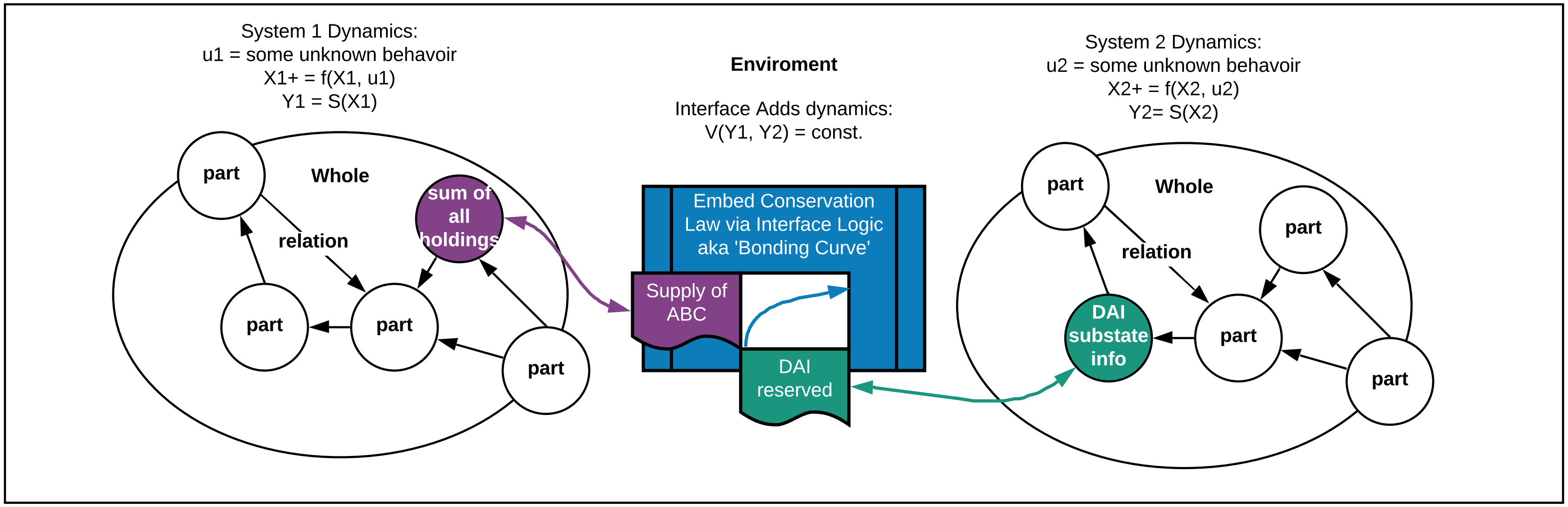
+
-
munity
gement





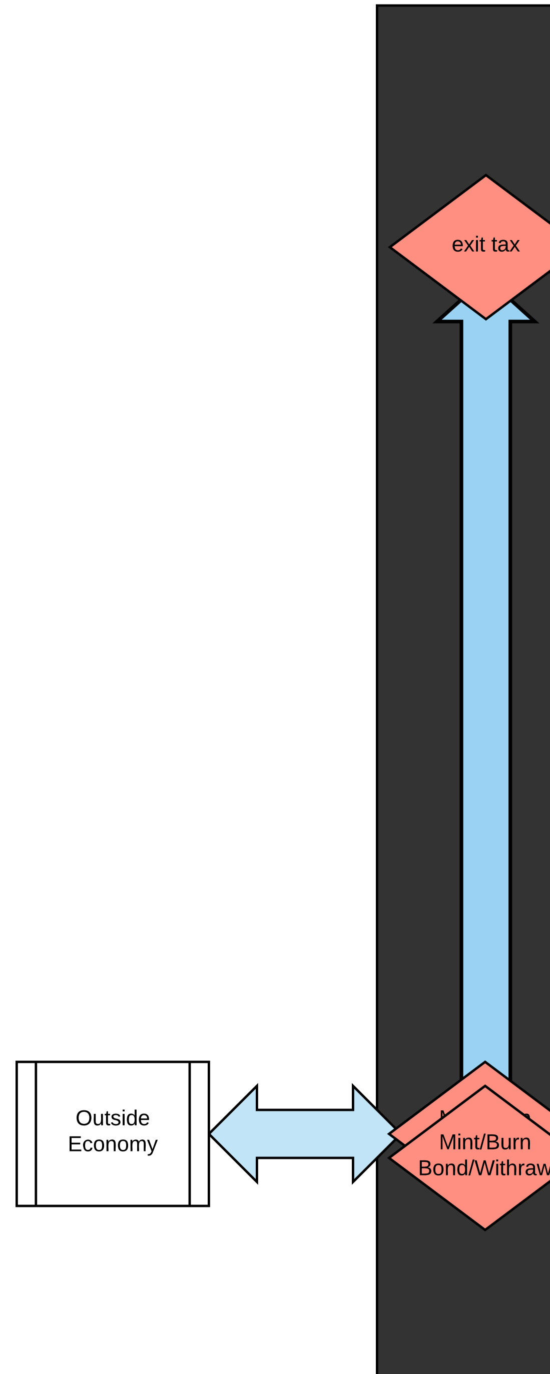
Local economy

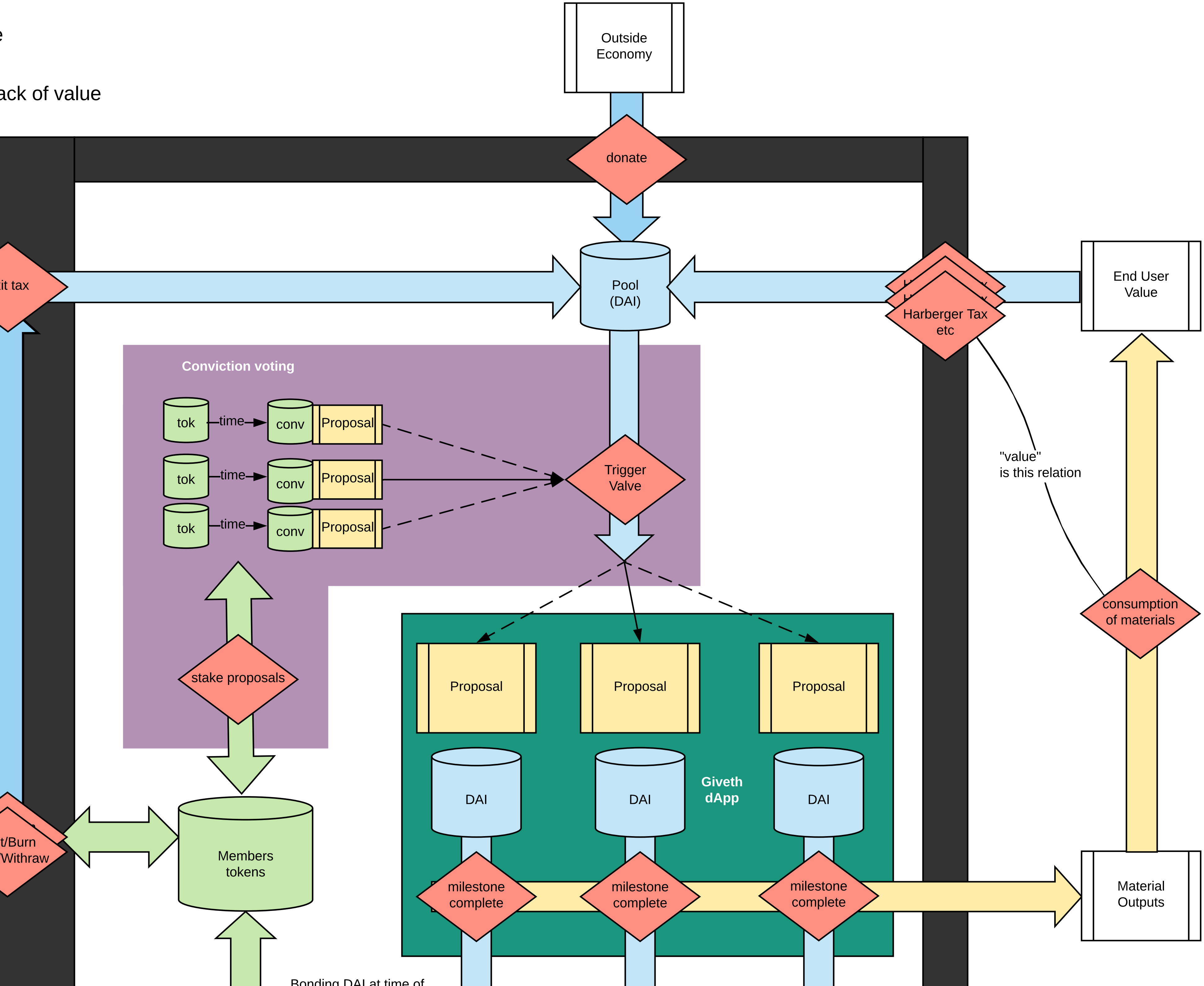
Outside Economy


System 1 Dynamics:
u1 = some unknown behavoir
X1+ = f(X1, u1)
Y1 = S(X1)
Whole
part
relation
sum of all holdings
part
part
part
Enviroment
Interface Adds dynamics:
V(Y1, Y2) = const.
Embed Conservation Law via Interface Logic aka 'Bonding Curve'
Supply of ABC
DAI reserved
System 2 Dynamics:
u2 = some unknown behavoir
X2+ = f(X2, u2)
Y2= S(X2)
Whole
part
relation
part
DAI substate info
part
part

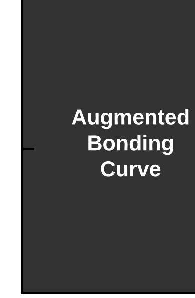
Cyber-Physical Commons

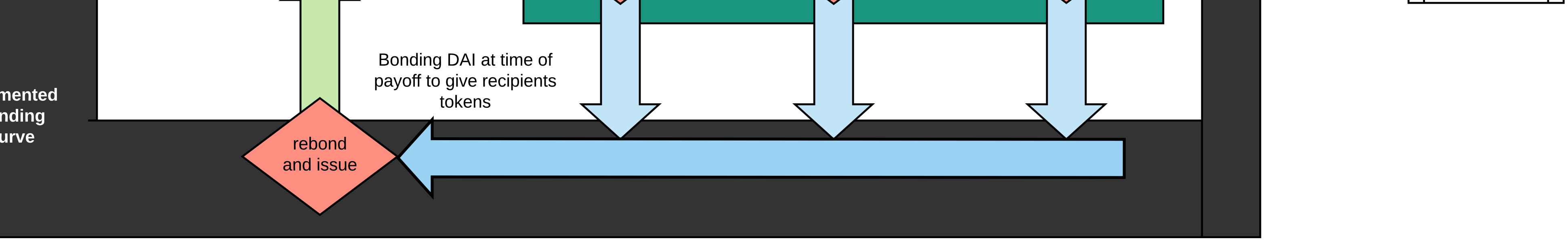
- Decentralized Governance
- Continuous Funding
- Regenerative returns/flowback of



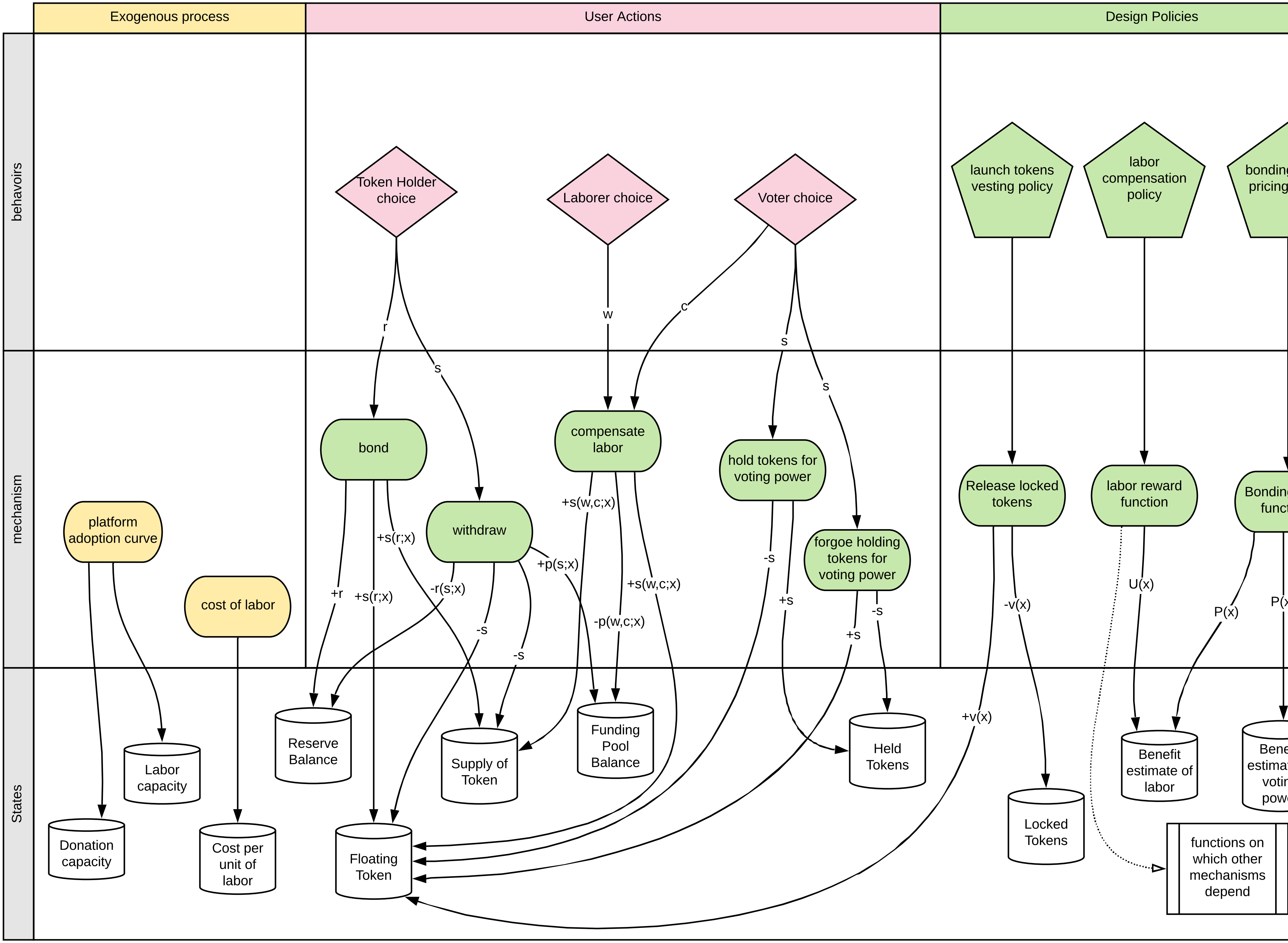


e
ack of value
Outside Economy
donate
it tax
Pool (DAI)
Harberger Tax etc
End User Value
Conviction voting
tok
time
conv
Proposal
tok
time
conv
Proposal
tok
time
conv
Proposal
Trigger Valve
"value" is this relation
consumption of materials
stake proposals
Proposal
Proposal
Proposal
DAI
DAI
Giveth dApp
DAI
t/Burn
Withraw
Members tokens
milestone complete
milestone complete
milestone complete
Material Outputs
Bonding DAI at time of

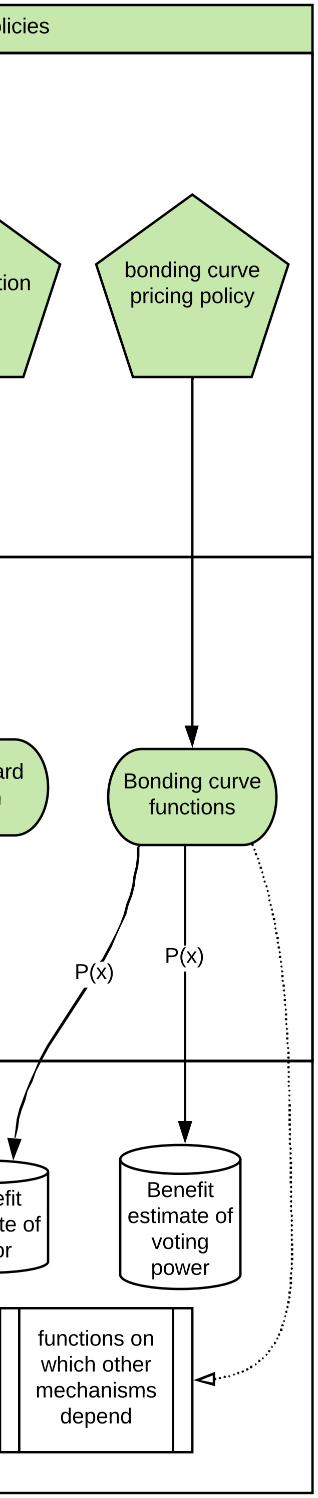

Augmented
Bonding
Curve



mented
nding
urve
Bonding DAI at time of payoff to give recipients tokens
rebond and issue



Exogenous process
User Actions
Design Policies
behavoirs
mechanism
States
Token Holder choice
Laborer choice
Voter choice
launch tokens vesting policy
labor compensation policy
r
s
w
c
s
s
bond
withdraw
compensate labor
hold tokens for voting power
forgoe holding tokens for voting power
Release locked tokens
labor reward function
platform adoption curve
cost of labor
+r
+s(r;x)
+s(r;x)
-r(s;x)
-s
-s
+p(s;x)
+s(w,c;x)
+s(w,c;x)
-p(w,c;x)
-s
+s
+s
-s
-v(x)
+v(x)
U(x)
P(x)
Reserve Balance
Supply of Token
Funding Pool Balance
Held Tokens
Locked Tokens
Benefit estimate of labor
Labor capacity
Donation capacity
Cost per unit of labor
Floating Token
functions on which other mechanisms depend



Price Consistency

$$s(r;x) = r \cdot P_b(r;x)$$
$$r(s;x) = s \cdot P_w(s;x)$$

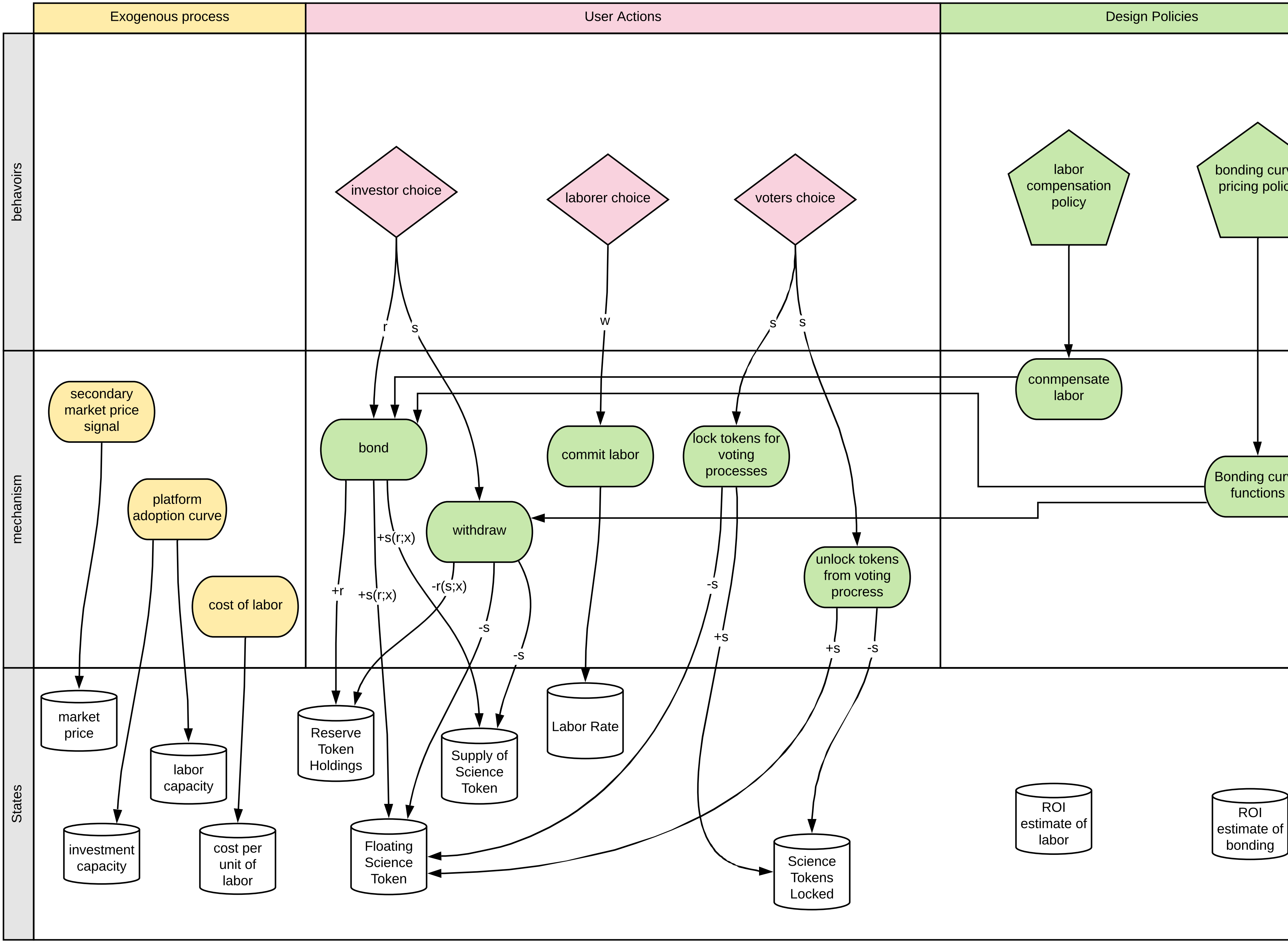
$$\forall x, (r, s) :$$
$$r(s;x) = r,$$
$$s(r;x) = s$$
$$P_b(r;x) = \frac{\phi}{P_w(s;x)}$$

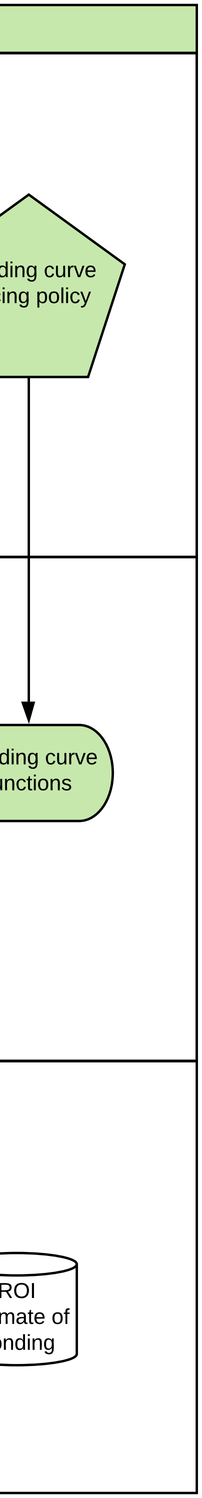
with widrawal friction

Labor compensation

$$s(w;x) = w \cdot U(w;x)$$

Must Exibit diminishing returns relative to the Reserve:Floating:Supply ratios to protect price from inflation


Exogenous process
User Actions
Design Policies
behavoirs
mechanism
States
investor choice
laborer choice
voters choice
labor compensation policy
bonding cur
pricing polic
r
s
w
s
s
secondary market price signal
platform adoption curve
cost of labor
bond
withdraw
commit labor
lock tokens for voting processes
unlock tokens from voting procress
conmpensate labor
Bonding cur
functions
+r
+s(r;x)
+s(r;x)
-r(s;x)
-s
-s
-s
+s
+s
-s
market price
labor capacity
investment capacity
cost per unit of labor
Reserve Token Holdings
Supply of Science Token
Floating Science Token
Labor Rate
Science Tokens Locked
ROI estimate of labor
ROI estimate of bonding



Labor compensation

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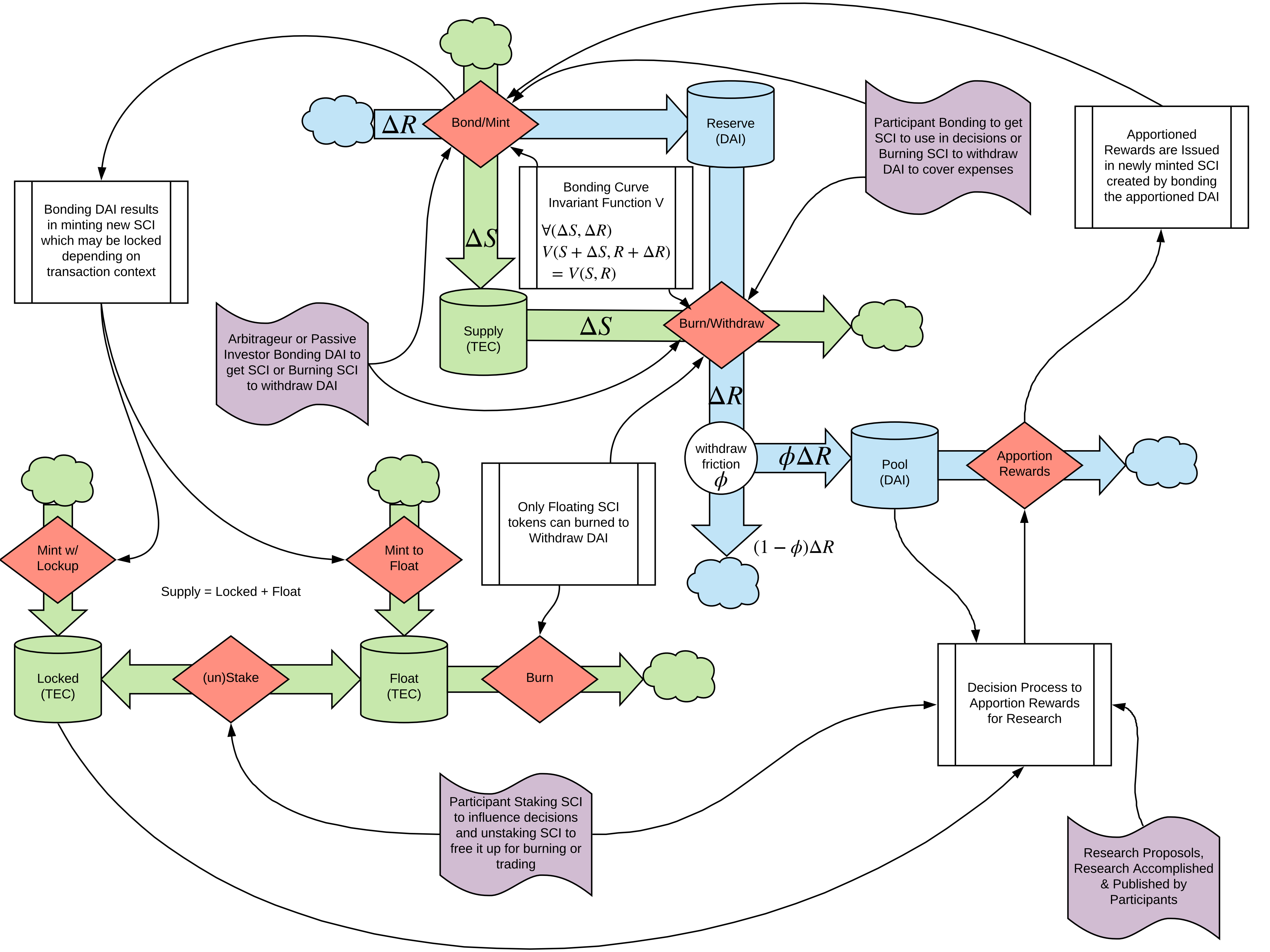
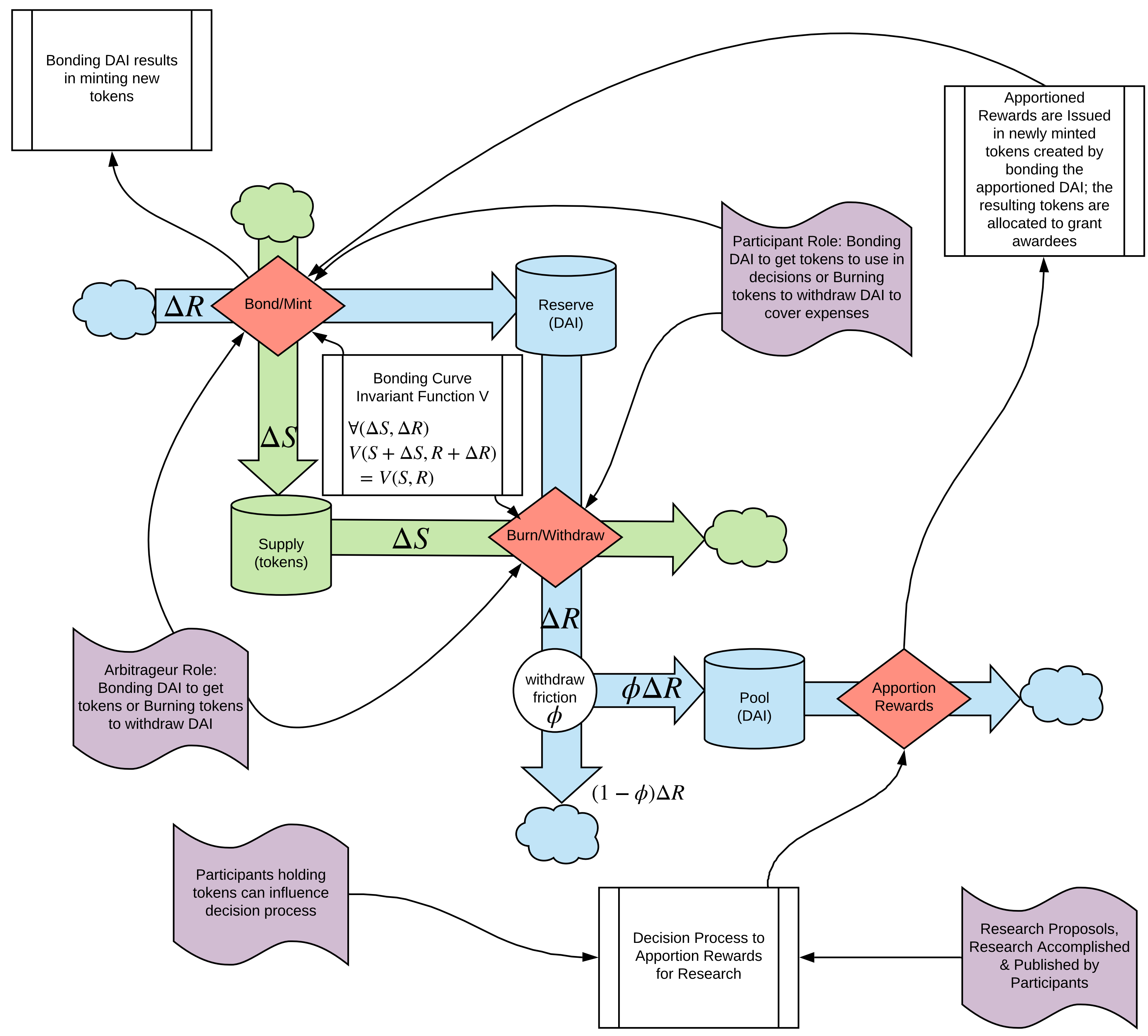
$$r(s;x) = s \cdot P_w(s;x)$$

$$\forall x, (r,s):$$
$$r(s;x) = r,$$
$$s(r;x) = s$$
$$P_b(r;x) = \frac{\phi}{P_w(s;x)}$$

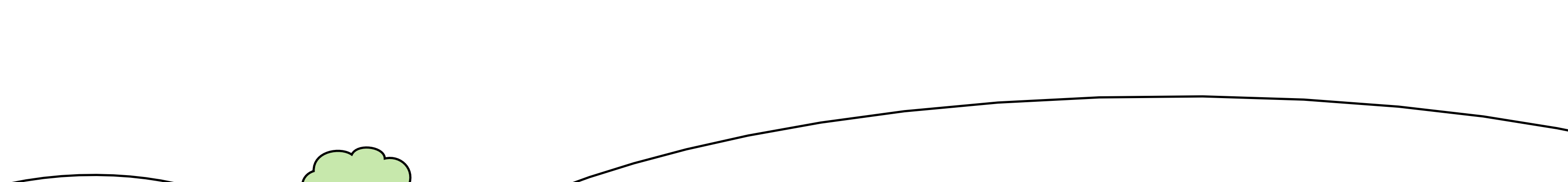
with widrawal friction

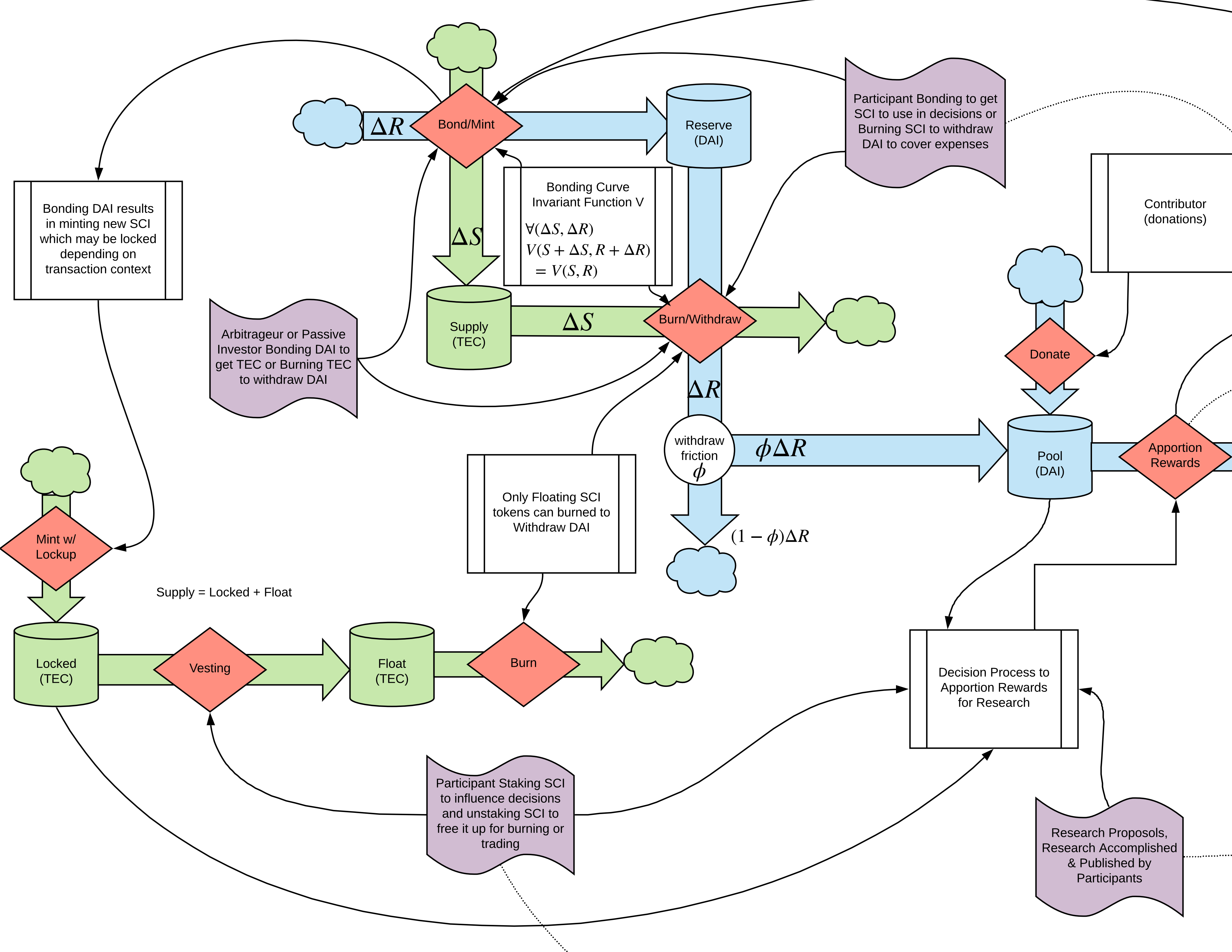


ΔR
Bond/Mint
Reserve (DAI)
Participant Bonding to get SCI to use in decisions or Burning SCI to withdraw DAI to cover expenses
Apportioned Rewards are Issued in newly minted SCI created by bonding the apportioned DAI
Bonding DAI results in minting new SCI which may be locked depending on transaction context
ΔS
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S + \Delta S, R + \Delta R)$
$= V(S, R)$
Arbitrageur or Passive Investor Bonding DAI to get SCI or Burning SCI to withdraw DAI
Supply (TEC)
ΔS
Burn/Withdraw
ΔR
withdraw friction ϕ
$\phi \Delta R$
Pool (DAI)
Apportion Rewards
$(1 - \phi)\Delta R$
Mint w/ Lockup
Mint to Float
Only Floating SCI tokens can burned to Withdraw DAI
Supply = Locked + Float
Locked (TEC)
(un)Stake
Float (TEC)
Burn
Decision Process to Apportion Rewards for Research
Participant Staking SCI to influence decisions and unstaking SCI to free it up for burning or trading
Research Proposols, Research Accomplished & Published by Participants



Bonding DAI results in minting new tokens
Apportioned Rewards are Issued in newly minted tokens created by bonding the apportioned DAI; the resulting tokens are allocated to grant awardees
Participant Role: Bonding DAI to get tokens to use in decisions or Burning tokens to withdraw DAI to cover expenses
ΔR
Bond/Mint
Reserve (DAI)
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S+\Delta S, R+\Delta R)$
$= V(S,R)$
ΔS
Supply (tokens)
ΔS
Burn/Withdraw
ΔR
withdraw friction ϕ
$\phi \Delta R$
Pool (DAI)
Apportion Rewards
$(1-\phi)\Delta R$
Arbitrageur Role: Bonding DAI to get tokens or Burning tokens to withdraw DAI
Participants holding tokens can influence decision process
Decision Process to Apportion Rewards for Research
Research Proposols, Research Accomplished & Published by Participants





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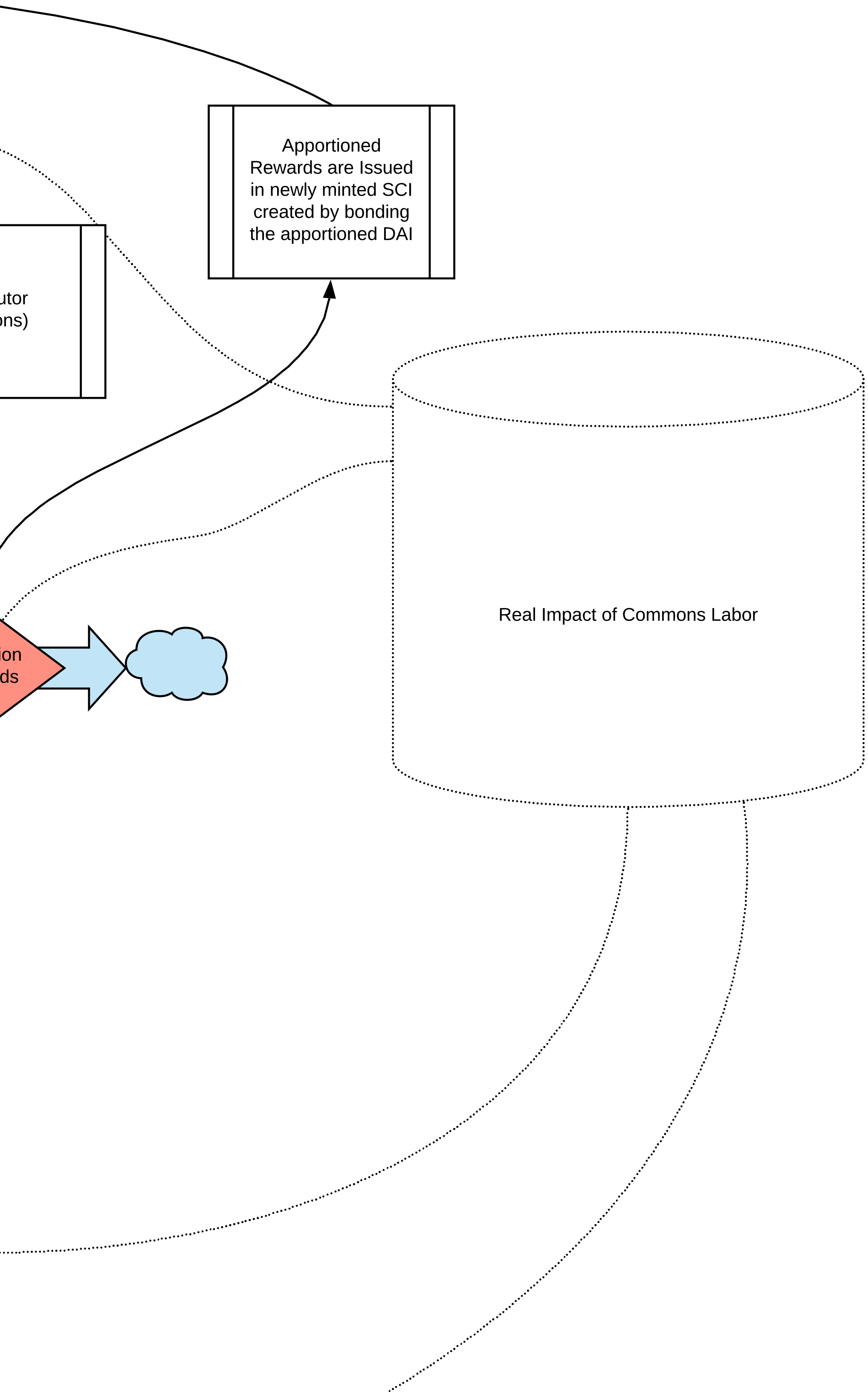
$V(S+\Delta S, R+\Delta R)$
$= V(S,R)$
Contributor (donations)
Supply (TEC)
ΔS
Burn/Withdraw
Arbitrageur or Passive Investor Bonding DAI to get TEC or Burning TEC to withdraw DAI
Donate
ΔR
withdraw friction ϕ
$\phi \Delta R$
Pool (DAI)
Apportion Rewards
Only Floating SCI tokens can burned to Withdraw DAI
$(1-\phi)\Delta R$
Mint w/ Lockup
Supply = Locked + Float
Locked (TEC)
Vesting
Float (TEC)
Burn
Decision Process to Apportion Rewards for Research
Participant Staking SCI to influence decisions and unstaking SCI to free it up for burning or trading
Research Proposals, Research Accomplished & Published by Participants



Apportioned Rewards are Issued in newly minted SCI created by bonding the apportioned DAI
utor
ons)
Real Impact of Commons Labor
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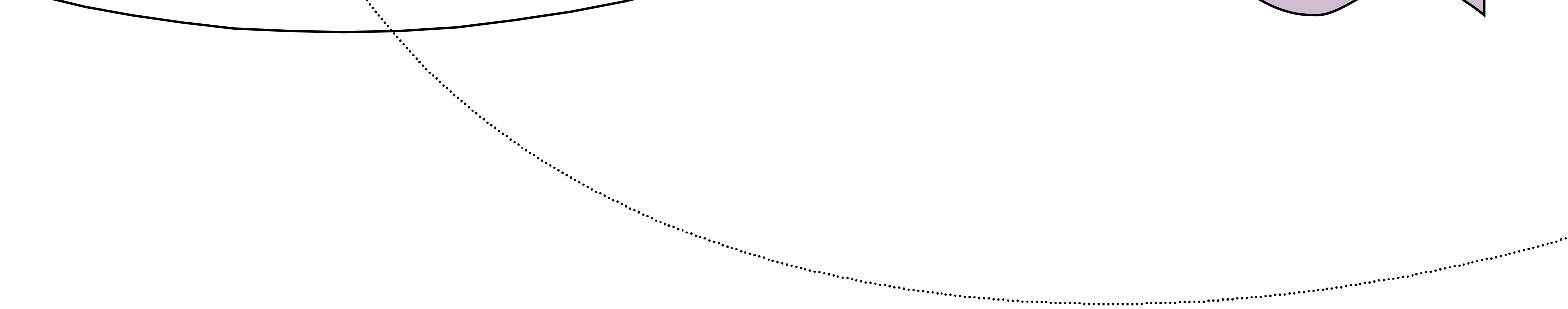
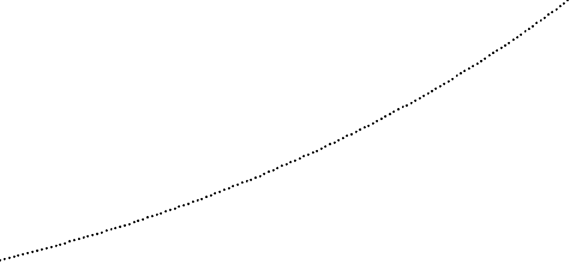
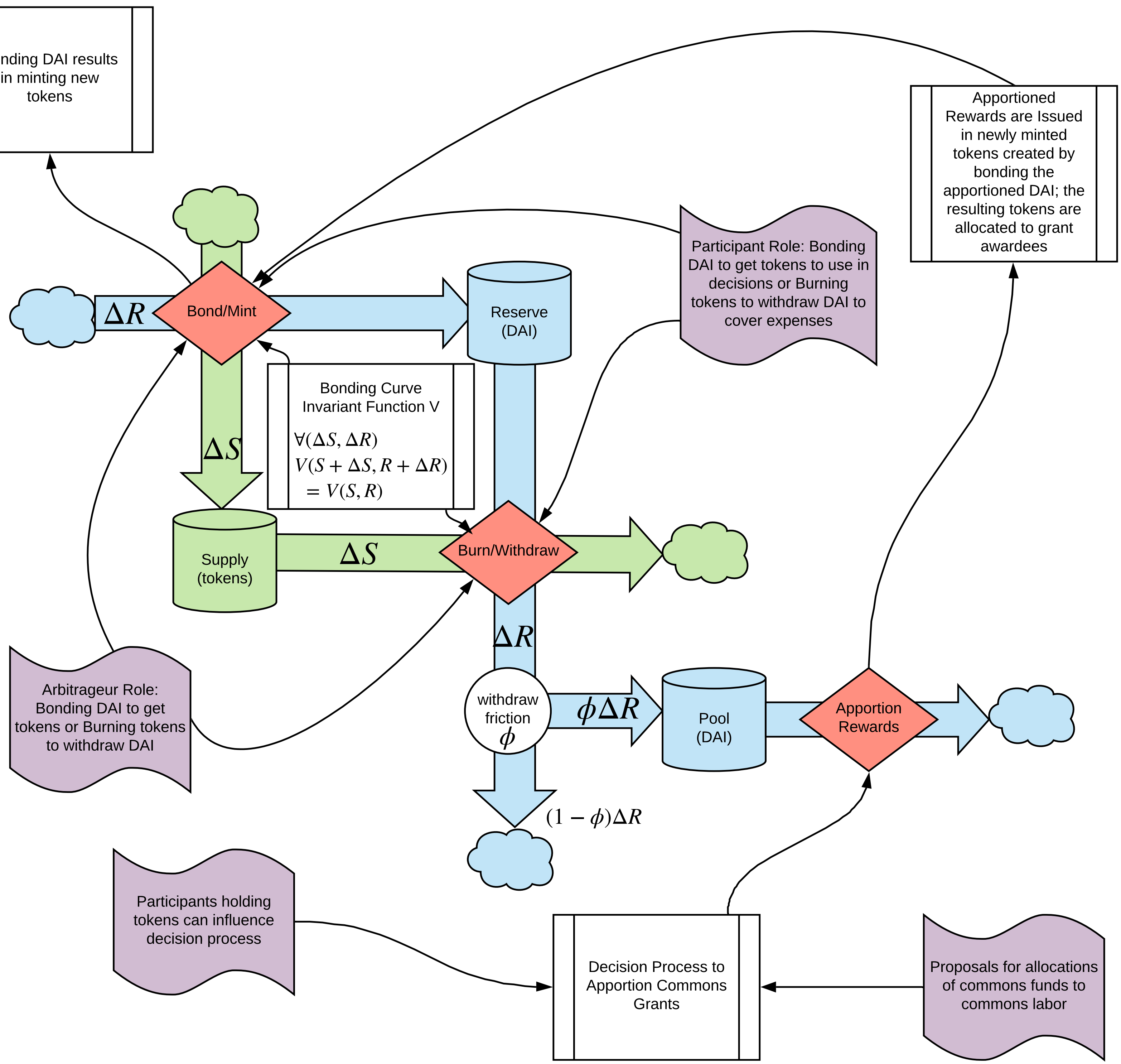
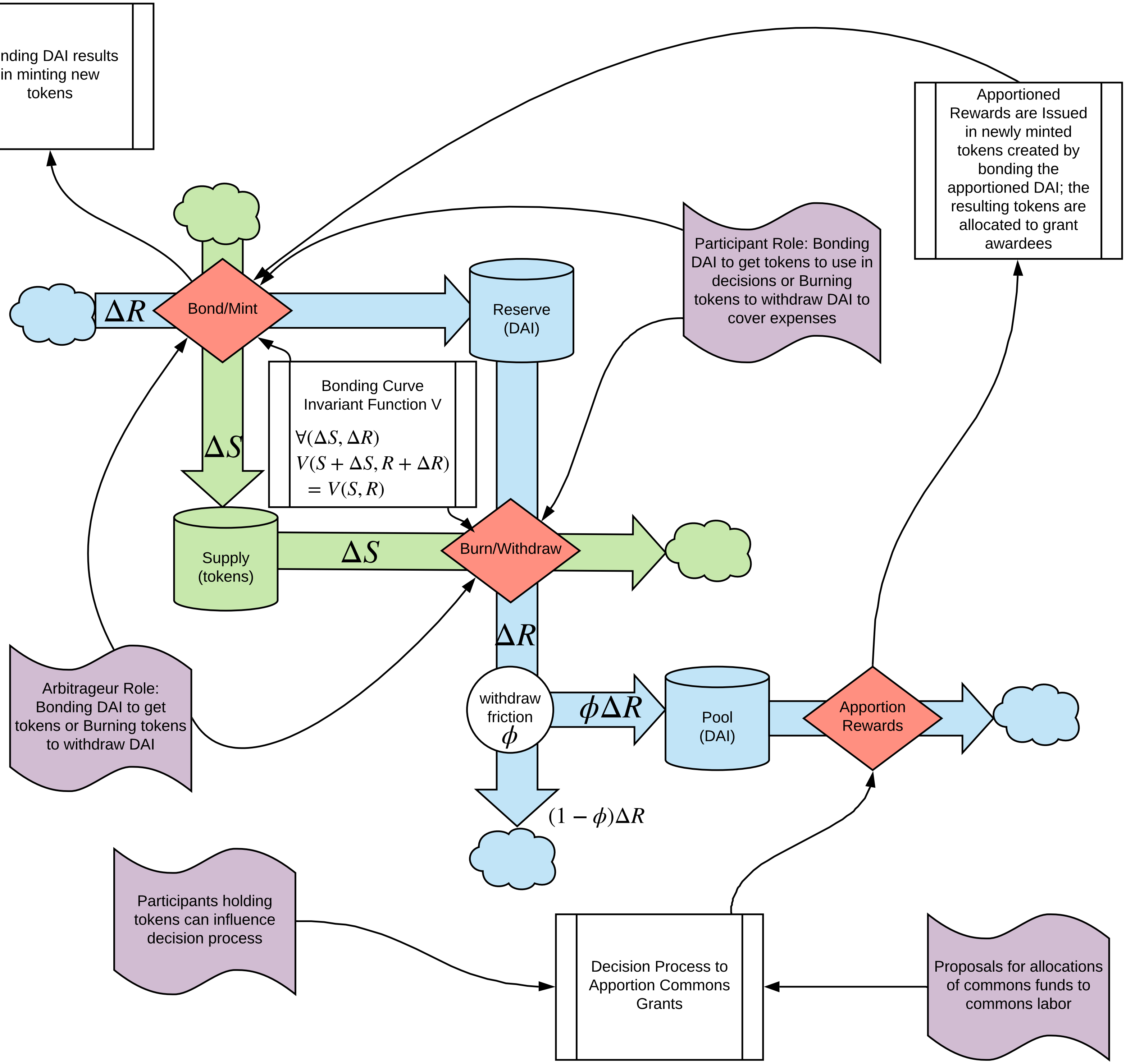





Bonding DAI results in minting new tokens
Apportioned Rewards are Issued in newly minted tokens created by bonding the apportioned DAI; the resulting tokens are allocated to grant awardees
Participant Role: Bonding DAI to get tokens to use in decisions or Burning tokens to withdraw DAI to cover expenses
ΔR
Bond/Mint
Reserve (DAI)
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S+\Delta S, R+\Delta R)$
$= V(S, R)$
ΔS
Supply (tokens)
ΔS
Burn/Withdraw
ΔR
withdraw friction ϕ
$\phi \Delta R$
Pool (DAI)
Apportion Rewards
$(1-\phi)\Delta R$
Arbitrageur Role: Bonding DAI to get tokens or Burning tokens to withdraw DAI
Participants holding tokens can influence decision process
Decision Process to Apportion Commons Grants
Proposals for allocations of commons funds to commons labor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$\forall(\Delta S, \Delta R)$
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ΔS
Supply (tokens)
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Burn/Withdraw
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withdraw friction ϕ
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Pool (DAI)
Apportion Rewards
$(1-\phi)\Delta R$
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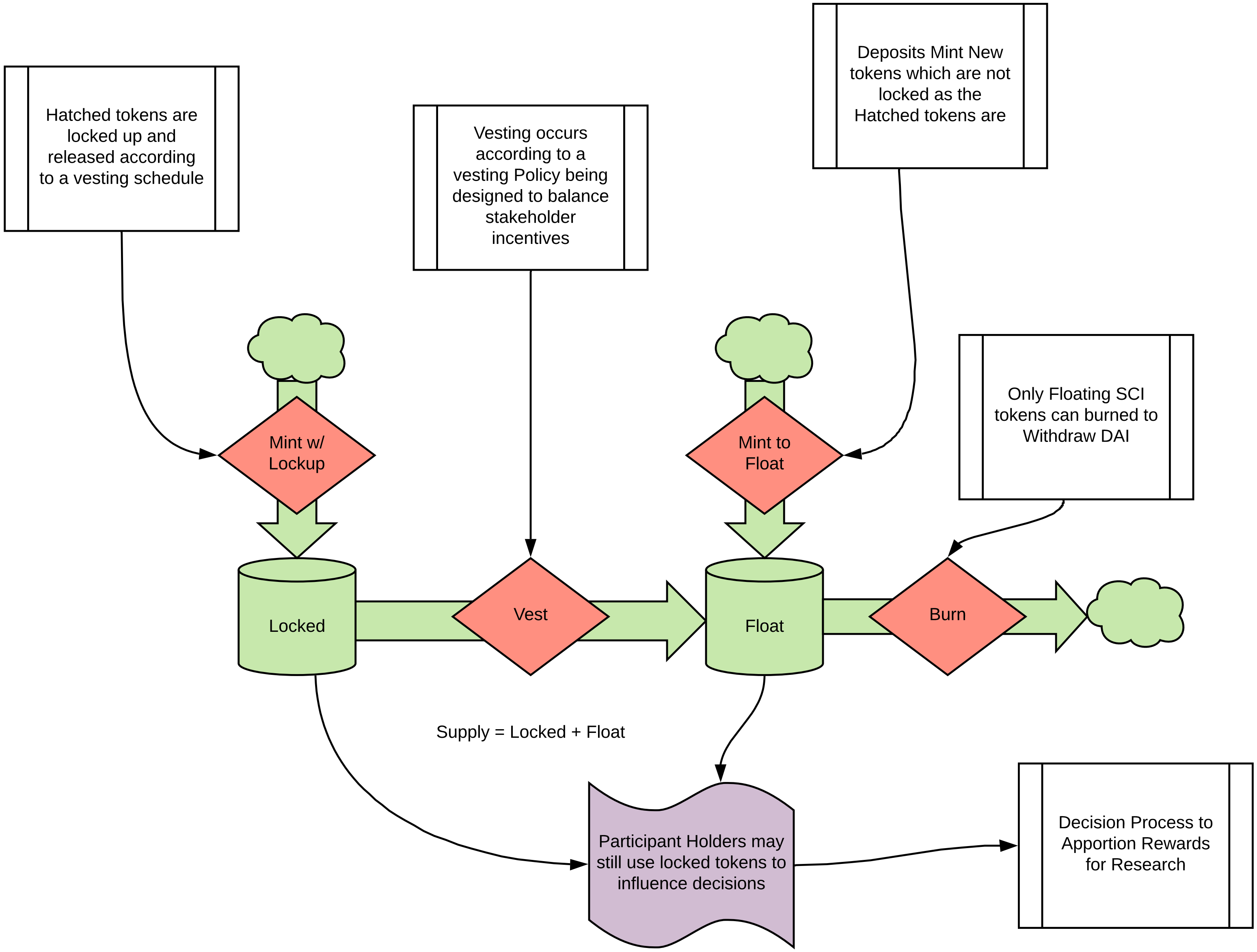
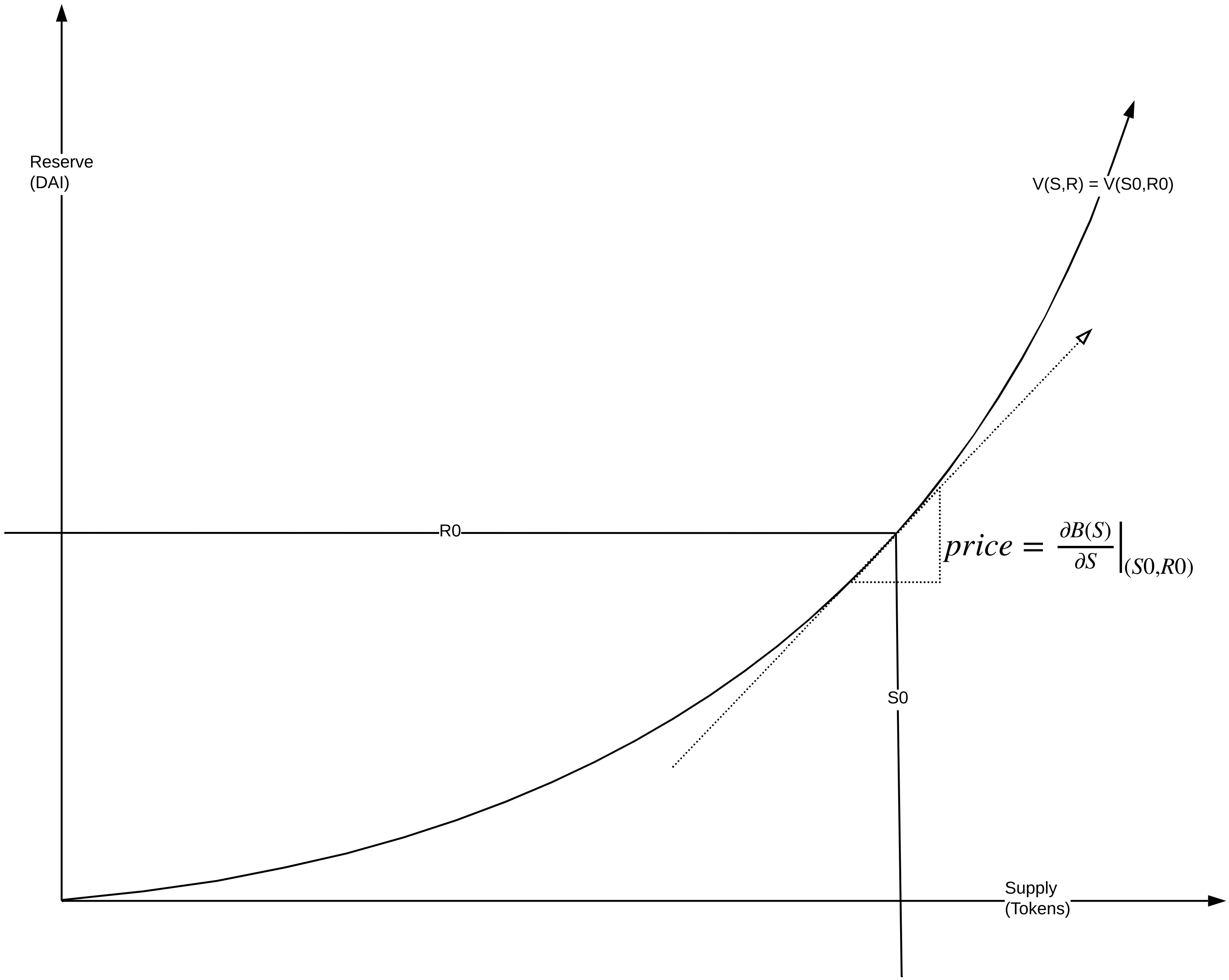
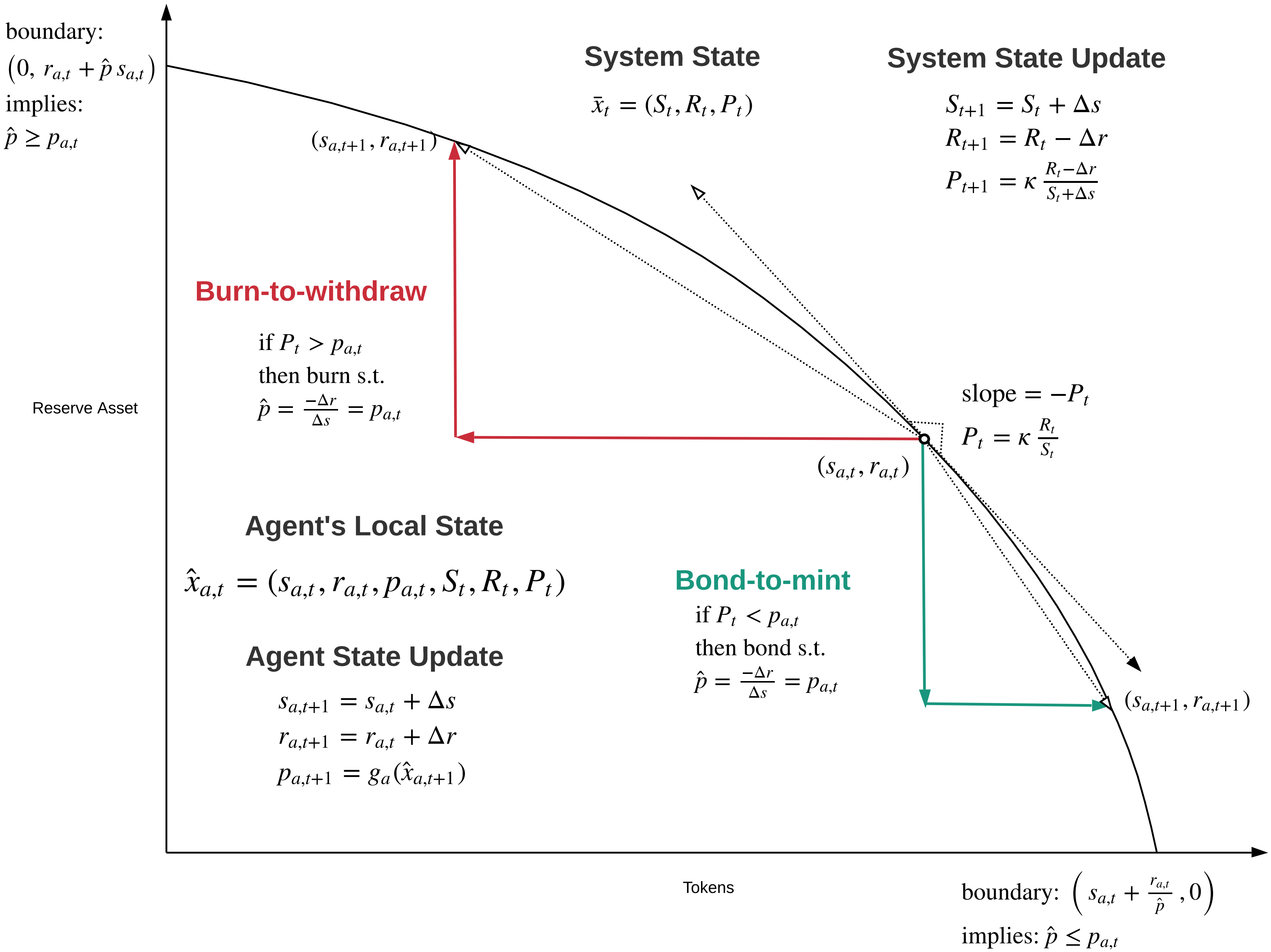
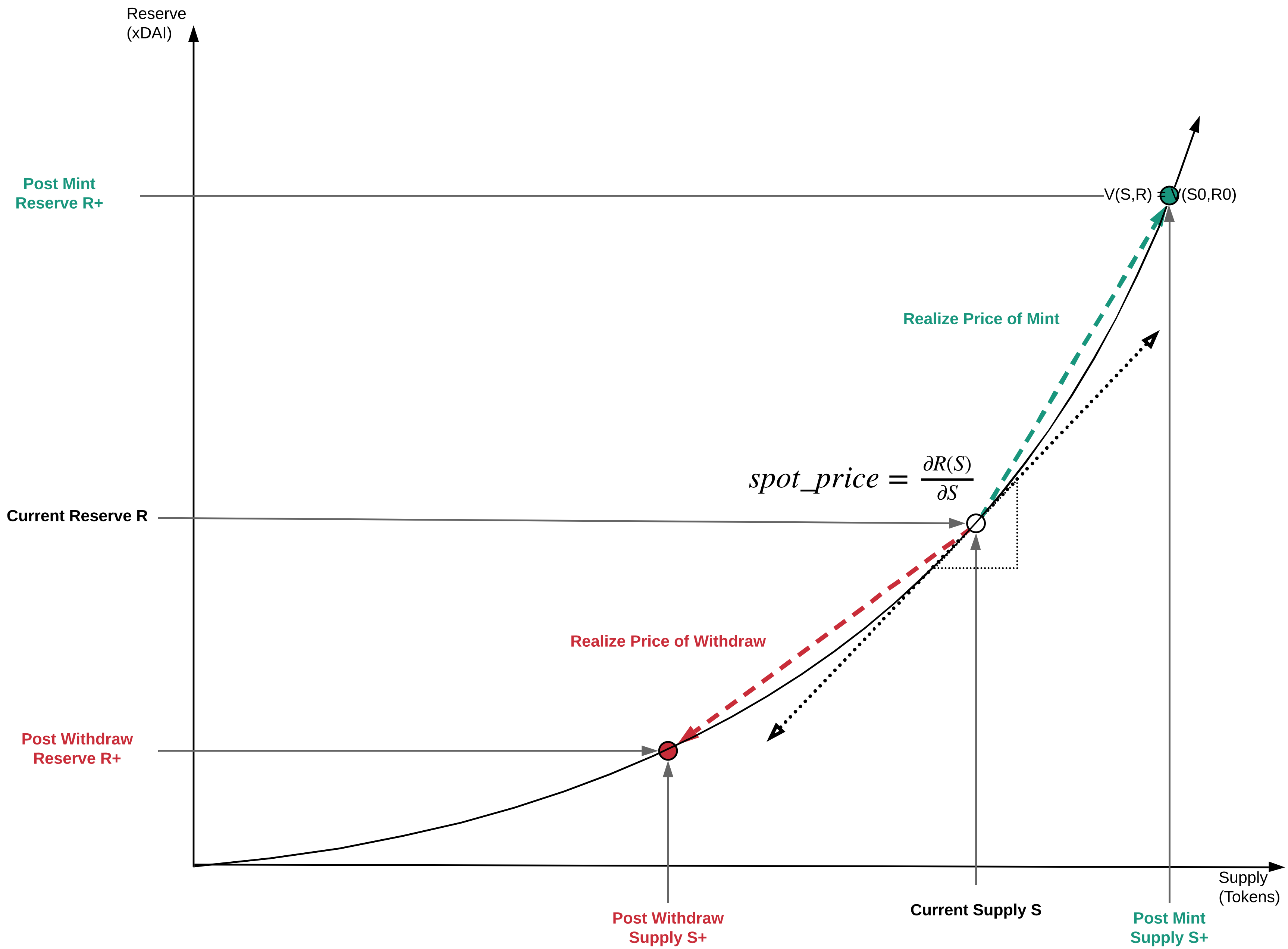

Hatched tokens are locked up and released according to a vesting schedule
Vesting occurs according to a vesting Policy being designed to balance stakeholder incentives
Deposits Mint New tokens which are not locked as the Hatched tokens are
Only Floating SCI tokens can burned to Withdraw DAI
Mint w/ Lockup
Mint to Float
Locked
Vest
Float
Burn
Supply = Locked + Float
Participant Holders may still use locked tokens to influence decisions
Decision Process to Apportion Rewards for Research



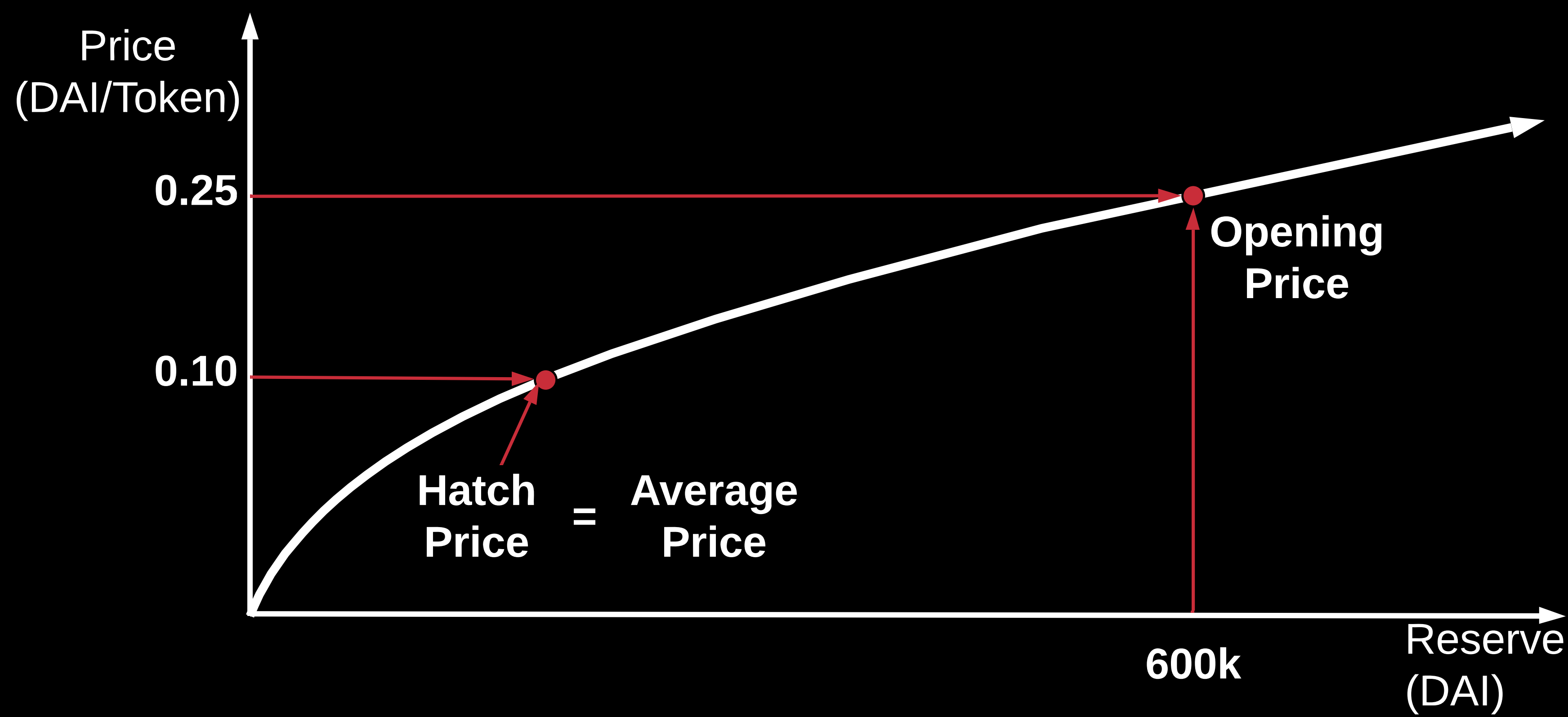
Reserve
(DAI)
V(S,R) = V(S0,R0)
R0
$price = \frac{\partial B(S)}{\partial S}\Big|_{(S0,R0)}$
S0
Supply
(Tokens)



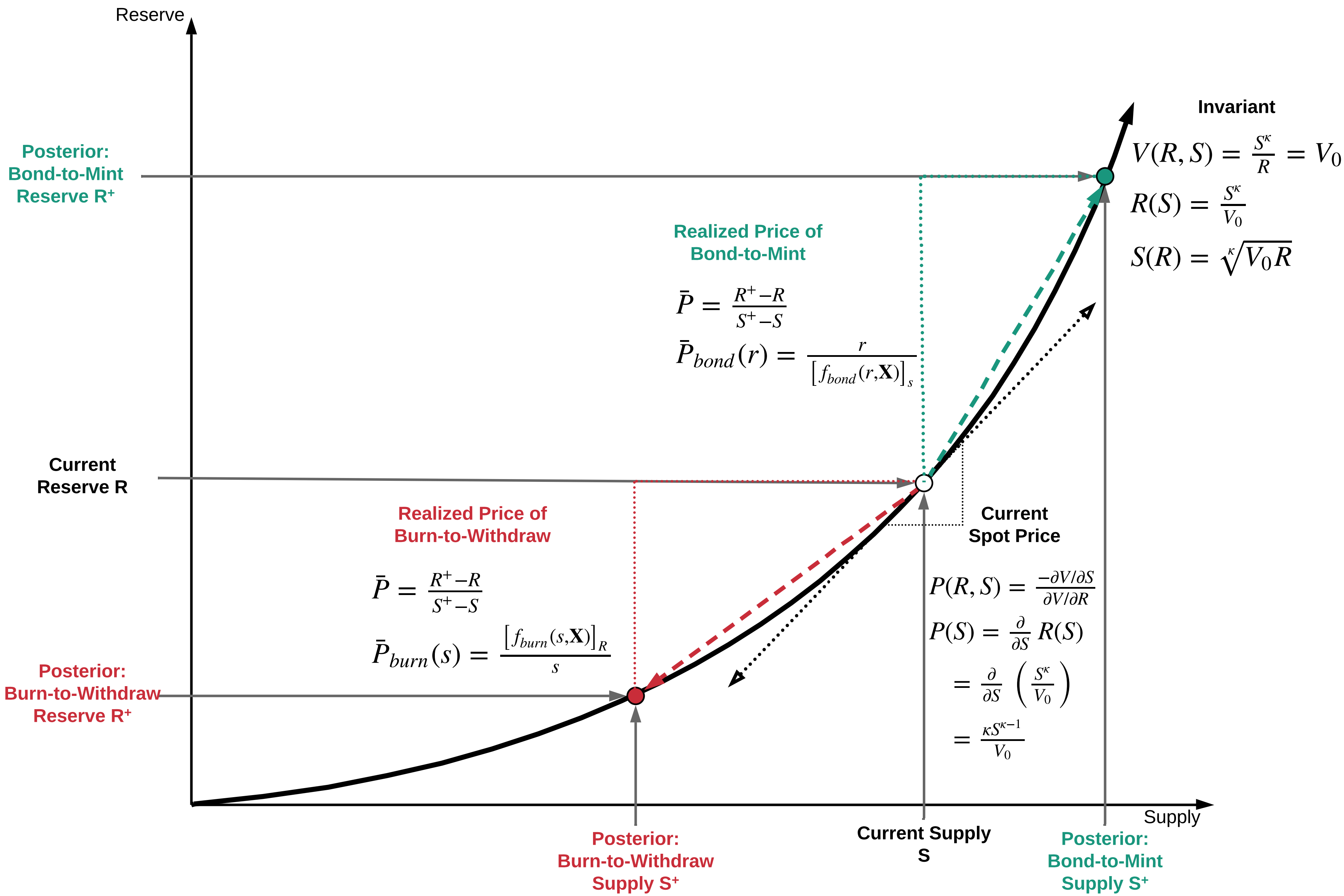
boundary:
$(0, r_{a,t} + \hat{p}\, s_{a,t})$
implies:
$\hat{p} \geq p_{a,t}$
System State
$\bar{x}_t = (S_t, R_t, P_t)$
System State Update
$S_{t+1} = S_t + \Delta s$
$R_{t+1} = R_t - \Delta r$
$P_{t+1} = \kappa \frac{R_t - \Delta r}{S_t + \Delta s}$
$(s_{a,t+1}, r_{a,t+1})$
Burn-to-withdraw
if $P_t > p_{a,t}$
then burn s.t.
$\hat{p} = \frac{-\Delta r}{\Delta s} = p_{a,t}$
Reserve Asset
slope $= -P_t$
$P_t = \kappa \frac{R_t}{S_t}$
$(s_{a,t}, r_{a,t})$
Agent's Local State
$\hat{x}_{a,t} = (s_{a,t}, r_{a,t}, p_{a,t}, S_t, R_t, P_t)$
Bond-to-mint
if $P_t < p_{a,t}$
then bond s.t.
$\hat{p} = \frac{-\Delta r}{\Delta s} = p_{a,t}$
Agent State Update
$s_{a,t+1} = s_{a,t} + \Delta s$
$r_{a,t+1} = r_{a,t} + \Delta r$
$p_{a,t+1} = g_a(\hat{x}_{a,t+1})$
$(s_{a,t+1}, r_{a,t+1})$
Tokens
boundary: $\left(s_{a,t} + \frac{r_{a,t}}{\hat{p}}, 0\right)$
implies: $\hat{p} \leq p_{a,t}$



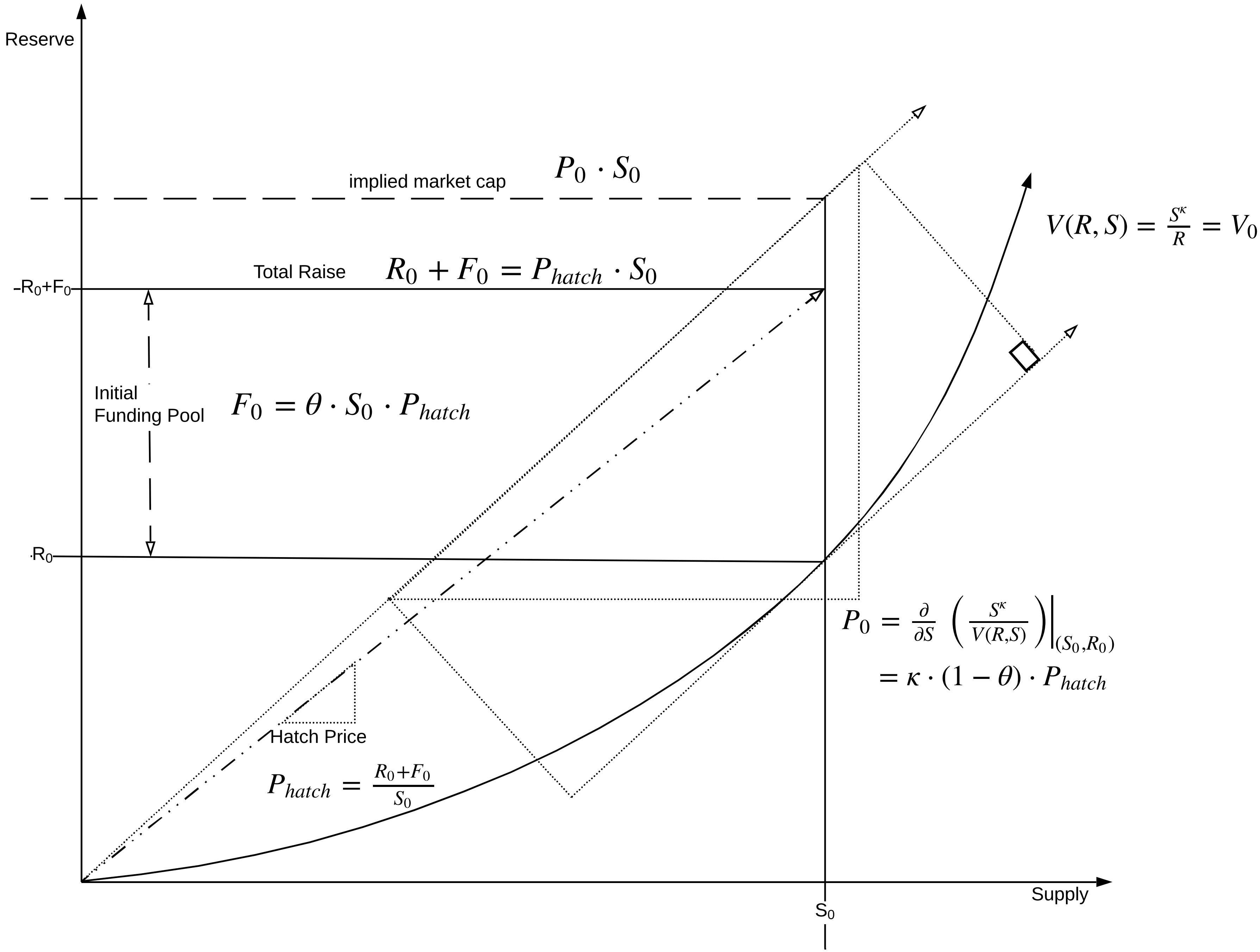
Reserve
(xDAI)
Post Mint
Reserve R+
V(S,R) = V(S0,R0)
Realize Price of Mint
$spot\_price = \frac{\partial R(S)}{\partial S}$
Current Reserve R
Realize Price of Withdraw
Post Withdraw
Reserve R+
Supply
(Tokens)
Post Withdraw
Supply S+
Current Supply S
Post Mint
Supply S+

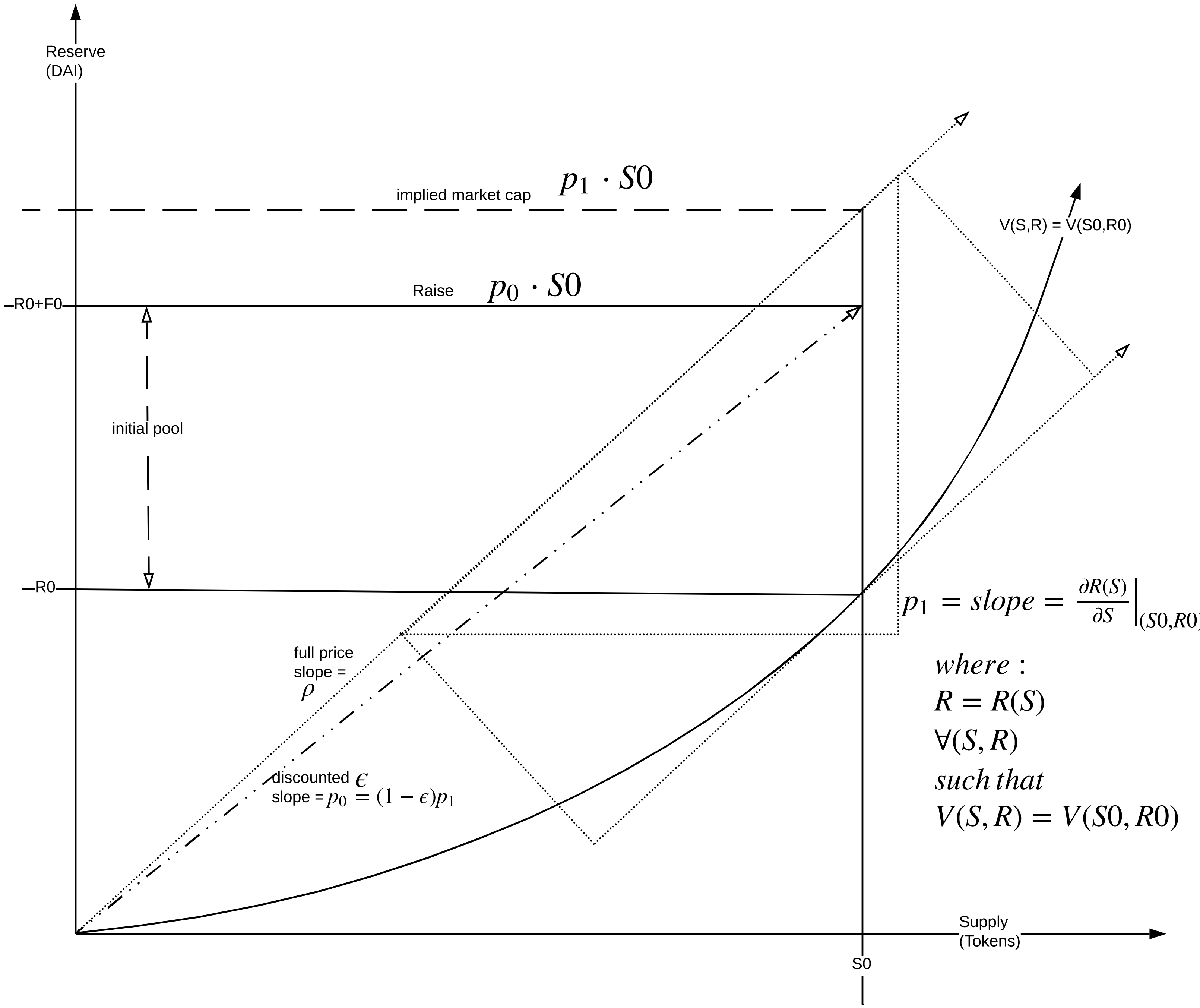


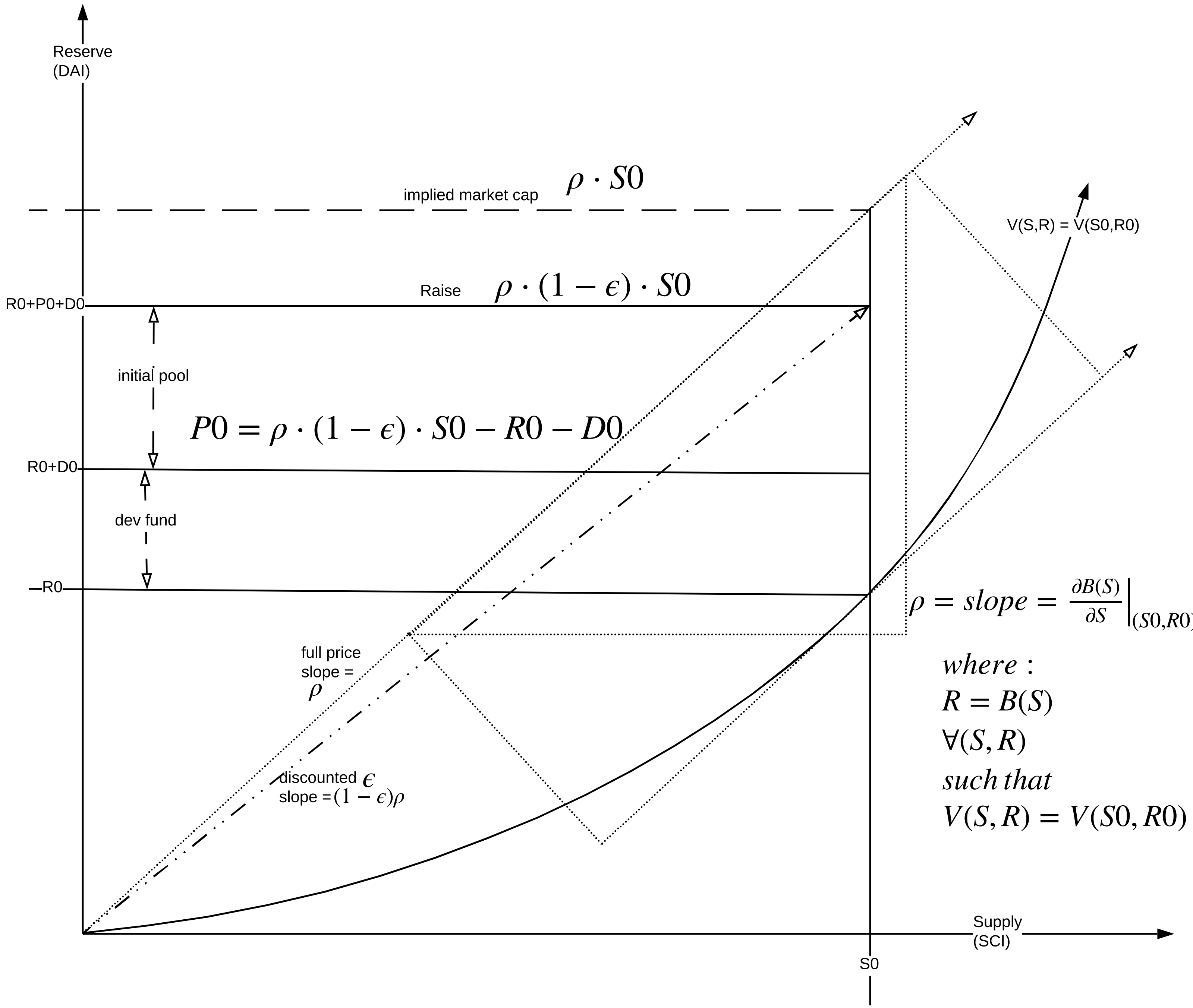
Price
(DAI/Token)
0.25
0.10
Opening
Price
Hatch
Price
=
Average
Price
600k
Reserve
(DAI)

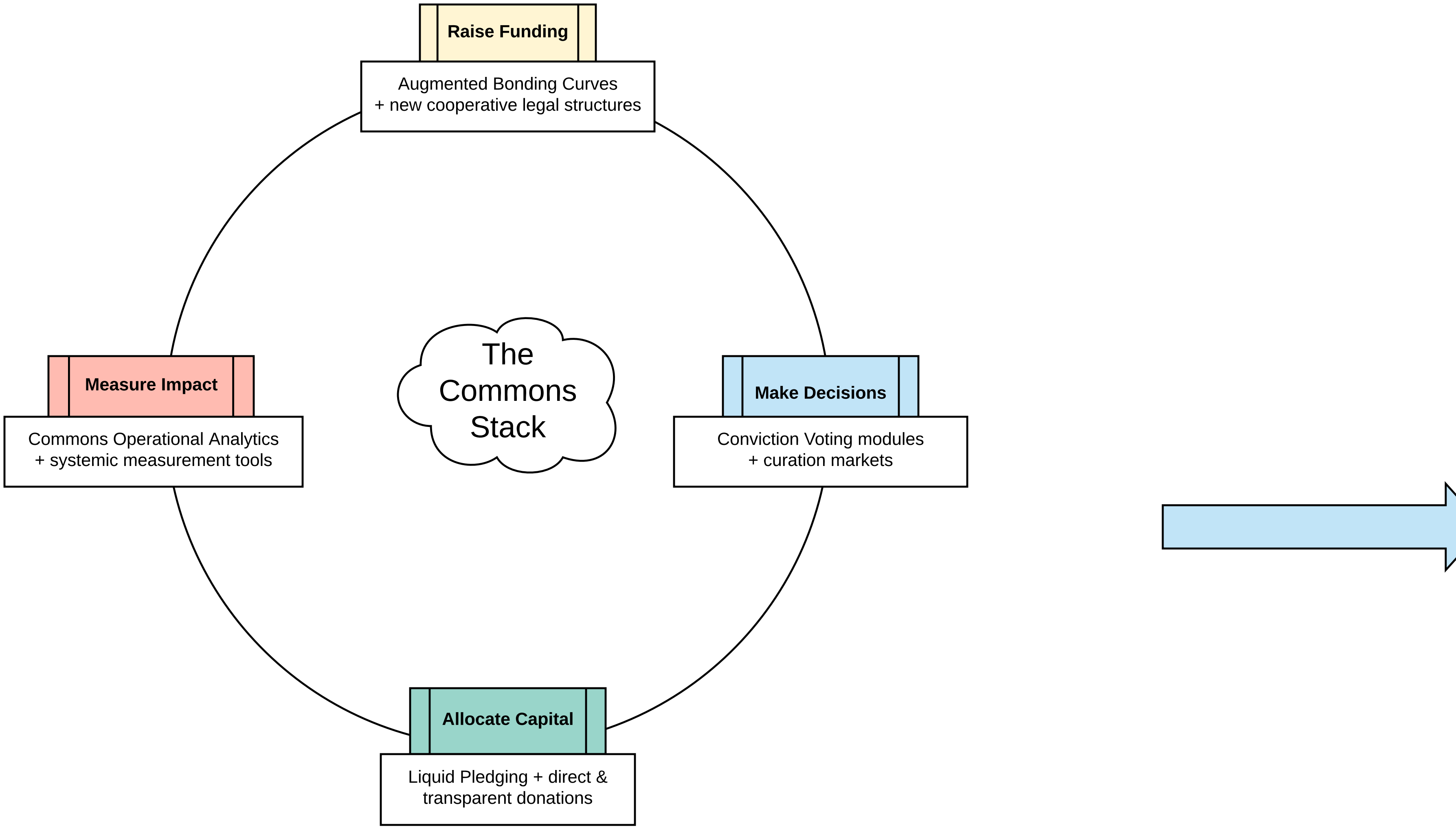


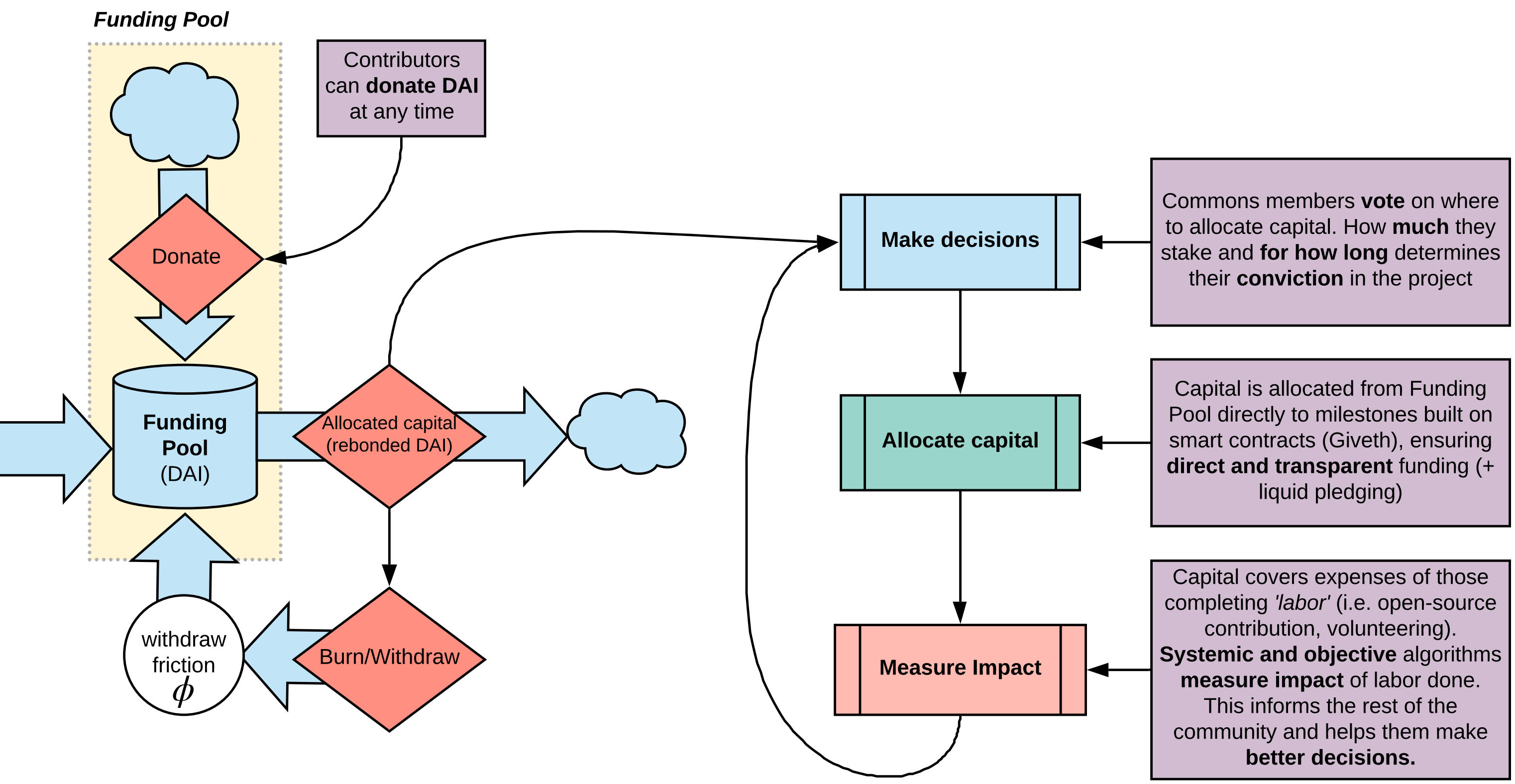
Reserve
Supply
Invariant
$V(R,S) = \frac{S^\kappa}{R} = V_0$
$R(S) = \frac{S^\kappa}{V_0}$
$S(R) = \sqrt[\kappa]{V_0 R}$
Posterior: Bond-to-Mint Reserve R⁺
Realized Price of Bond-to-Mint
$\bar{P} = \frac{R^+ - R}{S^+ - S}$
$\bar{P}_{bond}(r) = \frac{r}{\left[f_{bond}(r,\mathbf{X})\right]_s}$
Current Reserve R
Realized Price of Burn-to-Withdraw
$\bar{P} = \frac{R^+ - R}{S^+ - S}$
$\bar{P}_{burn}(s) = \frac{\left[f_{burn}(s,\mathbf{X})\right]_R}{s}$
Current Spot Price
$P(R,S) = \frac{-\partial V/\partial S}{\partial V/\partial R}$
$P(S) = \frac{\partial}{\partial S} R(S)$
$= \frac{\partial}{\partial S}\left(\frac{S^\kappa}{V_0}\right)$
$= \frac{\kappa S^{\kappa-1}}{V_0}$
Posterior: Burn-to-Withdraw Reserve R⁺
Posterior: Burn-to-Withdraw Supply S⁺
Current Supply S
Posterior: Bond-to-Mint Supply S⁺


Reserve
implied market cap $P_0 \cdot S_0$
Total Raise $R_0 + F_0 = P_{hatch} \cdot S_0$
R_0+F_0
$V(R,S) = \frac{S^\kappa}{R} = V_0$
Initial
Funding Pool
$F_0 = \theta \cdot S_0 \cdot P_{hatch}$
R_0
$P_0 = \frac{\partial}{\partial S}\left(\frac{S^\kappa}{V(R,S)}\right)\Big|_{(S_0,R_0)}$
$= \kappa \cdot (1-\theta) \cdot P_{hatch}$
Hatch Price
$P_{hatch} = \frac{R_0+F_0}{S_0}$
Supply
S_0

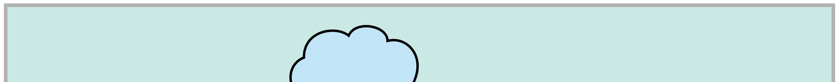

Reserve
(DAI)
implied market cap $p_1 \cdot S0$
Raise $p_0 \cdot S0$
R0+F0
initial pool
R0
V(S,R) = V(S0,R0)
$p_1 = slope = \left.\frac{\partial R(S)}{\partial S}\right|_{(S0,R0)}$
full price
slope =
ρ
discounted ϵ
slope = $p_0 = (1-\epsilon)p_1$
$where:$
$R = R(S)$
$\forall(S,R)$
$such\ that$
$V(S,R) = V(S0,R0)$
Supply
(Tokens)
S0

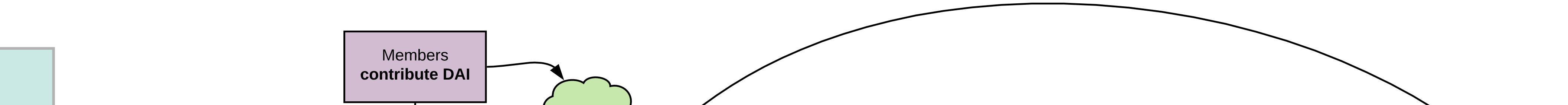

Reserve
(DAI)
implied market cap
$\rho \cdot S0$
Raise
$\rho \cdot (1-\epsilon) \cdot S0$
R0+P0+D0
initial pool
$P0 = \rho \cdot (1-\epsilon) \cdot S0 - R0 - D0$
R0+D0
dev fund
R0
V(S,R) = V(S0,R0)
$\rho = slope = \frac{\partial B(S)}{\partial S}\Big|_{(S0,R0)}$
full price
slope =
ρ
discounted ϵ
slope = $(1-\epsilon)\rho$
$where:$
$R = B(S)$
$\forall (S,R)$
$such\ that$
$V(S,R) = V(S0,R0)$
Supply
(SCI)
S0

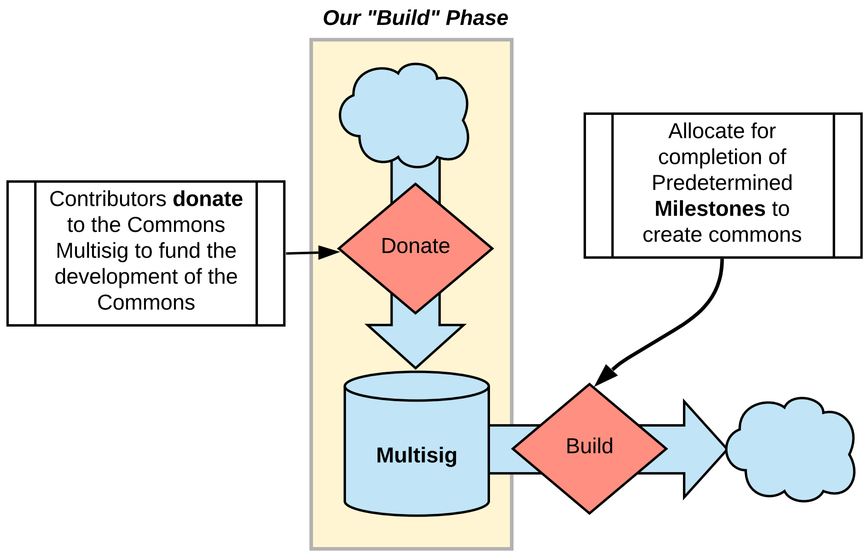

Raise Funding
Augmented Bonding Curves
+ new cooperative legal structures
Measure Impact
Commons Operational Analytics
+ systemic measurement tools
The Commons Stack
Make Decisions
Conviction Voting modules
+ curation markets
Allocate Capital
Liquid Pledging + direct &
transparent donations

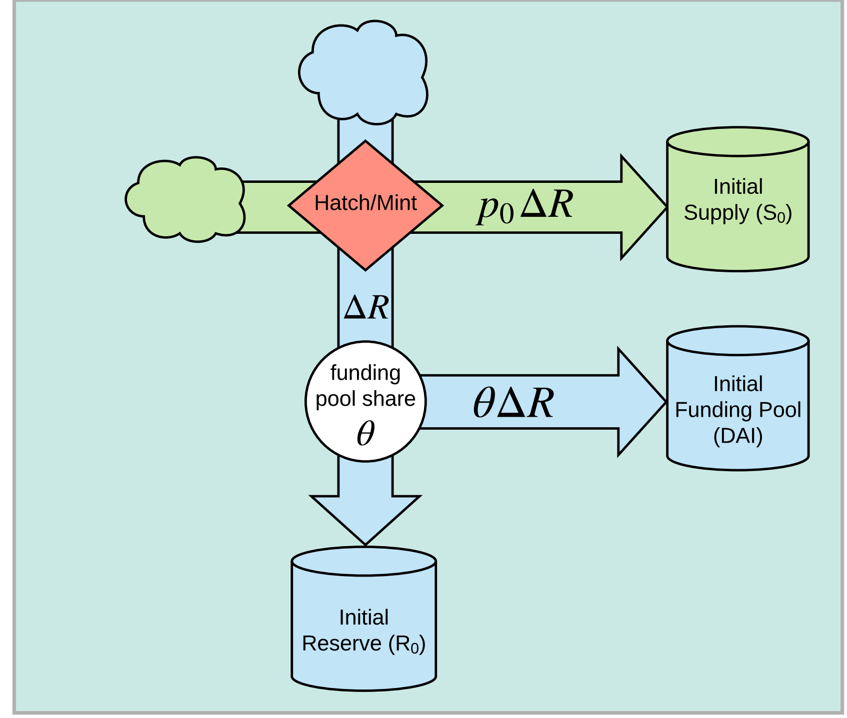

Funding Pool
Contributors can donate DAI at any time
Donate
Funding Pool (DAI)
Allocated capital (rebonded DAI)
withdraw friction ϕ
Burn/Withdraw
Make decisions
Allocate capital
Measure Impact
Commons members vote on where to allocate capital. How much they stake and for how long determines their conviction in the project
Capital is allocated from Funding Pool directly to milestones built on smart contracts (Giveth), ensuring direct and transparent funding (+ liquid pledging)
Capital covers expenses of those completing 'labor' (i.e. open-source contribution, volunteering). Systemic and objective algorithms measure impact of labor done. This informs the rest of the community and helps them make better decisions.

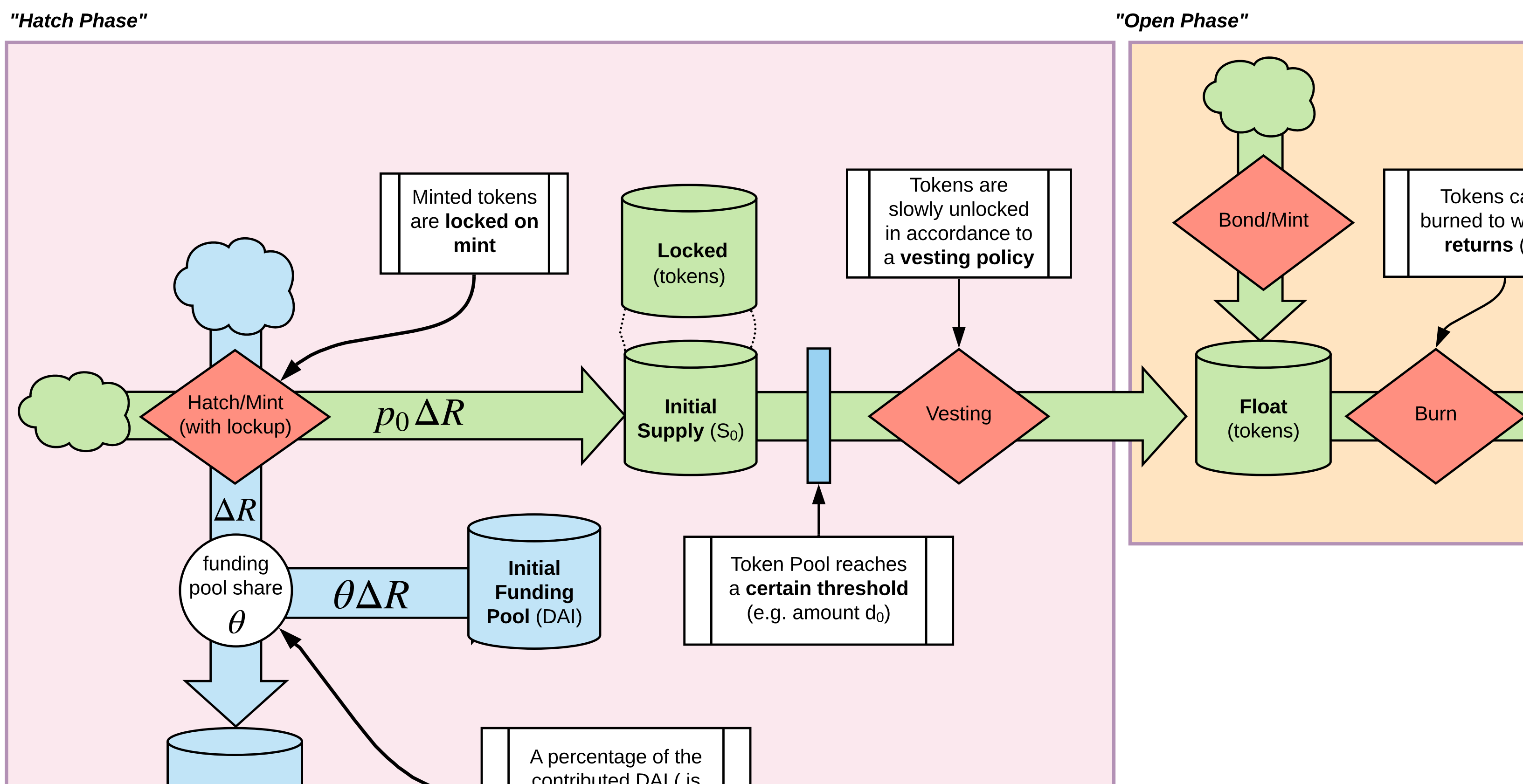
Hatch Sale

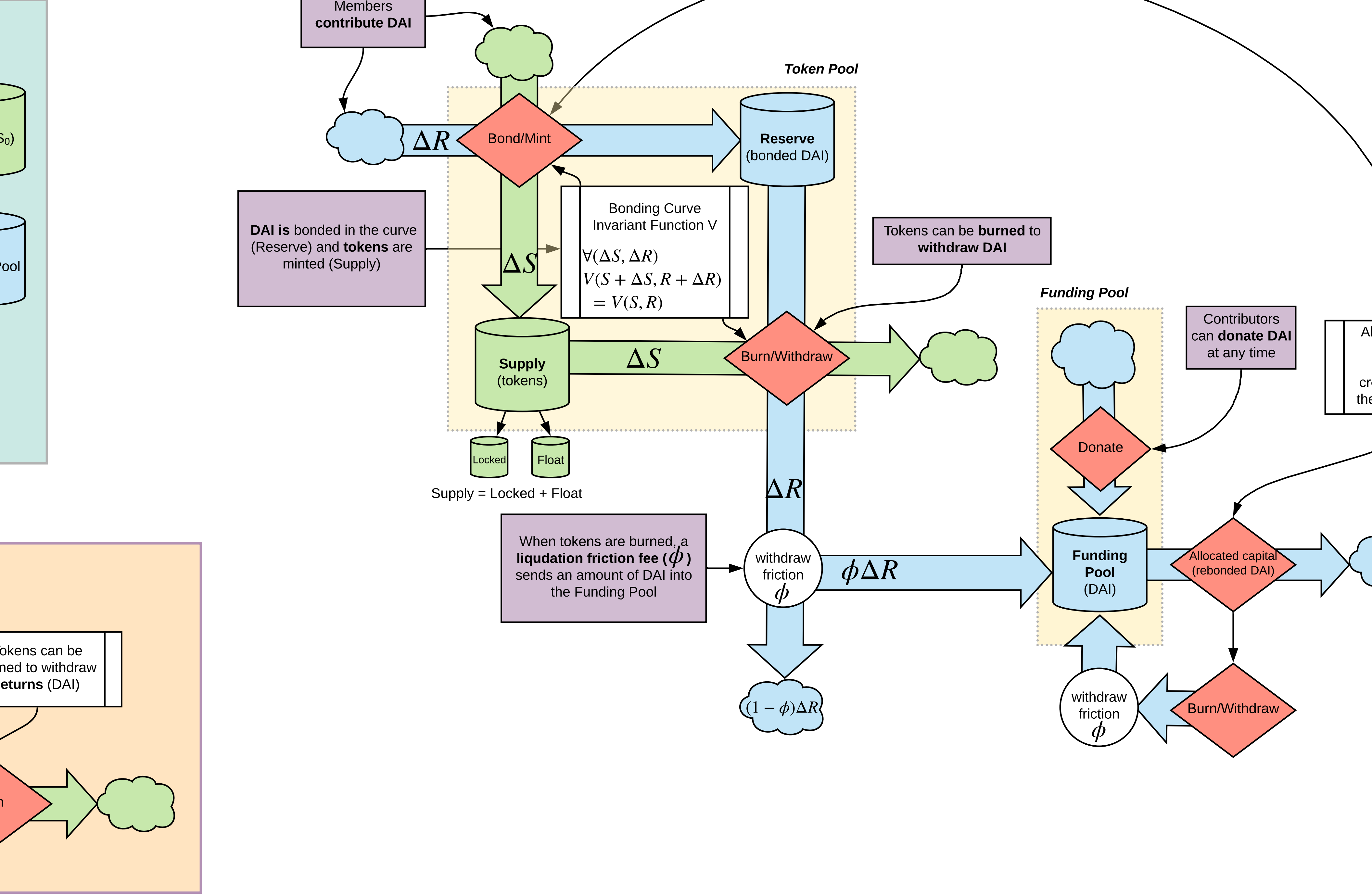



Members
contribute DAI

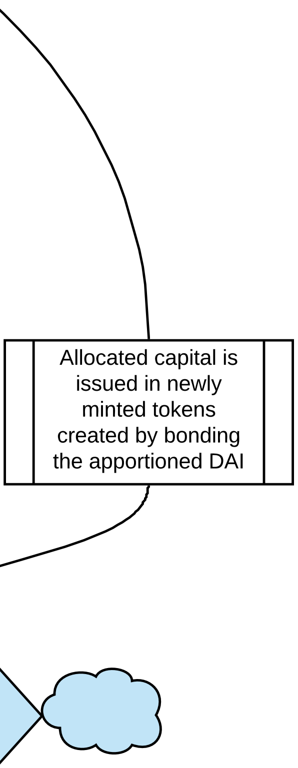

Our "Build" Phase
Contributors **donate** to the Commons Multisig to fund the development of the Commons
Donate
Multisig
Allocate for completion of Predetermined **Milestones** to create commons
Build

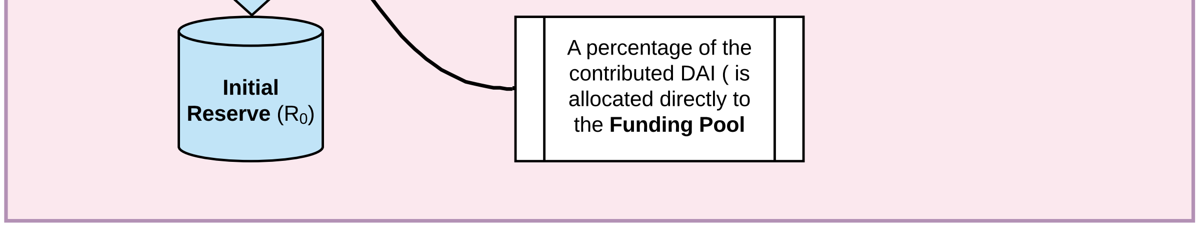

Hatch/Mint
$p_0 \Delta R$
Initial Supply (S_0)
ΔR
funding pool share θ
$\theta \Delta R$
Initial Funding Pool (DAI)
Initial Reserve (R_0)


"Hatch Phase"
"Open Phase"
Minted tokens are **locked on mint**
Hatch/Mint (with lockup)
$p_0 \Delta R$
Locked (tokens)
Initial Supply (S_0)
Tokens are slowly unlocked in accordance to a **vesting policy**
Vesting
ΔR
funding pool share θ
$\theta \Delta R$
Initial Funding Pool (DAI)
Token Pool reaches a **certain threshold** (e.g. amount d_0)
A percentage of the
Bond/Mint
Float (tokens)
Burn



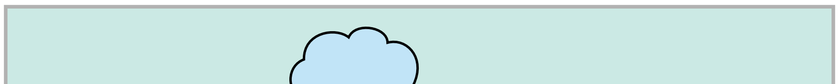
Members
contribute DAI
Token Pool
ΔR
Bond/Mint
Reserve
(bonded DAI)
S₀)
Pool
DAI is bonded in the curve (Reserve) and tokens are minted (Supply)
ΔS
Bonding Curve
Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S + \Delta S, R + \Delta R)$
$= V(S, R)$
Tokens can be burned to withdraw DAI
Funding Pool
Contributors can donate DAI at any time
Supply
(tokens)
ΔS
Burn/Withdraw
Donate
Locked
Float
Supply = Locked + Float
ΔR
When tokens are burned, a liqudation friction fee (ϕ) sends an amount of DAI into the Funding Pool
withdraw
friction
ϕ
$\phi \Delta R$
Funding
Pool
(DAI)
Allocated capital
(rebonded DAI)
$(1-\phi)\Delta R$
okens can be
ned to withdraw
eturns (DAI)
withdraw
friction
ϕ
Burn/Withdraw

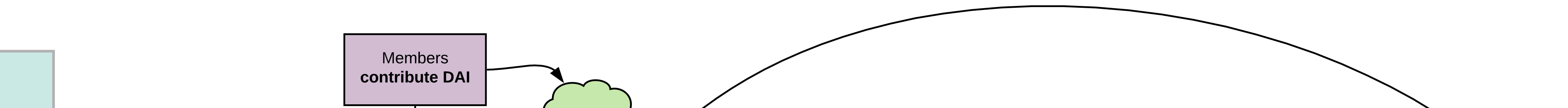

Allocated capital is issued in newly minted tokens created by bonding the apportioned DAI

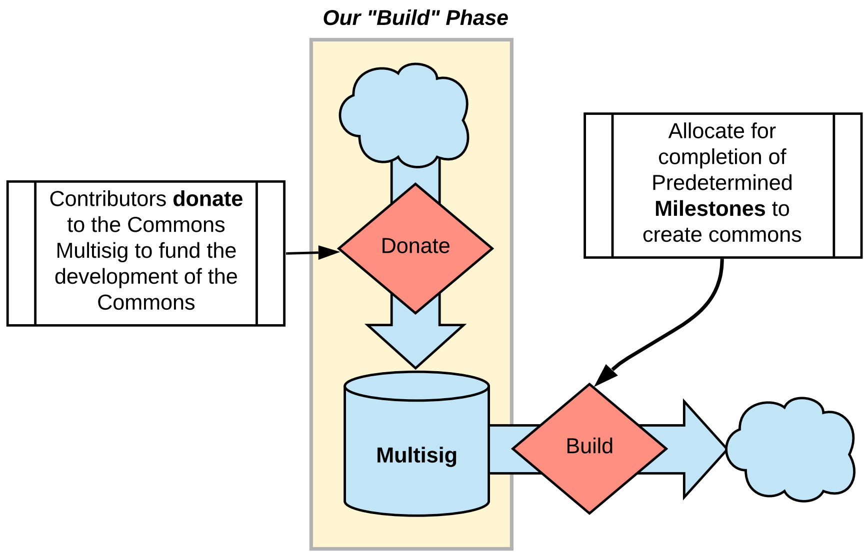


Initial Reserve (R_0)
A percentage of the contributed DAI (is allocated directly to the Funding Pool

Hatch Sale




Members
contribute DAI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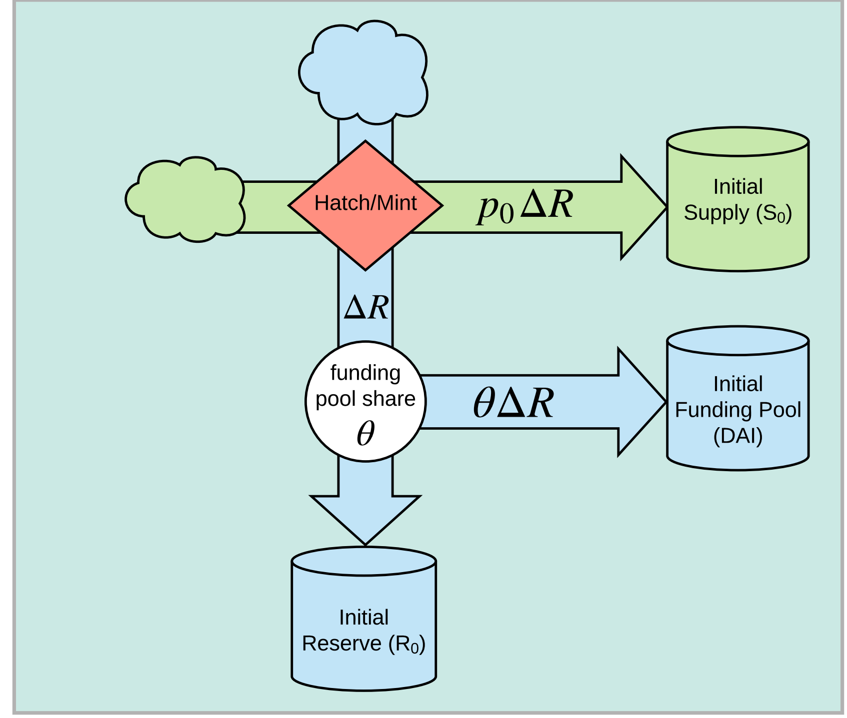
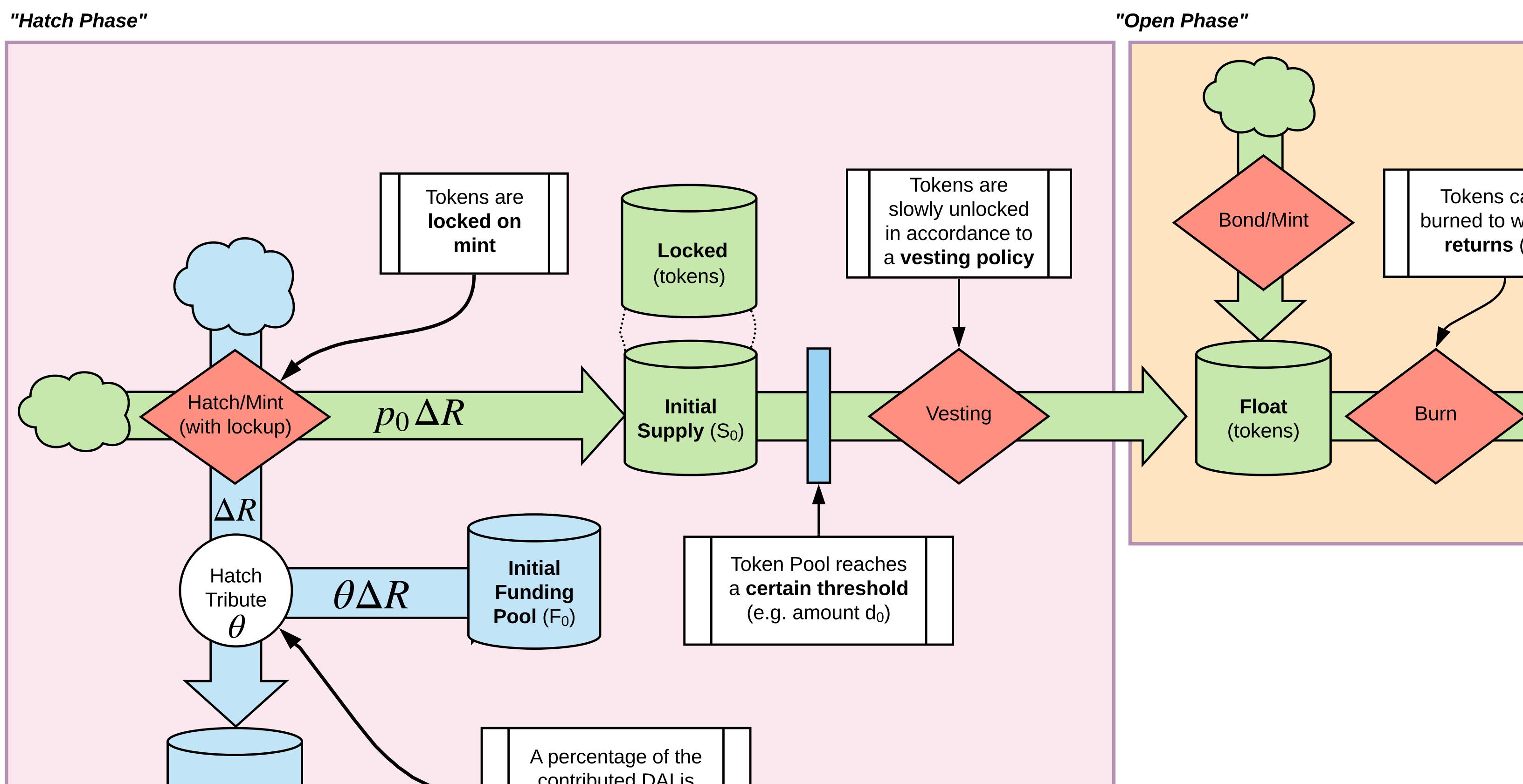
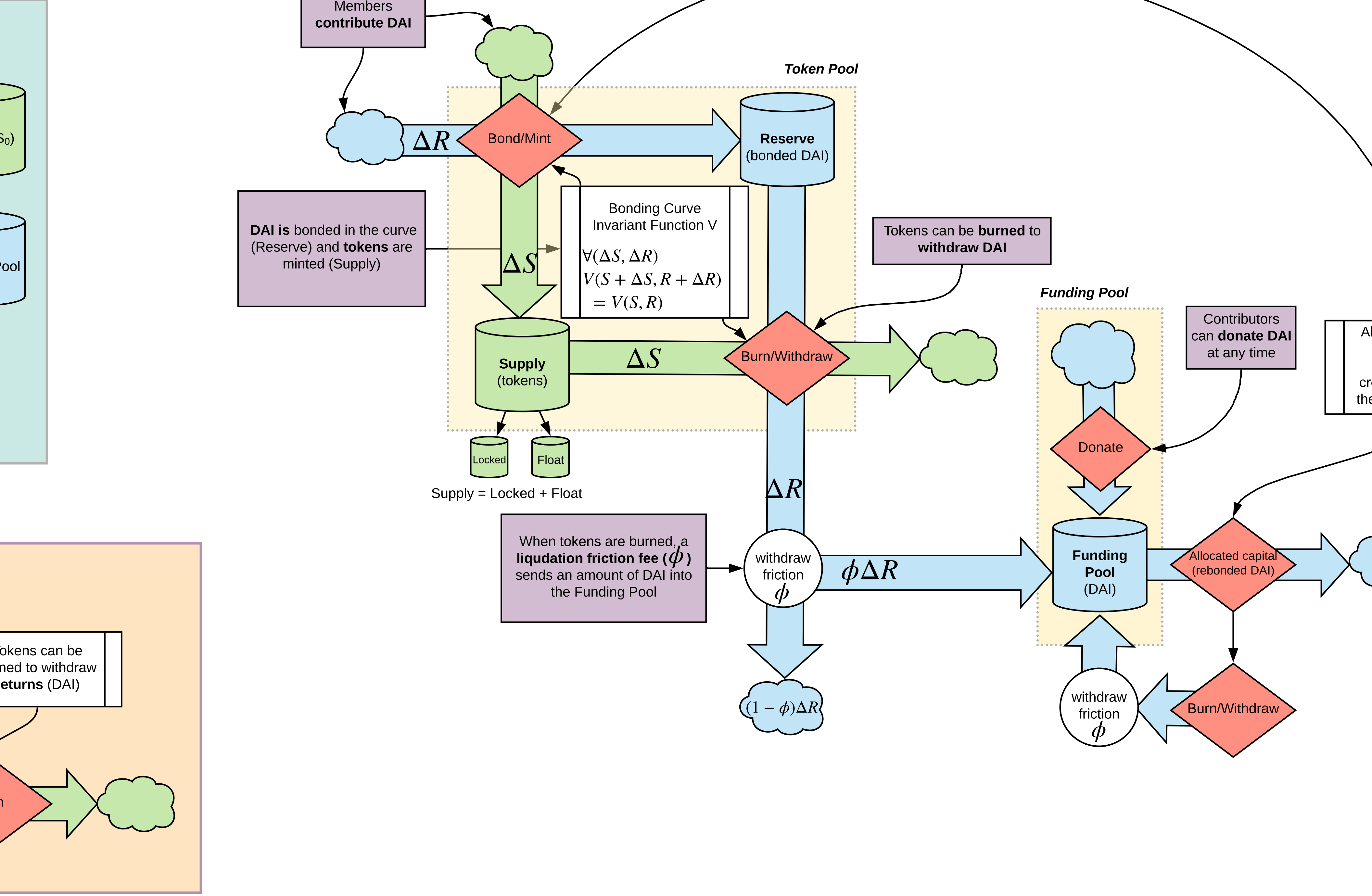
Contributors **donate** to the Commons Multisig to fund the development of the Commons
Donate
Multisig
Allocate for completion of Predetermined **Milestones** to create commons
Build


Hatch/Mint
$p_0 \Delta R$
Initial Supply (S_0)
ΔR
funding pool share θ
$\theta \Delta R$
Initial Funding Pool (DAI)
Initial Reserve (R_0)


"Hatch Phase"
"Open Phase"
Tokens are **locked on mint**
Hatch/Mint (with lockup)
$p_0 \Delta R$
Locked (tokens)
Initial Supply (S_0)
Tokens are slowly unlocked in accordance to a **vesting policy**
Vesting
Token Pool reaches a **certain threshold** (e.g. amount d_0)
ΔR
Hatch Tribute θ
$\theta \Delta R$
Initial Funding Pool (F_0)
A percentage of the
Bond/Mint
Float (tokens)
Burn



Members contribute DAI
Token Pool
ΔR
Bond/Mint
Reserve (bonded DAI)
DAI is bonded in the curve (Reserve) and tokens are minted (Supply)
ΔS
Bonding Curve Invariant Function V
$\forall(\Delta S, \Delta R)$
$V(S + \Delta S, R + \Delta R) = V(S, R)$
Tokens can be burned to withdraw DAI
Supply (tokens)
ΔS
Burn/Withdraw
Locked
Float
Supply = Locked + Float
Funding Pool
Contributors can donate DAI at any time
Donate
ΔR
When tokens are burned, a liqudation friction fee (ϕ) sends an amount of DAI into the Funding Pool
withdraw friction ϕ
$\phi \Delta R$
Funding Pool (DAI)
Allocated capital (rebonded DAI)
$(1 - \phi)\Delta R$
withdraw friction ϕ
Burn/Withdraw
okens can be
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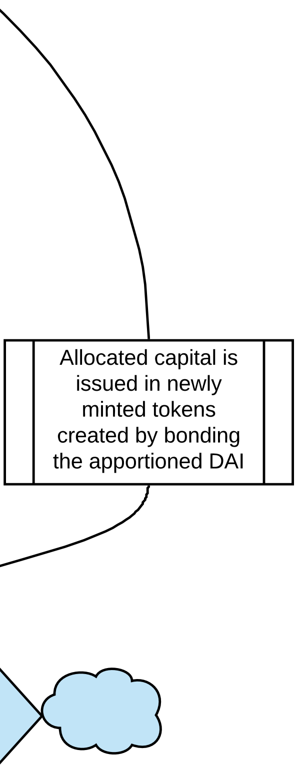
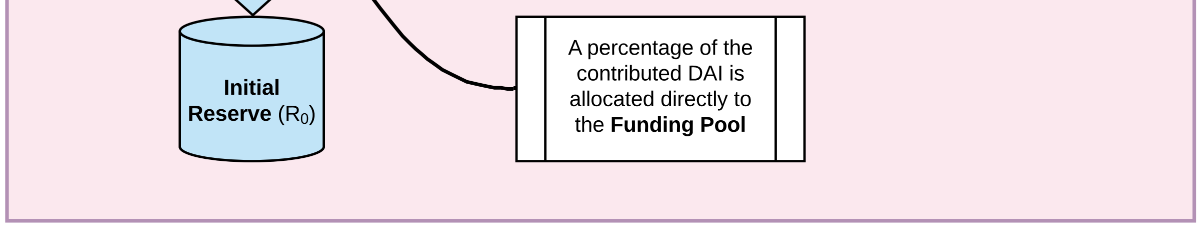
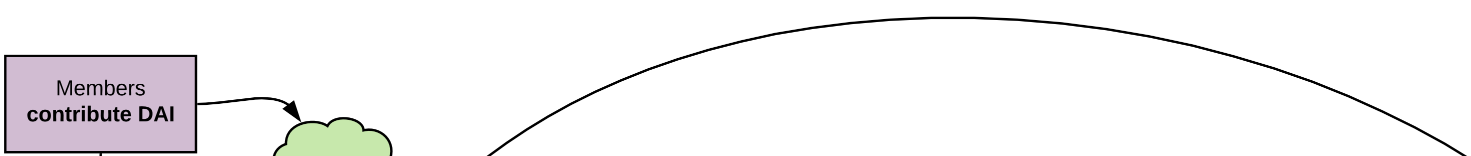
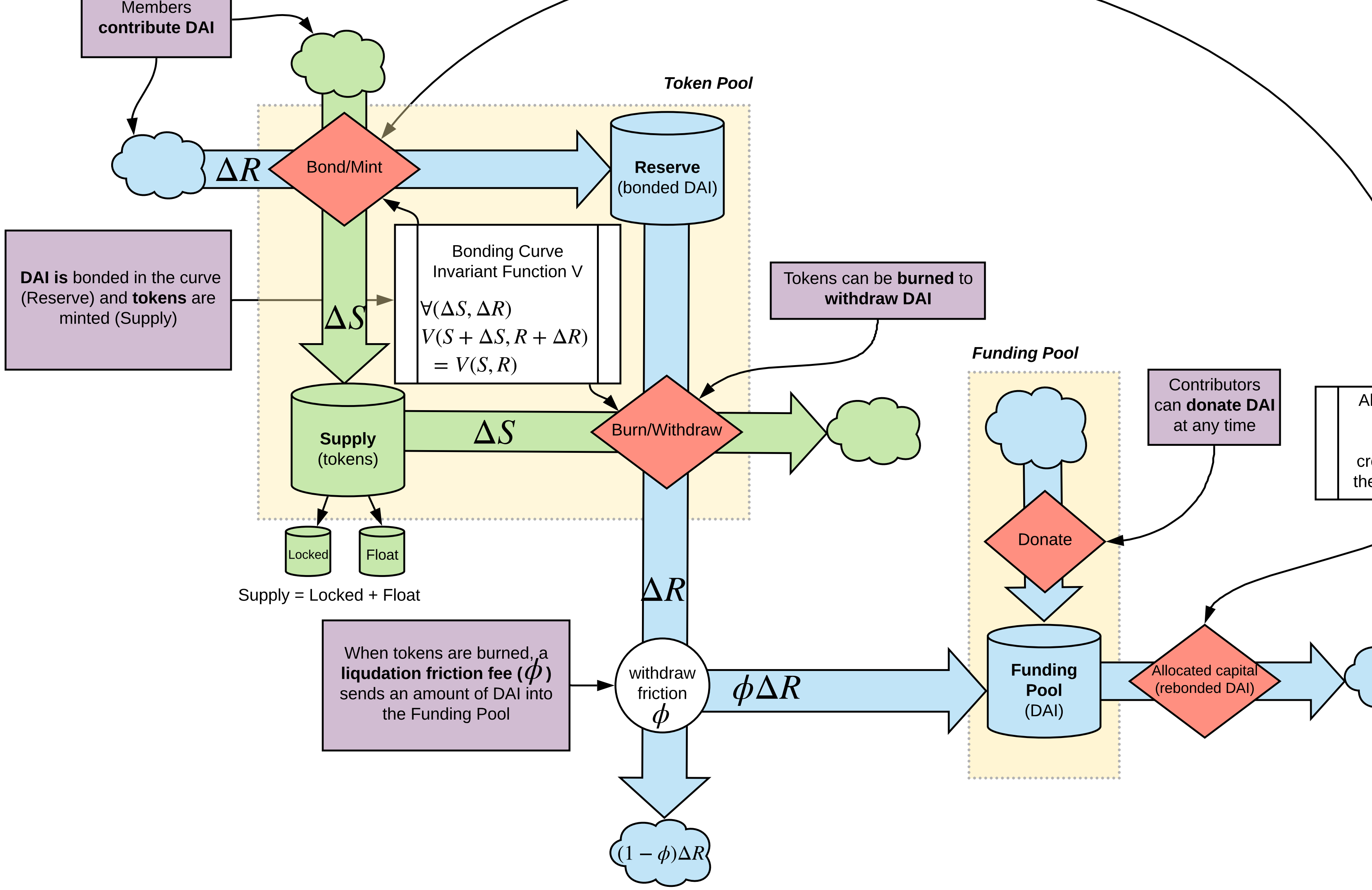
eturns (DAI)


Allocated capital is issued in newly minted tokens created by bonding the apportioned DAI



Initial Reserve (R_0)
A percentage of the contributed DAI is allocated directly to the Funding Pool


Members
contribute DAI



Members
contribute DAI
Token Pool
ΔR
Bond/Mint
Reserve
(bonded DAI)
DAI is bonded in the curve (Reserve) and tokens are minted (Supply)
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Bonding Curve
Invariant Function V
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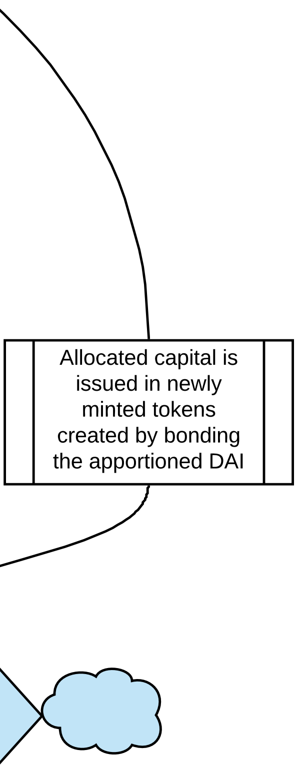
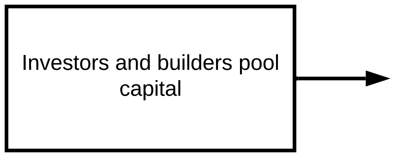
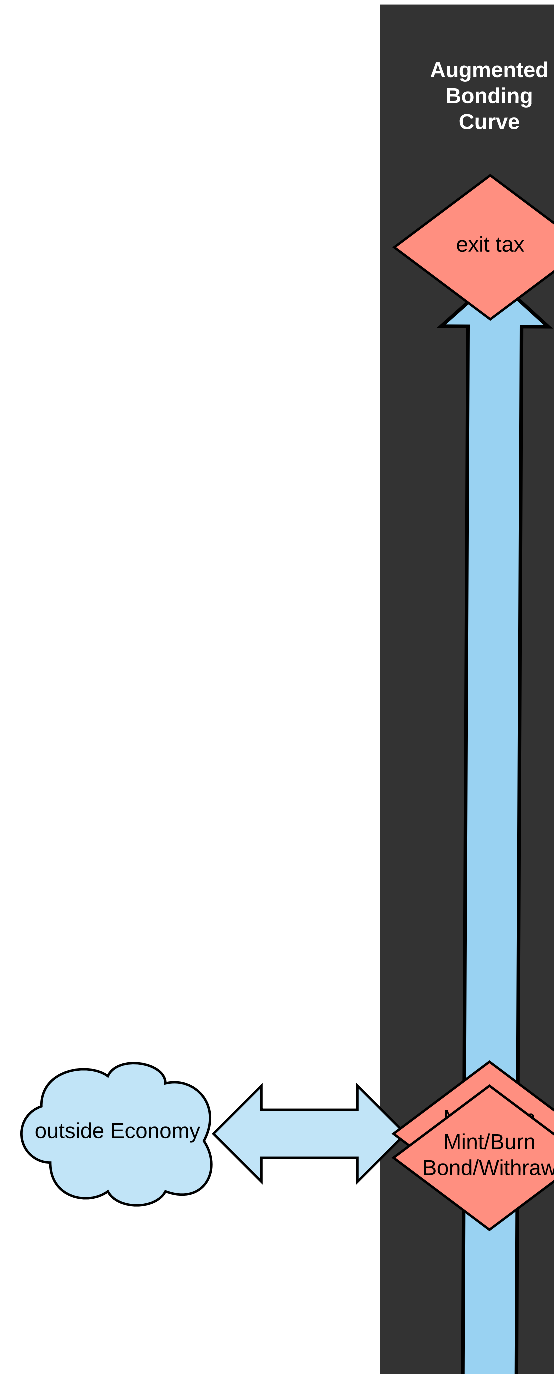
Tokens can be burned to withdraw DAI
Supply
(tokens)
ΔS
Burn/Withdraw
Funding Pool
Contributors can donate DAI at any time
Locked
Float
Supply = Locked + Float
Donate
ΔR
When tokens are burned, a liqudation friction fee (ϕ) sends an amount of DAI into the Funding Pool
withdraw
friction
ϕ
$\phi \Delta R$
Funding
Pool
(DAI)
Allocated capital
(rebonded DAI)
$(1-\phi)\Delta R$

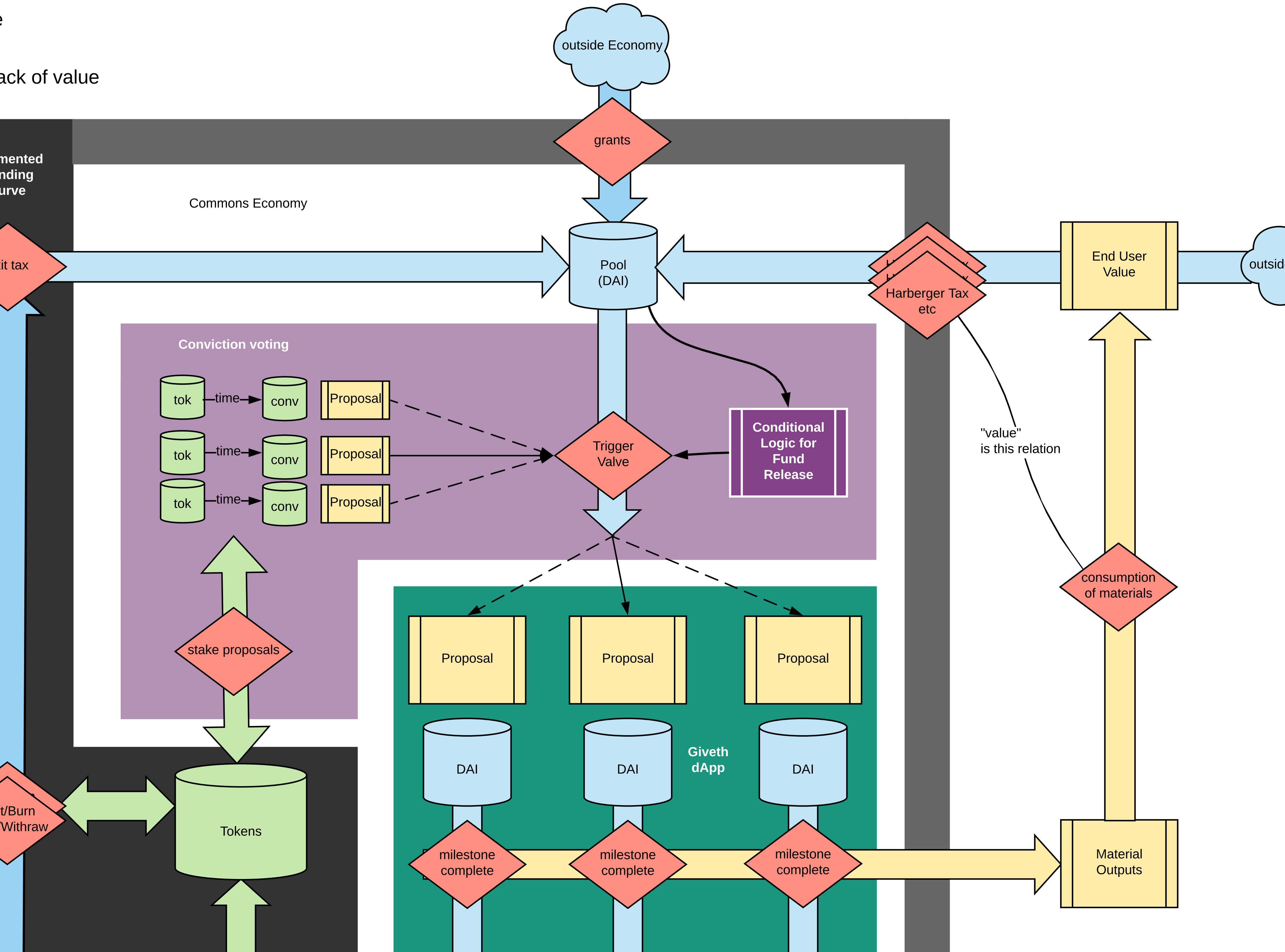

Allocated capital is issued in newly minted tokens created by bonding the apportioned DAI


Investors and builders pool capital

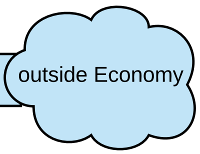
Cyber-Physical Commons

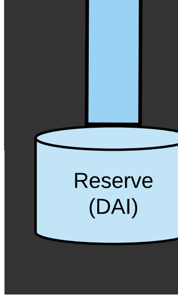
- Decentralized Governance
- Continuous Funding
- Regenerative returns/flowback of

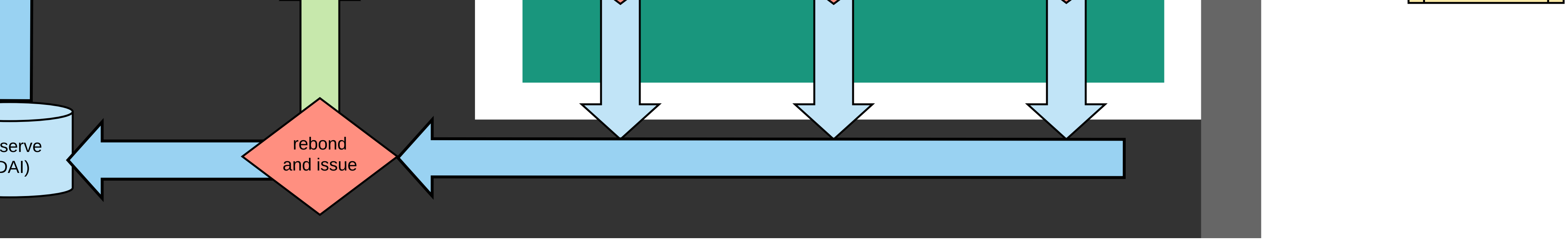




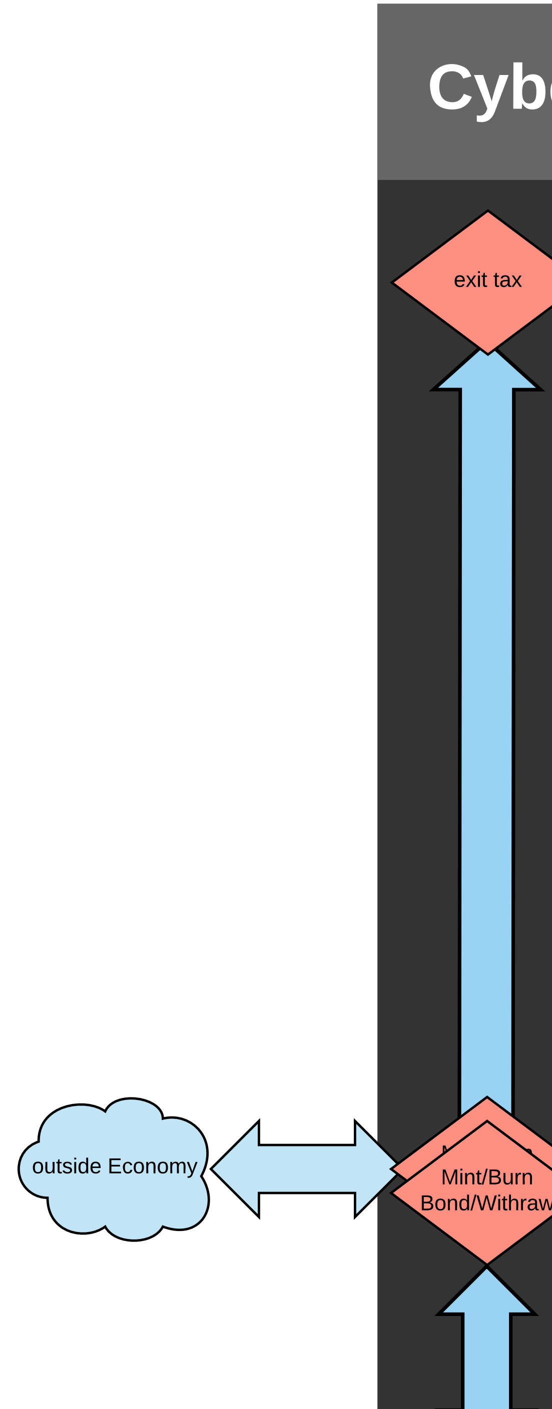
outside Economy
grants
Commons Economy
Pool (DAI)
Harberger Tax etc
End User Value
Conviction voting
tok
time
conv
Proposal
Trigger Valve
Conditional Logic for Fund Release
"value" is this relation
consumption of materials
stake proposals
Proposal
Proposal
Proposal
DAI
DAI
DAI
Giveth dApp
Tokens
milestone complete
milestone complete
milestone complete
Material Outputs


outside Economy

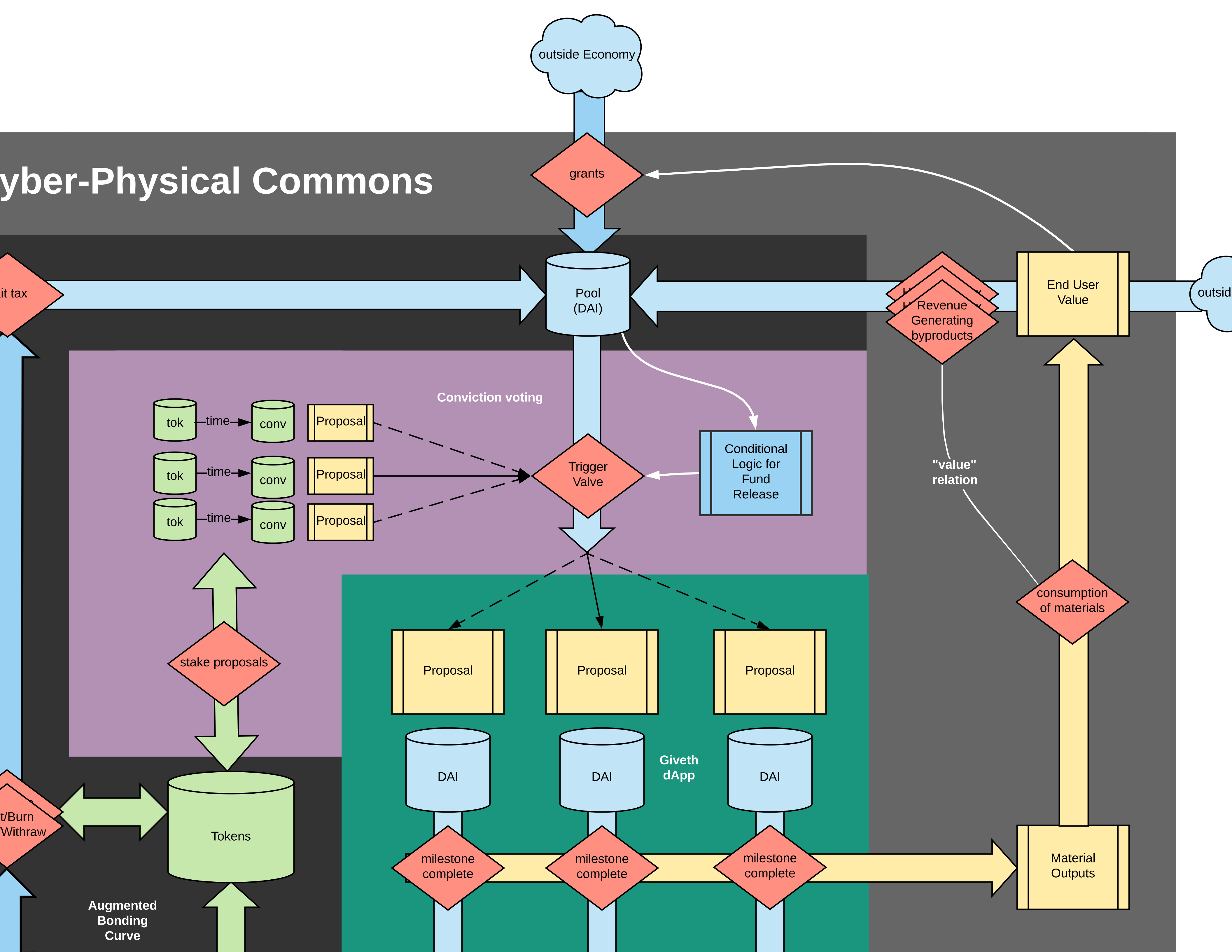

Reserve
(DAI)



Bonding DAI at time of
payoff to give
recipients tokens



Cyb
exit tax
outside Economy
Mint/Burn
Bond/Withraw



outside Economy
yber-Physical Commons
grants
it tax
Pool
(DAI)
End User
Value
outsid
Revenue
Generating
byproducts
Conviction voting
tok
time
conv
Proposal
Trigger
Valve
Conditional
Logic for
Fund
Release
"value"
relation
consumption
of materials
stake proposals
Proposal
Proposal
Proposal
DAI
DAI
DAI
Giveth
dApp
t/Burn
Withraw
Tokens
milestone
complete
milestone
complete
milestone
complete
Material
Outputs
Augmented
Bonding
Curve

outside Economy

Notes on Revenue

What have is a sort of mixture of the 3 types above

blockchain provenance really changes the relationship between club goods and common-pool resources. we should talk aboutit

Rivalrous

Non-rivalrous

https://en.wikipedia.org/wiki/Club_good

commons source software
(discount token model)

open source software

"Harberger Tax Model"
pay for use, possession, can flow to whomever needs it most

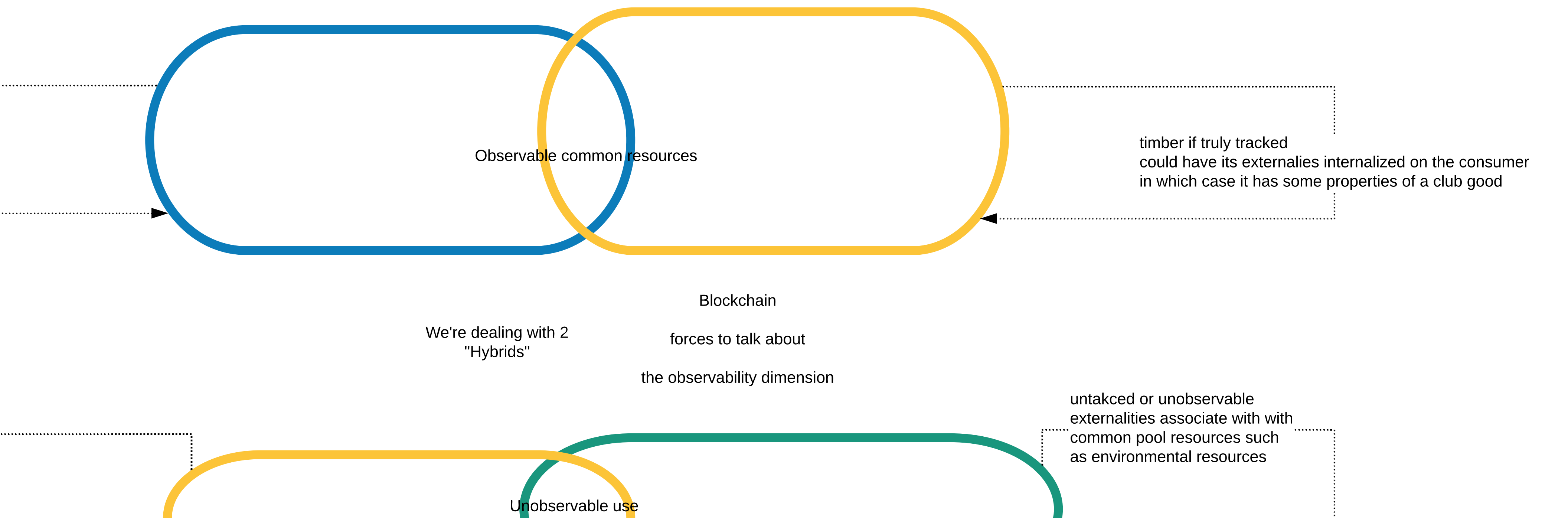
"Harberger Tax Model"
pay for use, possession, can flow to whomever needs it most

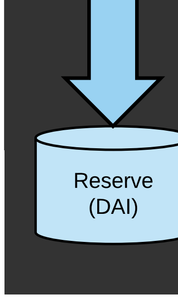
Excludable	Non-excludable
Private goods food, clothing, cars, parking spaces	**Common-pool resources** fish stocks, timber, coal
Club goods cinemas, private parks, satellite television	**Public goods** free-to-air television, air, national defense

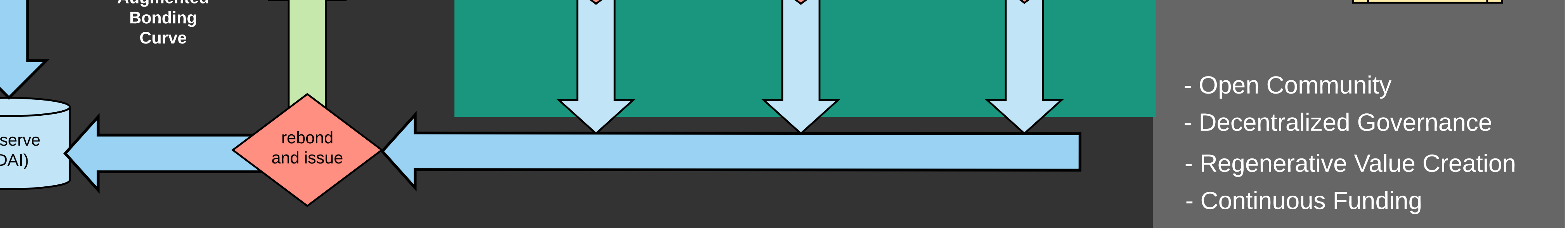
Grants from those who can afford it, likely free use by those who cannot

"Discount Token Model"
'Co-op' fee for license/use for non-members, that can be offset by members

Grants from those who can afford it, likely free use by those who cannot

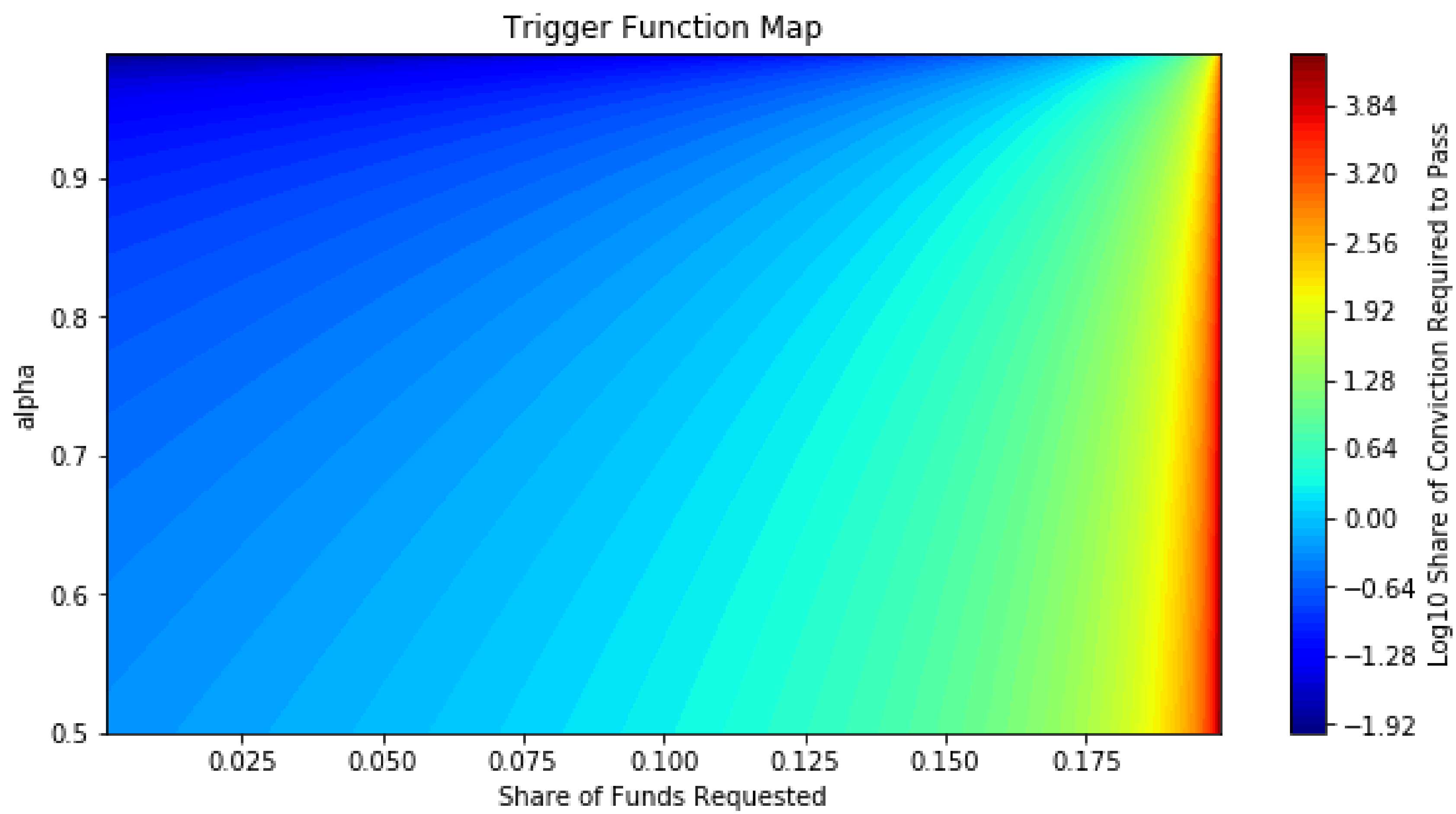


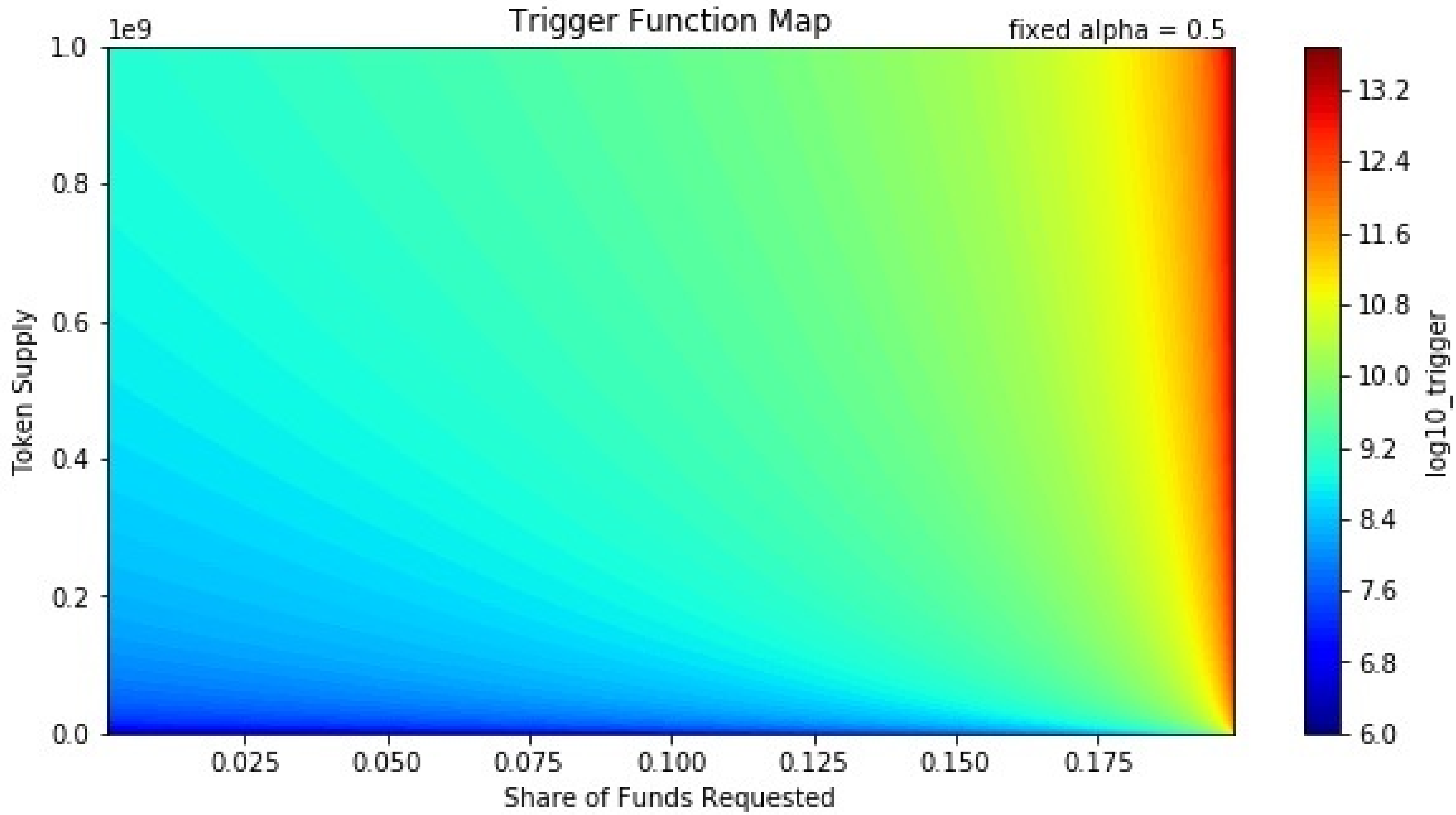

Reserve
(DAI)



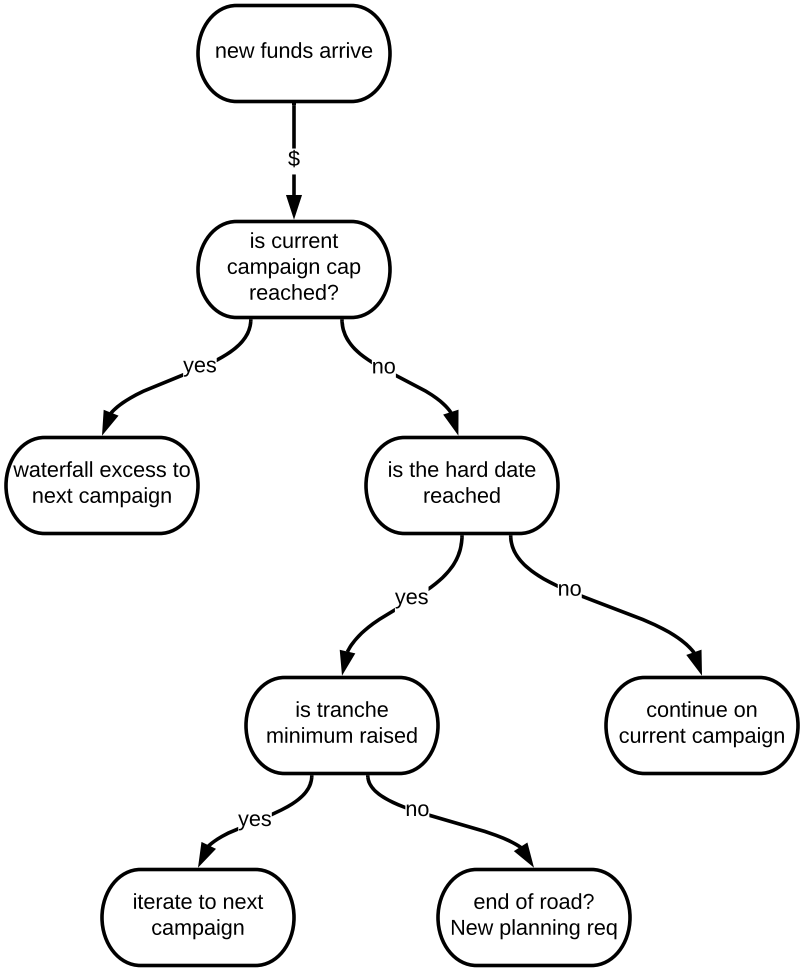
Bonding
Curve
rebond
and issue
- Open Community
- Decentralized Governance
- Regenerative Value Creation
- Continuous Funding

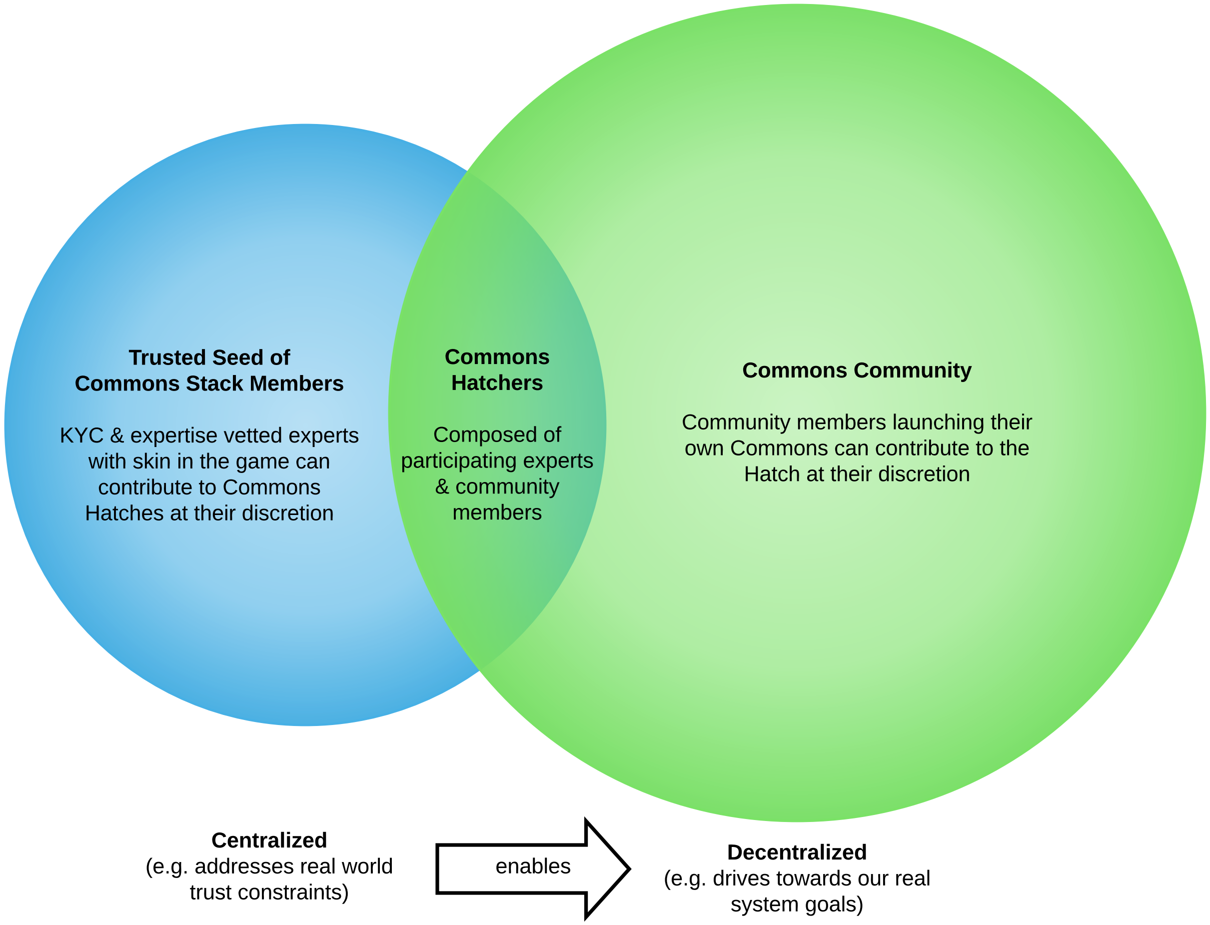

Unobservable use
common resources

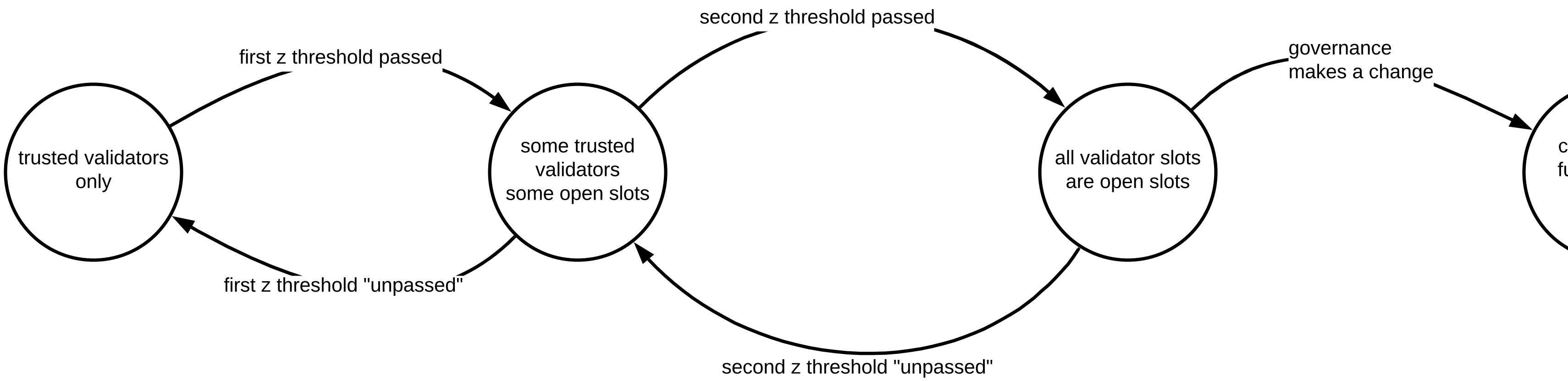

Trigger Function Map
alpha
0.9
0.8
0.7
0.6
0.5
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Share of Funds Requested
3.84
3.20
2.56
1.92
1.28
0.64
0.00
−0.64
−1.28
−1.92
Log10 Share of Conviction Required to Pass



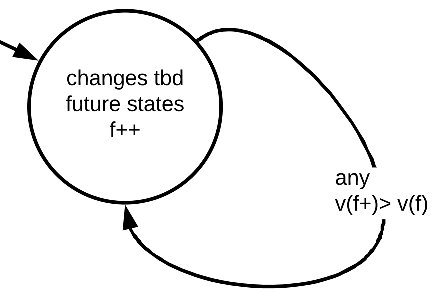
Trigger Function Map
fixed alpha = 0.5
1e9
1.0
0.8
0.6
0.4
0.2
0.0
Token Supply
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Share of Funds Requested
13.2
12.4
11.6
10.8
10.0
9.2
8.4
7.6
6.8
6.0
log10_trigger

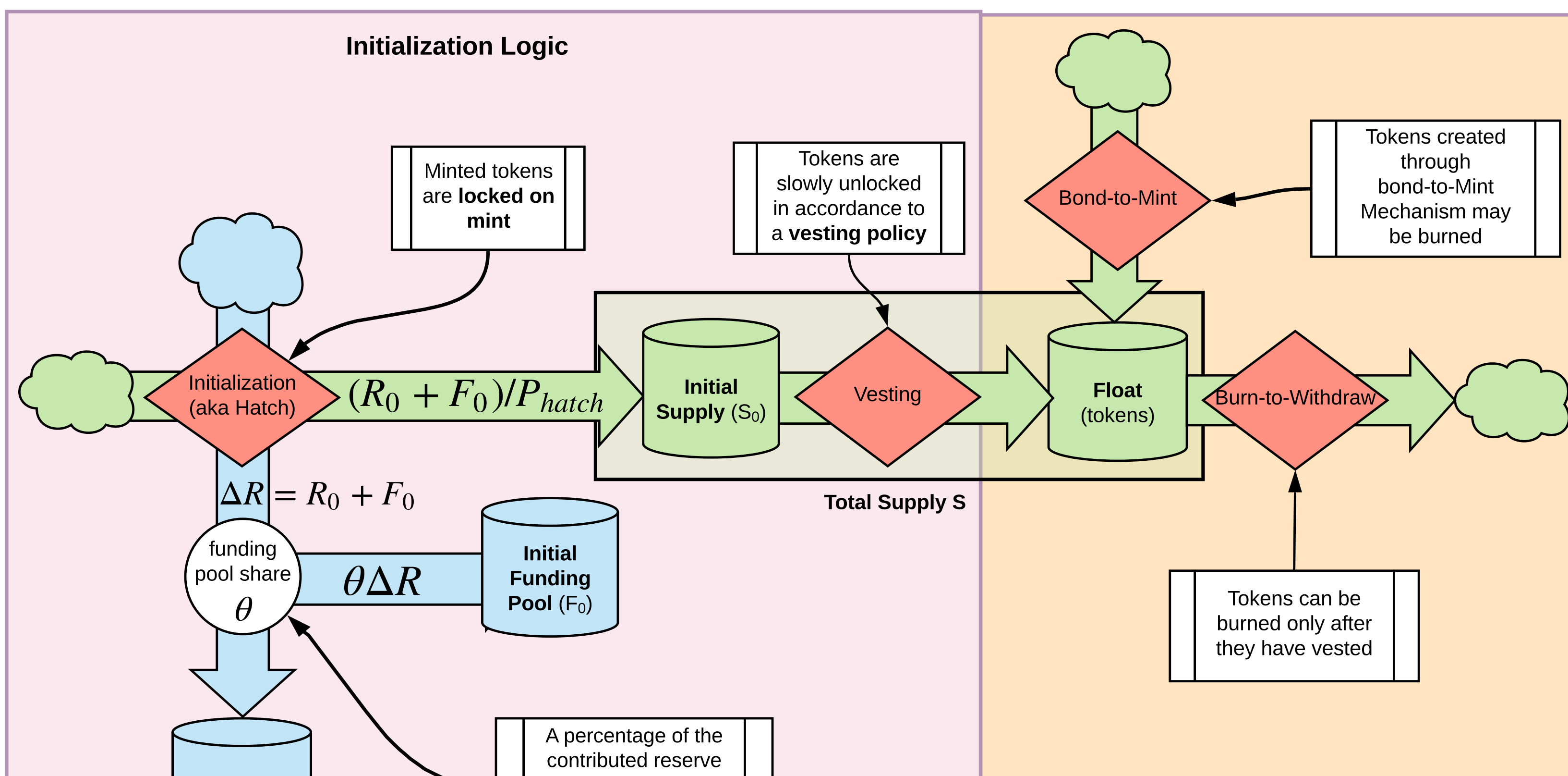

new funds arrive
$
is current campaign cap reached?
yes
no
waterfall excess to next campaign
is the hard date reached
yes
no
is tranche minimum raised
continue on current campaign
yes
no
iterate to next campaign
end of road? New planning req


Trusted Seed of
Commons Stack Members
KYC & expertise vetted experts with skin in the game can contribute to Commons Hatches at their discretion
Commons
Hatchers
Composed of participating experts & community members
Commons Community
Community members launching their own Commons can contribute to the Hatch at their discretion
Centralized
(e.g. addresses real world trust constraints)
enables
Decentralized
(e.g. drives towards our real system goals)

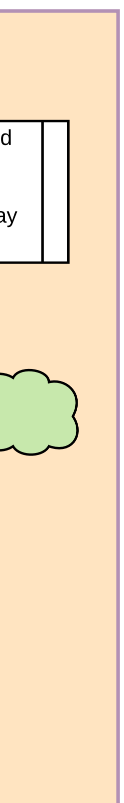


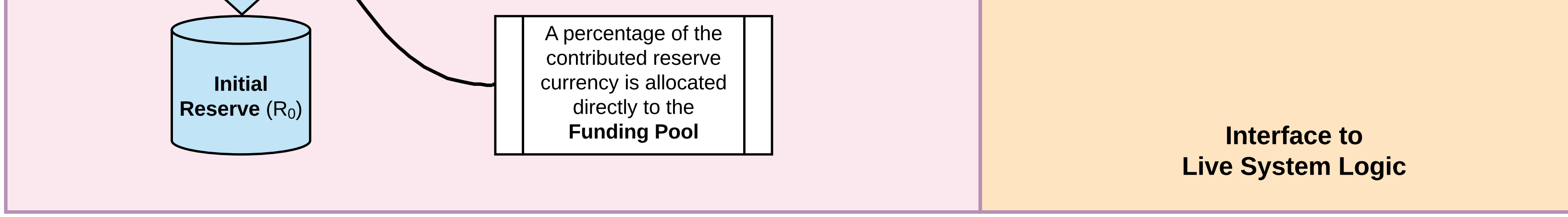
trusted validators only
some trusted validators some open slots
all validator slots are open slots
first z threshold passed
first z threshold "unpassed"
second z threshold passed
second z threshold "unpassed"
governance makes a change

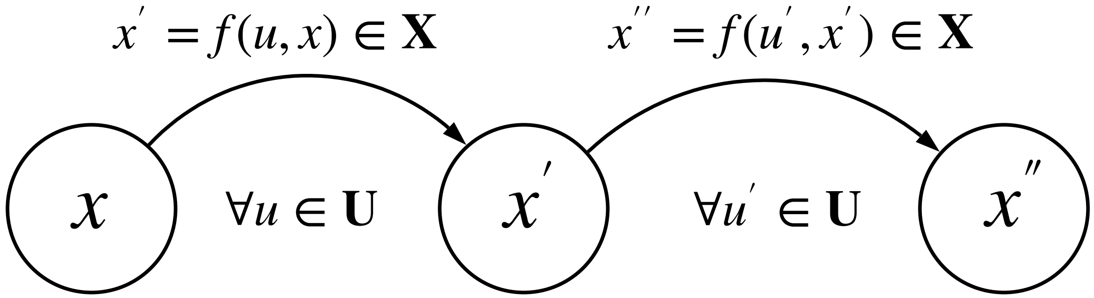

changes tbd
future states
f++
any
v(f+)> v(f)



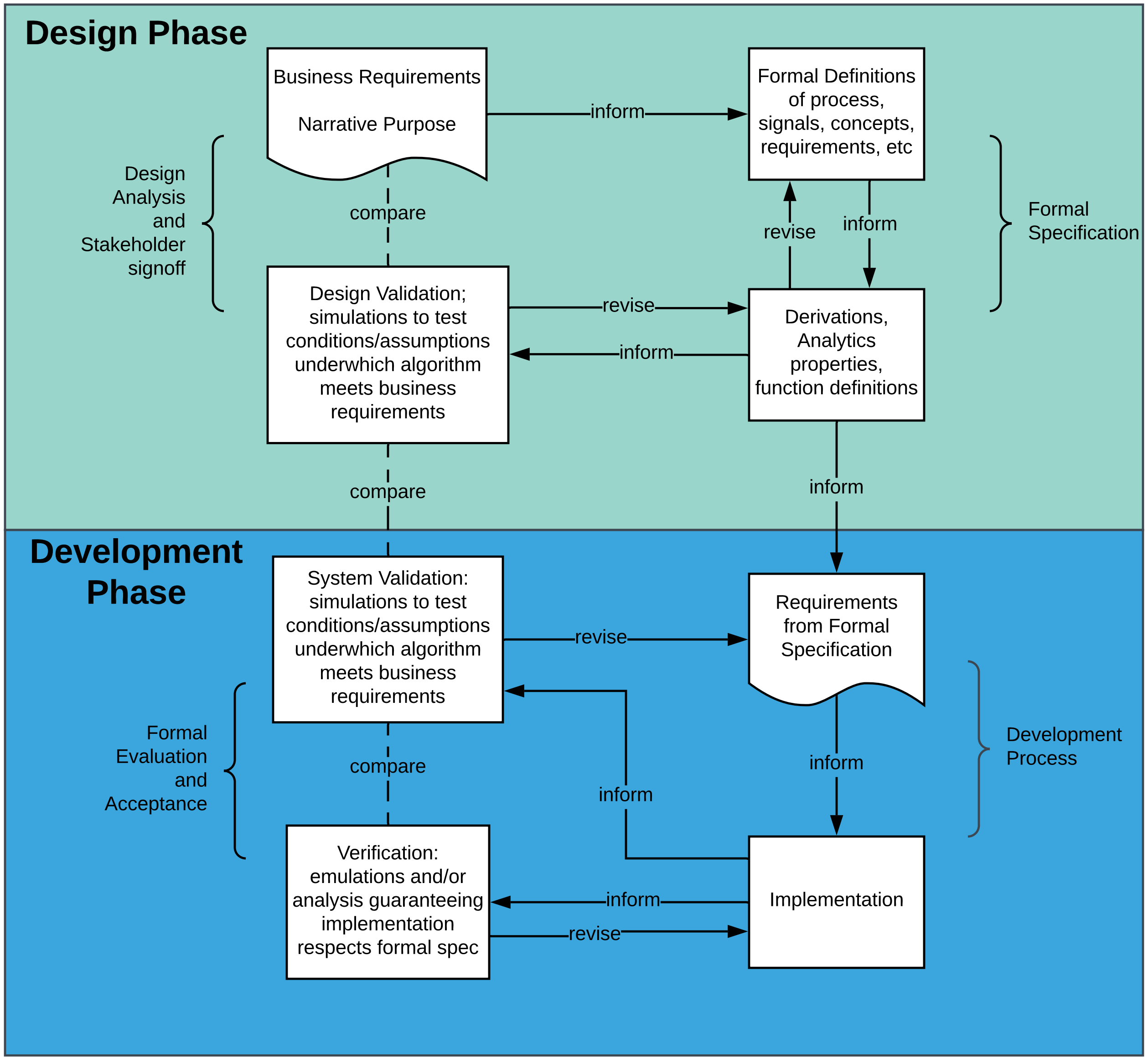
Initialization Logic
Minted tokens are **locked on mint**
Tokens are slowly unlocked in accordance to a **vesting policy**
Bond-to-Mint
Tokens created through bond-to-Mint Mechanism may be burned
Initialization (aka Hatch)
$(R_0 + F_0)/P_{hatch}$
Initial Supply (S_0)
Vesting
Float (tokens)
Burn-to-Withdraw
Total Supply S
$\Delta R = R_0 + F_0$
funding pool share θ
$\theta \Delta R$
Initial Funding Pool (F_0)
Tokens can be burned only after they have vested
A percentage of the contributed reserve


d
ay


Initial Reserve (R_0)
A percentage of the contributed reserve currency is allocated directly to the **Funding Pool**
Interface to Live System Logic



$$V\left(x\right) = V\left(f\left(u, x\right)\right) = V\left(x'\right) = V\left(f\left(u', x'\right)\right) = V\left(x''\right)$$


Design Phase
Business Requirements
Narrative Purpose
inform
Formal Definitions of process, signals, concepts, requirements, etc
Design Analysis and Stakeholder signoff
compare
revise
inform
Formal Specification
Design Validation; simulations to test conditions/assumptions underwhich algorithm meets business requirements
revise
inform
Derivations, Analytics properties, function definitions
compare
inform
Development Phase
System Validation: simulations to test conditions/assumptions underwhich algorithm meets business requirements
revise
Requirements from Formal Specification
Formal Evaluation and Acceptance
compare
inform
inform
Development Process
Verification: emulations and/or analysis guaranteeing implementation respects formal spec
inform
revise
Implementation

Potential Future Work